5G RAN

Multi-Frequency Smart Selection Feature Parameter Description

Issue	Draft B
Date	2026-01-30

HUAWEI TECHNOLOGIES CO., LTD.

Trademarks and Permissions

Notice

Huawei Technologies Co., Ltd.

Address:	Huawei Industrial Base Bantian, Longgang Shenzhen 518129 People's Republic of China
Website:	https://www.huawei.com
Email:	support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:
https://www.huawei.com/en/psirt/vul-response-process
For vulnerability information, enterprise customers can visit the following web page:
https://securitybulletin.huawei.com/enterprise/en/security-advisory

Contents

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes

  Revised descriptions of the principles of multi-frequency beam coordination. For details, see **5.1 Principles**.

  Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 06 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added LTE-NR joint evaluation of large-packet service status for the "protection of UEs under weak uplink coverage" function. For details, see: • **4.2.1.1 Decision on Whether to Initiate an Uplink-Coverage-based Inter-Frequency Handover** • **4.2.3.2.1 Prerequisite Functions** • **4.2.4.1.1 Data Preparation** • **4.2.4.1.2 Using MML Commands**	Modified parameter: Added the **LNR_LRG_PKT_UL_WEAK_COV_SW** option to the **NRCellMobilityConfig**.***InterFreqHoExtSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added a measurement timer for protection against weak uplink coverage. For details, see: • **4.2.1.2 Uplink-Coverage-based Inter-Frequency Handover Execution** • **4.2.4.1.1 Data Preparation** • **4.2.4.1.2 Using MML Commands**	Added the **NRCellMobilityConfig**.***UlWeakCovPrtMeasTimer*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for latency-based optimal carrier selection by the UMPThb. For details, see **7.3.3 Hardware**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for prohibition of multi-frequency smart selection for NSA UEs whose RRC connections are reestablished. For details, see: • **4.1.1.1.2 Triggering of Experience-based Smart Carrier Selection** • **4.1.2.1 Benefits** • **4.1.2.2 Impacts** • **4.1.4.1.1 Data Preparation** • **4.1.4.1.2 Using MML Commands**	Modified NR parameter: Added the **NSA_REEST_FORB_MUL_FREQ_SEL_SW** option to the **gNodeBParam**.***NetworkingOptionOptSw*** parameter. Modified LTE parameter: Added the **NSA_REEST_NOTIFY_TO_NR_SW** option to the **EnodebAlgoExtSwitch**.***NsaDcAlgoExtSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

  Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-061223	Experience Boosting based on Multi-Band Coordination	4.1 Experience-based Smart Carrier Selection 4.2 Protection of UEs Under Weak Uplink Coverage 4.3 User-Experience-based Coverage Expansion 4.4 User-Experience-based NSA SCG Coverage Expansion 4.5 SCC Coverage Expansion
FOFD-070201	Multi-Frequency Spatial Smart Carrier Selection	5 Multi-Frequency Beam Coordination
FOFD-071901	Macro-Micro Smart Carrier Selection	6 Macro-Micro Smart Carrier Selection
FOFD-091209	Latency-based Optimal Carrier Selection (NR)	7 Latency-based Optimal Carrier Selection

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Experience-based smart carrier selection	Supported in both NSA and SA networking. For NSA UEs, experience-based smart carrier selection can be performed on the NR side only if the anchor on the LTE side remains unchanged.	4.1 Experience-based Smart Carrier Selection
Protection of UEs under weak uplink coverage	None	4.2 Protection of UEs Under Weak Uplink Coverage

Function Name	Difference	Chapter/Section
User-experience-based coverage expansion	Supported in SA networking	4.3 User-Experience-based Coverage Expansion
User-experience-based NSA SCG coverage expansion	Supported in NSA networking	4.4 User-Experience-based NSA SCG Coverage Expansion
SCC coverage expansion	None	4.5 SCC Coverage Expansion
Multi-frequency beam coordination	Supported in both NSA and SA networking. For NSA UEs, multi-frequency beam coordination can be performed on the NR side only if the anchor on the LTE side remains unchanged.	5 Multi-Frequency Beam Coordination
Macro-micro intra-frequency optimal carrier selection	None	6.1.1 Macro-Micro Intra-Frequency Optimal Carrier Selection
Latency-based optimal carrier selection	Supported only in SA networking	7 Latency-based Optimal Carrier Selection

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Experience-based smart carrier selection	Supported only in low frequency bands	4.1 Experience-based Smart Carrier Selection
Protection of UEs under weak uplink coverage	Supported only in low frequency bands	4.2 Protection of UEs Under Weak Uplink Coverage
User-experience-based coverage expansion	Supported only in low frequency bands	4.3 User-Experience-based Coverage Expansion

Function Name	Difference	Chapter/Section
User-experience-based NSA SCG coverage expansion	Supported only in low frequency bands	**4.4 User-Experience-based NSA SCG Coverage Expansion**
SCC coverage expansion	Supported only in low frequency bands	**4.5 SCC Coverage Expansion**
Multi-frequency beam coordination	Supported only in low frequency bands	**5 Multi-Frequency Beam Coordination**
Macro-micro intra-frequency optimal carrier selection	Supported only in low frequency bands	**6.1.1 Macro-Micro Intra-Frequency Optimal Carrier Selection**
Latency-based optimal carrier selection	Supported only in low frequency bands	**7 Latency-based Optimal Carrier Selection**

3 Overview

iHashBand is a 5G intelligent coordination solution proposed by Huawei for multi-frequency networks. This solution is used to efficiently coordinate various resources (including spectrums, channels, beams, and base station models) of multi-frequency networks and to implement intelligent convergence and mutual assistance among frequency bands. By fully exploiting the characteristics and advantages of each frequency band, it can provide optimal network performance and user experience even with limited spectrum resources, thereby maximizing spectrum value. The iHashBand solution includes functions in the following categories:

- **Multi-frequency smart selection: Functions in this category can intelligently select carriers or carrier combinations that provide the optimal user experience for UEs based on factors such as the spectral efficiency, bandwidth, and load, thereby improving user experience.** These functions are detailed in this document. The application scenarios of the functions are briefed in **Table 3-1**.
- **Multi-frequency smart aggregation: Functions in this category can quickly and intelligently aggregate time-frequency resources of multiple cells operating on different frequencies for individual UEs, thereby increasing UE or cell throughput.** These functions are detailed in *Multi-Frequency Smart Aggregation* and *CA Experience Improvement*. Examples of the functions are as follows:
  - 3CC aggregation: This function aggregates three intra-band or inter-band carriers in the downlink to provide higher bandwidth. As a UE in the 3CC aggregation state can use three carriers at the same time, it can enjoy a higher peak data rate. For details about 3CC aggregation, see *CA Experience Improvement* .
  - Inter-carrier frequency selective scheduling: This function enables the base station to allocate resources of the carriers with high spectral efficiency and low load to individual downlink CA UEs whenever possible based on the spectral efficiency difference and millisecond-level load difference among carriers. This achieves appropriate downlink resource allocation for CA UEs and thereby increases the average downlink UE throughput on the entire network. For details about inter-carrier frequency selective scheduling, see *Multi-Frequency Smart Aggregation*.

Table 3-1 Functions of multi-frequency smart selection and their application scenarios

Feature	Function	Application Scenario	Chapter/ Section
FOFD-061223 Experience Boosting based on Multi-Band Coordination	Experience-based smart carrier selection	On a multi-carrier network, a base station triggers handovers or carrier changes for UEs mainly based on cell load and UE-reported reference signal received power (RSRP) and selects the optimal carriers for UEs. As user experience difference is not considered, the carriers selected by the base station for some UEs are unable to provide the optimal experience. To adaptively select carriers capable of providing the optimal experience for UEs, the experience-based smart carrier selection function is introduced.	**4.1 Experience-based Smart Carrier Selection**
	Protection of UEs under weak uplink coverage	The application scenario of this function is complementary to that of experience-based smart carrier selection. In multi-carrier coverage areas with weak uplink coverage, if measurement reports from UEs fail to reach the base station, experience-based smart carrier selection cannot take effect. To ensure user experience of UEs under weak uplink coverage, the "protection of UEs under weak uplink coverage" function is introduced.	**4.2 Protection of UEs Under Weak Uplink Coverage**

Feature	Function	Application Scenario	Chapter/ Section
	User-experience-based coverage expansion	On a multi-carrier network where the frequency bands have varying coverage capabilities and bandwidths, if the frequency band with good uplink and downlink coverage and a small bandwidth is congested because a large number of UEs have accessed this frequency band, the experience of some UEs will be affected. To reduce the load of cells on frequency bands with good coverage and small bandwidths, the user-experience-based coverage expansion function is introduced. This function works on SA networks.	**4.3 User-Experience-based Coverage Expansion**
	User-experience-based NSA SCG coverage expansion	The application scenario of this function is the same as that of user-experience-based coverage expansion except that this function works on NSA networks.	**4.4 User-Experience-based NSA SCG Coverage Expansion**
	SCC coverage expansion	In multi-carrier scenarios where the coverage of different frequency bands differs greatly, if there is discontinuous coverage, CA cannot be performed for cell-edge UEs, affecting user experience. To increase the average downlink throughput of cell-edge UEs through CA, SCC coverage expansion is introduced.	**4.5 SCC Coverage Expansion**

Feature	Function	Application Scenario	Chapter/ Section
FOFD-070201 Multi-Frequency Spatial Smart Carrier Selection	Multi-frequency beam coordination	In multi-carrier scenarios, UEs are densely distributed on some sounding reference signal (SRS) beams, which restricts the MU pairing capability and consequently hinders the average downlink UE throughput from increasing. The multi-frequency beam coordination function adjusts the downlink traffic distribution on SRS beams, thereby increasing the average downlink UE throughput on the entire network.	**5 Multi-Frequency Beam Coordination**
FOFD-071901 Macro-Micro Smart Carrier Selection	Macro-micro intra-frequency optimal carrier selection	When a UE moves to an overlapping area between outdoor and indoor cells respectively served by macro and micro base stations deployed on the same frequency, the UE may not obtain the optimal downlink experience after selecting the strongest cell to camp on because the macro and micro base stations differ greatly in aspects such as power and antenna gains. To address this issue, the macro-micro intra-frequency optimal carrier selection function is introduced with the aim of improving downlink user experience.	**6.1.1 Macro-Micro Intra-Frequency Optimal Carrier Selection**
FOFD-091209 Latency-based Optimal Carrier Selection (NR)	Latency-based optimal carrier selection	Cloud X services have high requirements on air interface latency. In multi-carrier scenarios where the coverage varies with carriers, the air interface latency also varies with carriers at a given UE location. To reduce the downlink air interface latency for UEs, the latency-based optimal carrier selection function is introduced.	**7 Latency-based Optimal Carrier Selection**

NOTE

- In NSA networking or NSA and SA hybrid networking, when the gNodeB selects carriers for UEs, it filters carrier combinations based on UE capabilities and LTE-NR frequency relationships. For details, see section "Intra-Band EN-DC" in *EPC-based NSA Basics*.
- For details about basic mobility functions such as measurement configuration delivery and handover execution, see *SA Mobility Management in Connected Mode*.
- For basic functions of CA, see *Carrier Aggregation*.

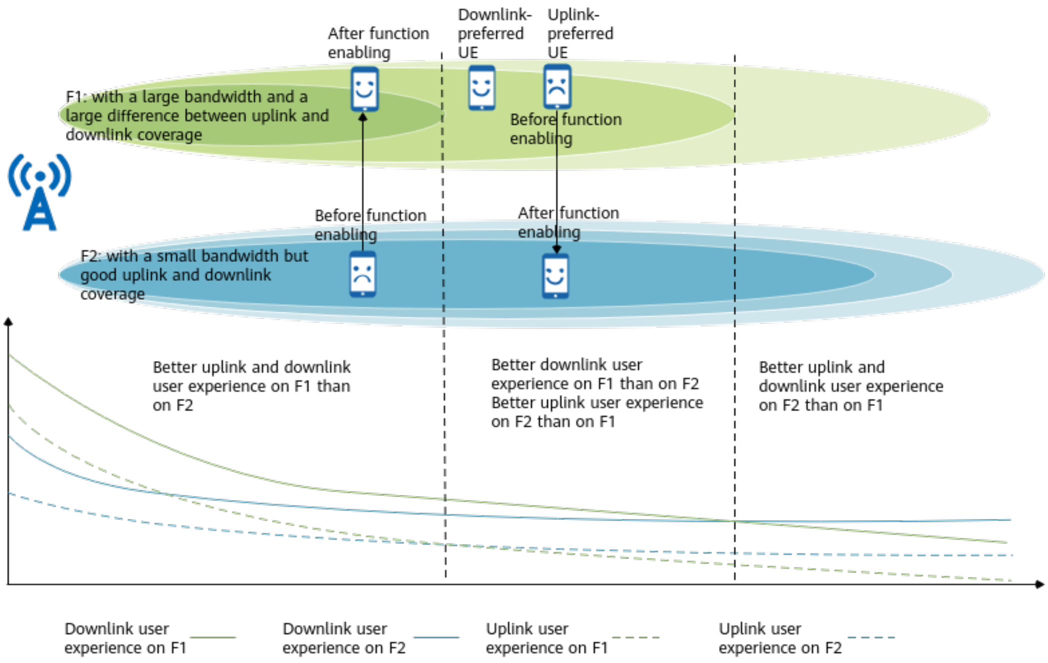
4 Experience Boosting based on Multi-Band Coordination

4.1 Experience-based Smart Carrier Selection

4.1.1 Principles

On a multi-carrier network, a base station triggers handovers or carrier changes for UEs mainly based on cell load and UE-reported RSRP and selects the optimal carriers for UEs. As user experience difference is not considered, the carriers selected by the base station for some UEs are unable to provide the optimal experience. To adaptively select carriers capable of providing the optimal experience for UEs, the experience-based smart carrier selection function illustrated in **Figure 4-1** is introduced. This function supports different smart carrier selection policies, and the optimal carrier or carrier combination selection policy varies depending on the configured smart carrier selection policy.

Figure 4-1 Schematic diagram of experience-based smart carrier selection



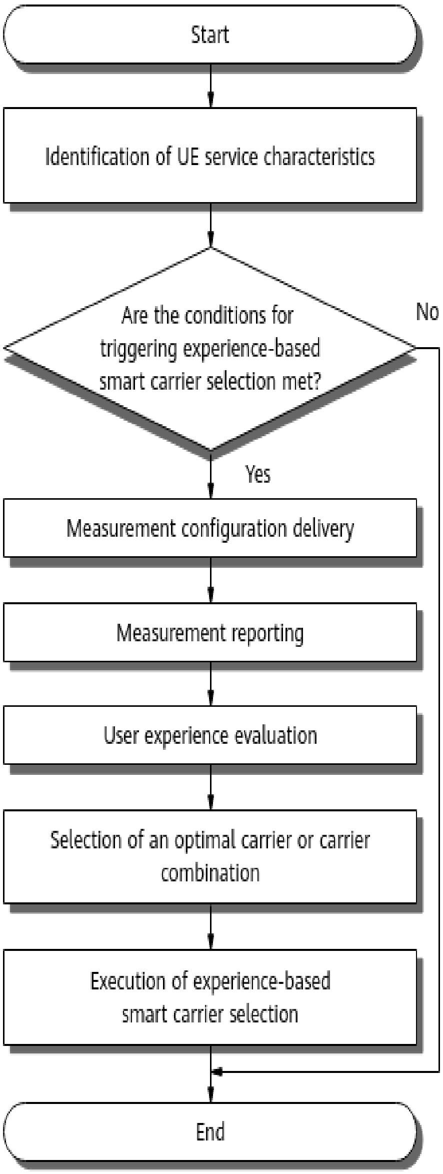
4.1.1.1 Basic Functions of Experience-based Smart Carrier Selection

With the experience-based smart carrier selection function, the gNodeB considers factors including cell coverage, bandwidth, and load when it selects carriers or carrier combinations that provide the optimal user experience for UEs.

This function is controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter.

Figure 4-2 shows the basic process of experience-based smart carrier selection.

Figure 4-2 Basic process of experience-based smart carrier selection



1. The method for identifying UE service characteristics is described in **4.1.1.1.1 UE Service Characteristics Identification**.
2. The scenarios where experience-based smart carrier selection can be triggered are described in **4.1.1.1.2 Triggering of Experience-based Smart Carrier Selection**.
3. How the gNodeB selects target frequencies for measurement is described in **4.1.1.1.3 Measurement Configuration Delivery**.
4. Measurement reporting from UEs to the gNodeB is described in **4.1.1.1.4 Measurement Reporting**.
5. How the gNodeB evaluates user experience is described in **4.1.1.1.5 User Experience Evaluation**.
6. The method for selecting an optimal carrier or carrier combination is described in **4.1.1.1.6 Optimal Carrier or Carrier Combination Selection**.

7. How the gNodeB determines whether to perform an inter-frequency handover or carrier change is described in **4.1.1.1.7 Execution of Experience-based Smart Carrier Selection**.

4.1.1.1.1 UE Service Characteristics Identification

Service characteristics vary with UEs. For a given UE, a carrier or carrier combination may provide the optimal experience in only uplink or downlink. As such, the service characteristics of UEs need to be identified. The gNodeB identifies the service characteristics of UEs based on their uplink and downlink traffic volumes, as described in **Table 4-1**.

Table 4-1 UE service characteristics identification

Absolute Value of Traffic Volume		Relative Value of Traffic Volume	Policy on Multi-Carrier Capability Combination Selection	Identification Result
Uplink Traffic Volume	Downlink Traffic Volume			
Large	Small	Not involved	Not involved	Uplink-preferred UEs
Small	Large	Not involved	Not involved	Downlink-preferred UE
Small	Small	Not involved	Not involved	The experience-based smart carrier selection function does not take effect.
Large	Large	Uplink traffic volume ≥ 30 times the downlink traffic volume	Not involved	Uplink-preferred UEs
		Downlink traffic volume ≥ 30 times the uplink traffic volume	Not involved	Downlink-preferred UE
		Others	Preference based on the uplink peak rate	Uplink-preferred UEs
			Preference based on the downlink bandwidth or downlink peak rate	Downlink-preferred UE

The policy on multi-carrier capability combination selection is specified by the **NRCellCaMgmtConfig**.***CaBandCombSelectionStrat*** parameter:

- When this parameter is set to **UL_PEAK_RATE_FIRST**, the uplink peak rate is preferentially considered.
- When this parameter is set to **BANDWIDTH_FIRST**, the downlink bandwidth is preferentially considered.
- When this parameter is set to **PEAK_RATE_FIRST**, the downlink peak rate is preferentially considered.

NOTE

- Immediately after a UE initially accesses a cell or is admitted to a cell through an incoming handover, an incoming radio resource control (RRC) connection reestablishment, or incoming RRC connection resumption, both the uplink and downlink traffic volumes of the UE are considered small by default. Therefore, experience-based smart carrier selection does not take effect for the UE.
- The experience-based smart carrier selection function does not take effect for UEs handed over to the local cell through this function until the inter-frequency measurement delay timer (specified by the **NRCellMobilityConfig**.***InterFreqMeasDelay*** parameter) expires.

4.1.1.1.2 Triggering of Experience-based Smart Carrier Selection

Experience-based smart carrier selection can be triggered based on traffic models, periodically triggered, or triggered based on weak uplink coverage. Whether this function can be successfully triggered for a UE depends on the current UE status, the multi-frequency smart selection mode configured for the cell, and the setting of the switch controlling differentiated downlink handling for CA-incapable UEs.

- The current UE status refers to whether the UE is an uplink- or downlink-preferred UE at the moment. For details, see **4.1.1.1.1 UE Service Characteristics Identification**.
- The multi-frequency smart selection mode is specified by the **NRCellAlgoSwitch**.***MultiFreqSmartSelMode*** parameter. Whether experience-based smart carrier selection can be triggered in a specific multi-frequency smart selection mode is described in **Table 4-2**.

Table 4-2 Relationship between multi-frequency smart selection mode and experience-based smart carrier selection

Setting of MultiFreqSmartSel-Mode	Current UE Status	Whether Experience-based Smart Carrier Selection Can Be Triggered
UL_DL_MODE	Uplink-preferred UEs	Yes. Experience-based smart carrier selection can be performed for the UE based on its uplink services.

Setting of MultiFreqSmartSel-Mode	Current UE Status	Whether Experience-based Smart Carrier Selection Can Be Triggered
	Downlink-preferred UE	Yes. Experience-based smart carrier selection can be performed for the UE based on its downlink services.
UL_ONLY_MODE	Uplink-preferred UEs	Yes. Experience-based smart carrier selection can be performed for the UE based on its uplink services.
	Downlink-preferred UE	No
DL_ONLY_MODE	Uplink-preferred UEs	No
	Downlink-preferred UE	Yes. Experience-based smart carrier selection can be performed for the UE based on its downlink services.

- When differentiated downlink handling for CA-incapable UEs is enabled, experience-based smart carrier selection does not take effect for downlink-preferred CA-incapable UEs. Differentiated downlink handling for CA-incapable UEs is controlled by the **CA_INCAP_UE_DL_DIFF_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter.

The specific triggering scenarios of experience-based smart carrier selection and the scenarios where this function takes effect are described in **Table 4-3**.

Table 4-3 Triggering scenarios of experience-based smart carrier selection and scenarios where this function takes effect

Triggering Mode	Triggering Scenario	Scenario Where the Function Takes Effect
By traffic model	• After a UE accesses the network, both the uplink and downlink traffic volumes of the UE are small. When the traffic volume in either direction increases, experience-based smart carrier selection will be triggered for the UE. • When the service characteristics (uplink- or downlink-preferred) of a UE change due to changes in the uplink and downlink traffic volumes, experience-based smart carrier selection will be triggered for the UE in the newly preferred direction. For example, if a UE changes from an uplink-preferred UE to a downlink-preferred UE, experience-based smart carrier selection will be triggered for the UE based on its downlink services.	• If a UE is currently an uplink-preferred UE, experience-based smart carrier selection can take effect for the UE when the multi-frequency smart selection mode is **UL_DL_MODE** or **UL_ONLY_MODE**, regardless of whether differentiated downlink handling for CA-incapable UEs is enabled. • If a UE is currently a downlink-preferred UE and differentiated downlink handling for CA-incapable UEs is enabled: – If the UE is a CA-incapable UE, experience-based smart carrier selection cannot take effect for the UE, regardless of the multi-frequency smart selection mode configuration. – If the UE is a CA-capable UE, experience-based smart carrier selection can take effect for the UE under either the **UL_DL_MODE** or **DL_ONLY_MODE** multi-frequency smart selection mode configuration. • If a UE is currently a downlink-preferred UE and differentiated downlink handling for CA-incapable UEs is disabled, experience-based smart carrier selection can take effect for the UE under either the **UL_DL_MODE** or **DL_ONLY_MODE** multi-frequency smart selection mode configuration.

Triggering Mode	Triggering Scenario	Scenario Where the Function Takes Effect
Periodically	UE mobility, cell load changes, and other factors affect user experience. As such, the gNodeB periodically determines whether an optimal carrier or carrier combination needs to be selected for individual UEs. The length of the timer specifying the interval for periodic selection is determined by the **NRCellCaMgmtConfig**.***CarrSelMinInterval*** parameter. If the air interface capability for an uplink- or downlink-preferred UE degrades in the corresponding direction by over 30% when the timer expires, compared with when smart carrier selection was last performed for the UE, experience-based smart carrier selection will be triggered for the UE. For details about air interface capability evaluation, see **4.1.1.1.5 User Experience Evaluation**.	
By weak uplink coverage	The gNodeB checks the SRS signal to interference plus noise ratio (SINR) of each UE with heavy uplink traffic in its serving cell at 15-second intervals. If the SRS SINR of such a UE is less than the SRS SINR threshold, which is equal to the **NRDUCellSrsMeas**.***NrToEutranSinrLowThld*** parameter value plus 8 dB, the UE will be regarded as an uplink-preferred UE and experience-based smart carrier selection will be triggered for the UE based on its uplink services.	Experience-based smart carrier selection can take effect when the multi-frequency smart selection mode configuration is **UL_DL_MODE** or **UL_ONLY_MODE**, regardless of whether differentiated downlink handling for CA-incapable UEs is enabled.

In NSA networking, the SCG service drop rate may increase due to carrier combination changes or handover-triggered NR cell changes for UEs whose RRC connections are reestablished on the LTE side. To reduce the SCG service drop rate, prohibition of multi-frequency smart selection for NSA UEs whose RRC connections are reestablished is introduced. After this function is enabled, experience-based smart carrier selection does not take effect for NSA UEs whose RRC connections are reestablished on the LTE side, and there will be decreases in the number of handovers triggered by experience-based smart carrier selection and the number of SCC additions.

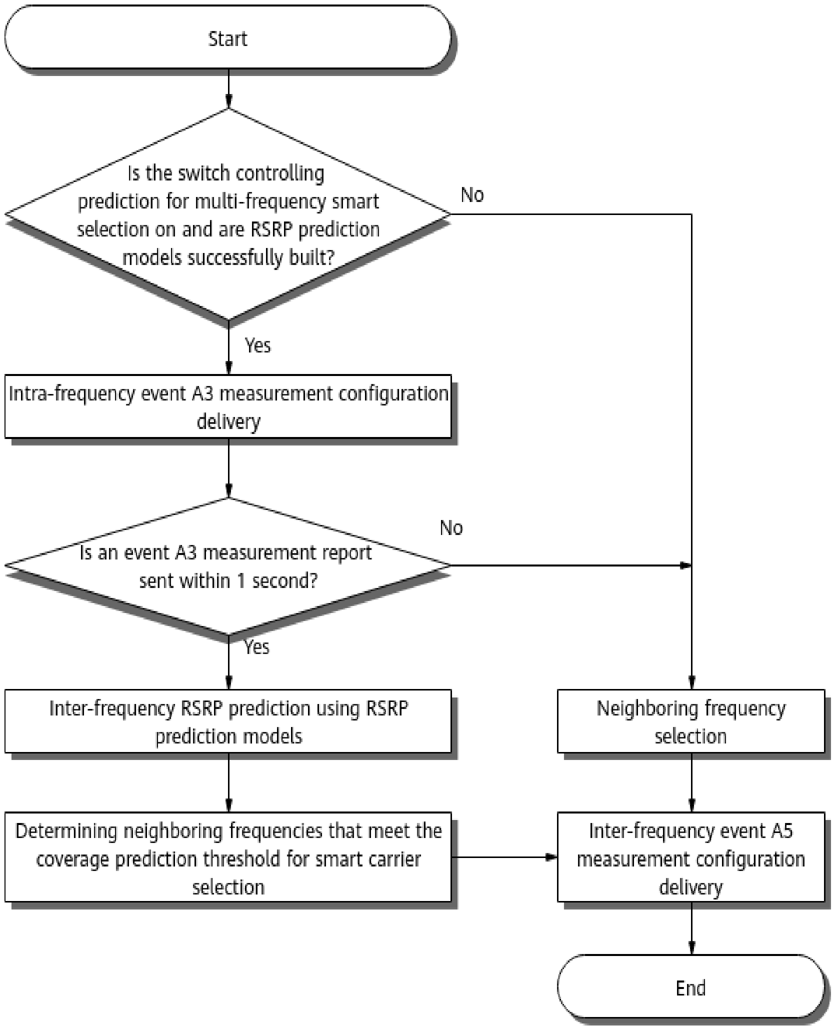
Specifically:

- A UE that has sent an RRCConnectionReestablishmentRequest message to the eNodeB is determined as a UE whose RRC connection is reestablished on the LTE side.
- The function is controlled by the following options:
  - NR side: **NSA_REEST_FORB_MUL_FREQ_SEL_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter
  - LTE side: **NSA_REEST_NOTIFY_TO_NR_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoExtSwitch*** parameter

4.1.1.1.3 Measurement Configuration Delivery

When experience-based smart carrier selection is triggered for a UE meeting QCI requirements, the gNodeB selects neighboring frequencies for measurement based on the setting of the switch controlling prediction for multi-frequency smart selection and the status of RSRP prediction models. It then delivers inter-frequency event A5 measurement configurations related to the frequencies to be measured all together to the UE. (Threshold 1 and threshold 2 for event A5 are –43 dBm and –140 dBm, respectively.) **Figure 4-3** shows the process. The switch controlling prediction for multi-frequency smart selection is specified by the **MULTI_FREQ_SMART_SEL_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter. The QCI requirements are described in **Policy for Determining Whether the Function Can Take Effect**.

Figure 4-3 Process of measurement configuration delivery



The process is as follows:

1. Checking the setting of the switch controlling prediction for multi-frequency smart selection and the building status of RSRP prediction models
   - If the switch is on and RSRP prediction models are successfully built, the gNodeB delivers intra-frequency event A3 measurement configurations and starts a 1-second timer.
   - If the switch is off or no RSRP prediction model is successfully built, the process proceeds to **3**.
2. Checking whether the UE sends an event A3 measurement report within 1 second
   - If the UE sends an event A3 measurement report within 1 second, the gNodeB inputs the RSRP values of the serving cell and intra-frequency neighboring cells in the measurement report into the corresponding RSRP prediction models to predict the RSRP values of neighboring frequencies. The gNodeB then filters out the neighboring frequencies of cells whose RSRP values are less than the coverage prediction threshold for smart carrier selection. The gNodeB can select a maximum of four neighboring frequencies for measurement. After the selection, the process proceeds to

4. For details about the coverage prediction threshold for smart carrier selection, which varies with UE capabilities, see **Coverage Prediction Threshold for Smart Carrier Selection**. For details about RSRP prediction models, see *Virtual Grid-based Multi-Frequency Coordination*.

– If the UE does not send an event A3 measurement report within 1 second, the process proceeds to 3.

3. Selecting neighboring frequencies based on certain principles

   The principles based on which the gNodeB selects neighboring frequencies for measurement configuration delivery vary with UEs. For details about the principles, see **Frequency Selection Principles**.

4. Delivering inter-frequency event A5 measurement configurations

   The gNodeB delivers inter-frequency event A5 measurement configurations related to the frequencies to be measured all together to the UE.

NOTE

- In experience-based smart carrier selection, the neighboring frequencies selected for event A5 measurement configuration delivery do not include those for which the **NRCellFreqRelation**.***InterFreqPrdcMrType*** parameter is set to **PERIODIC_MR_ONLY** or **FORBID_MEAS**.
- When a UE meets the conditions for triggering experience-based smart carrier selection, the gNodeB delivers measurement event configurations related to neighboring frequencies all together to the UE. Before delivering configurations of inter-frequency measurement events to a UE, the gNodeB performs UE-level filtering and frequency-level filtering. The UE-level filtering is to determine whether the UE supports inter-frequency measurement. In frequency-level filtering, the gNodeB selects the frequencies for which **NRCellFreqRelation**.***MeasActvFlag*** is set to **ACTIVE** and **NRCellFreqRelation**.***InterFreqPrdcMrType*** is set to **NORMAL** for measurement configuration delivery. These filtering actions help ensure that the experience-based smart carrier selection function takes effect properly.
- During gap-assisted inter-frequency measurements, if the gap conflicts with the time-domain positions of NR resources such as resources for sending scheduling requests (SRs) on the physical uplink control channel (PUCCH), KPIs such as the service drop rate, RRC connection reestablishment success rate, and contention-based random access success rate will be affected. To prevent the impacts, operators can enable the "non-overlapping between gap and NR resources" function, which is controlled by the **GAP_AVOID_NR_RES_SW** option of the **gNBMobilityCommParam**.***MeasAlgoSwitch*** parameter. For details about the "non-overlapping between gap and NR resources" function, see *SA Mobility Management in Connected Mode*.

Policy for Determining Whether the Function Can Take Effect

Before delivering measurement configurations to a UE, the gNodeB needs to determine whether experience-based smart carrier selection can take effect for the UE based on the settings of **gNBQciBearer**.***NetworkingOptionOptFlag*** corresponding to the QCIs of services running on the UE. If the services running on a UE include at least one service with a QCI for which the parameter is set to **FORBID** or all services running on the UE are services with QCIs for which the parameter is set to **FOLLOW**, experience-based smart carrier selection does not take effect for the UE. In other scenarios, experience-based smart carrier selection can take effect for the UE.

Coverage Prediction Threshold for Smart Carrier Selection

The coverage prediction threshold for smart carrier selection varies with UE capabilities and is related to the PCC's and SCC's thresholds. (PCC and SCC are short for primary component carrier and secondary component carrier). For details, see **Table 4-4**.

Table 4-4 Coverage prediction threshold for smart carrier selection

UE Capability	Coverage Prediction Threshold for Smart Carrier Selection	PCC's Threshold	SCC's Threshold
The UE does not support CA.	The coverage prediction threshold for smart carrier selection is equal to the PCC's threshold.	The PCC's threshold varies depending on whether the user-experience-based coverage expansion function (controlled by the **COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter) is enabled. ● If the user-experience-based coverage expansion function is disabled: – If the **NRCellInterFHoMeaGrp**.***MultiFreqSmtSelA5RsrpThld2*** parameter (specifying RSRP threshold 2 for event A5 related to multi-frequency smart selection) is set to **255**, the PCC's threshold is equal to the sum of **NRCellInterFHoMeaGrp**.***IntrfInterFreqHoRsrpThld*** and **NRCellFreqRelation**.***ExpBasedPCCFreqSelRsrpOfs***. – If the **NRCellInterFHoMeaGrp**.***MultiFreqSmtSelA5RsrpThld2*** parameter is set to a value other	Not involved

UE Capability	Coverage Prediction Threshold for Smart Carrier Selection	PCC's Threshold	SCC's Threshold
		than **255**, the PCC's threshold is equal to the sum of **NRCellInterFHoMeaGrp**.***MultiFreqSmtSelA5RsrpThld2*** and **NRCellFreqRelation**.***ExpBasedPCCFreqSelRsrpOfs***. ● If the user-experience-based coverage expansion function is enabled, the PCC's threshold is the same as the RSRP threshold for the target cell described in **4.3.1.2 Threshold Adaptation for Inter-Frequency Handovers Triggered by Experience-based Smart Carrier Selection**.	

UE Capability	Coverage Prediction Threshold for Smart Carrier Selection	PCC's Threshold	SCC's Threshold
The UE supports CA.	The coverage prediction threshold for smart carrier selection is equal to the PCC's threshold or SCC's threshold, whichever is smaller.	The PCC's threshold is the same as the PCC's threshold used for UEs that do not support CA.	The SCC's threshold is threshold 2 for event A5 related to SCell configuration for CA (sum of **NRCellCaMgmtConfig**.***CaA5RsrpThld2*** and **gNBCaFrequency**.***CaSccA5RsrpThld2Offset***). The **gNBCaFrequency**.***CaSccA5RsrpThld2Offset*** parameter takes the smallest value among the values of this parameter in the **gNBCaFrequency** MOs in which the specified frequencies are the same as those of the SCCs.

Frequency Selection Principles

The principles based on which the gNodeB selects frequencies for inter-frequency measurement configuration delivery vary with UEs, as described in **Table 4-5**.

Table 4-5 Frequency selection principles

UE Capability	Frequency Selection Principle
Single-carrier UEs or multi-carrier UEs supporting super uplink	• The base station selects the NR frequencies specified by the **NRCellFreqRelation**.***SsbDescMethod*** and **NRCellFreqRelation**.***SsbFreqPos*** parameters for measurement configuration delivery. • The base station selects a maximum of four neighboring frequencies based on the frequency priorities for RRC_CONNECTED UEs specified by the **NRCellFreqRelation**.***ConnFreqPriority*** parameter.
Multi-carrier UEs supporting CA	• The base station preferentially selects the NR frequencies specified by the **NRCellFreqRelation**.***SsbDescMethod*** and **NRCellFreqRelation**.***SsbFreqPos*** parameters. • If fewer than four frequencies can be delivered, the base station additionally delivers other frequencies in the CA frequency group. CA frequencies are specified in the **gNBCaFrequency** MOs.

4.1.1.1.4 Measurement Reporting

A UE performs measurements based on the event configuration delivered to it and reports measurement results.

- If the UE sends measurement reports for all neighboring frequencies contained in the measurement configurations within the time in which the gNodeB waits for inter-frequency event A5 measurement reports, user experience evaluation will start. For details about user experience evaluation, see **4.1.1.1.5 User Experience Evaluation**.
- When the measurement report waiting time ends:
  - If no cell on a measured neighboring frequency meets conditions, the gNodeB deletes the inter-frequency event A5 measurement configuration for this neighboring frequency so that the UE stops the current inter-frequency event A5 measurements. Then, user experience evaluation (see **4.1.1.1.5 User Experience Evaluation**) is started for the UE.
  - If no cell on all measured neighboring frequencies meets conditions, the gNodeB deletes the inter-frequency event A5 measurement

configurations for all the neighboring frequencies so that the UE stops the current inter-frequency event A5 measurements and the process ends.

NOTE

The time in which the gNodeB waits for inter-frequency A5 measurement reports is related to the number of measured neighboring frequencies.

4.1.1.1.5 User Experience Evaluation

The gNodeB selects cells based on the received inter-frequency event A5 measurement reports. That is, it selects the cells that meet specific conditions from the cells each with the highest RSRP value on the corresponding neighboring frequencies. The gNodeB then evaluates the user experience in the serving cell and in the selected cells and selects an optimal carrier or carrier combination based on the evaluation results.

Cells that meet the following criteria can be selected as candidate carriers or added to a candidate carrier combination:

- Assume that **RSRP threshold 2 for event A5 related to multi-frequency smart selection** is set to **255**. If the RSRP of the cell with the largest RSRP on a neighboring frequency is greater than the sum of the **RSRP threshold for interference-based inter-frequency handovers** and the **RSRP offset for experience-based PCC frequency selection**, this cell can be selected as a candidate carrier or as a PCC in a candidate carrier combination for user experience evaluation.
- Assume that **RSRP threshold 2 for event A5 related to multi-frequency smart selection** is set to a value other than **255**. If the RSRP of the cell with the largest RSRP on a neighboring frequency is greater than the sum of **RSRP threshold 2 for event A5 related to multi-frequency smart selection** and the **RSRP offset for experience-based PCC frequency selection**, this cell can be selected as a candidate carrier or as a PCC in a candidate carrier combination for user experience evaluation.
- If the RSRP of the cell with the largest RSRP value on a neighboring frequency is greater than or equal to **threshold 2 for event A5 related to secondary cell (SCell) configuration for CA**, the cell can be selected as an SCC in a candidate carrier combination for user experience evaluation.
- A cell failing to meet the preceding conditions will not be selected for user experience evaluation.

Table 4-6 lists the parameters involved in the preceding cell selection.

Table 4-6 Parameters related to candidate carrier or candidate carrier combination selection

Item	Parameter
RSRP threshold 2 for event A5 related to multi-frequency smart selection	**NRCellInterFHoMeaGrp**.*MultiFreqSmtSelA5RsrpThld2*

Item	Parameter
RSRP threshold for interference-based inter-frequency handovers	**NRCellInterFHoMeaGrp**.***IntrfInterFreqHoRsrpThld***
RSRP offset for experience-based PCC frequency selection	**NRCellFreqRelation**.***ExpBasedPCCFreqSelRsrpOfs***
Threshold 2 for event A5 related to SCell configuration for CA	**NRCellCaMgmtConfig**.***CaA5RsrpThld2*** + **gNBCaFrequency**.***CaSccA5RsrpThld2Offset***

Aspects Considered in Experience Evaluation

User experience evaluation is performed for candidate carriers or cells in candidate carrier combinations in the following aspects:

- Downlink air interface capability

  For downlink-preferred UEs, the downlink air interface capabilities of candidate carriers or candidate carrier combinations must be considered. For a multi-carrier UE, the downlink air interface capability of a candidate carrier combination is equal to the sum of the downlink air interface capabilities of all carriers in the combination. The downlink air interface capability of each carrier needs to be calculated and evaluated based on the following four factors:

  – UE-specific downlink spectral efficiency: The efficiency is considered in the downlink air interface capability calculation only if the **switch controlling spectral efficiency prediction for downlink experience evaluation** is on and the gNodeB can predict the downlink spectral efficiency of each frequency using virtual grid models (downlink intra-frequency spectral efficiency prediction models or downlink-frequency spectral efficiency models). The gNodeB retrieves the synchronization signal and PBCH block (SSB) beam information of the cell with the largest RSRP value on a frequency from the corresponding measurement report and inputs the information into the virtual grid model for this frequency to obtain the UE-specific downlink spectral efficiency in the cell.

  – Equivalent downlink cell bandwidth: This bandwidth is calculated based on factors such as the downlink cell bandwidth, slot configuration, and UE-supported downlink bandwidth.

  – Downlink load factor: This factor is calculated based on the number of UEs to be scheduled in the downlink per second.

  – Downlink experience coefficient factor: When both the downlink intra-frequency spectral efficiency prediction model and the downlink-frequency spectral efficiency model are unavailable and the **switch for evaluation based on multi-frequency experience coefficient** is on, this factor is calculated based on the **NR cell downlink experience coefficient**. In other cases, this factor is not considered for downlink air interface capability evaluation.

In inter-gNodeB CA, inter-gNodeB transmission delay affects uplink and downlink service experience of UEs. The impacts can be reduced by setting the **inter-gNodeB experience evaluation coefficient** for individual cells.

- Uplink air interface capability

  For uplink-preferred UEs, the uplink air interface capabilities of candidate carriers or candidate carrier combinations must be considered. For a multi-carrier UE, the uplink air interface capability of a candidate carrier combination is equal to the sum of the uplink air interface capabilities of all carriers in the combination. The uplink air interface capability of each carrier needs to be calculated and evaluated based on the following two factors:

  – UE-specific uplink spectral efficiency: The uplink spectral efficiency of a UE in an inter-frequency candidate cell is calculated based on the uplink spectral efficiency of the UE in its serving cell and the differences in the path loss, interference, and number of uplink receive antennas between the serving and candidate cells.

  – Cell-specific number of remaining uplink RBs: This number is calculated based on the accumulated number of remaining RBs on the physical uplink shared channel (PUSCH) per second in the cell involved.

  In inter-gNodeB CA, inter-gNodeB transmission delay affects uplink and downlink service experience of UEs. The impacts can be reduced by setting the **inter-gNodeB experience evaluation coefficient** for individual cells.

- Equivalent cell bandwidth: This bandwidth is calculated based on factors including the cell bandwidth, slot configuration, and UE-supported bandwidth.
- CCE allocation failure rate: This rate is calculated based on the total number of CCE allocation times and the number of successful CCE allocation times. CCE allocation failures affect uplink and downlink service experience of UEs.

  – If the downlink CCE allocation failure rate exceeds 8%, the corresponding candidate carrier or carrier combination will not be selected as the optimal carrier or carrier combination in the downlink.

  – If the uplink CCE allocation failure rate exceeds 20%, the corresponding candidate carrier or carrier combination will not be selected as the optimal carrier or carrier combination in the uplink.

  In the preceding information:

  – Downlink CCE allocation failure rate = [**N.CCE.DL.AllocReq.Num** – (**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)]/**N.CCE.DL.AllocReq.Num**

  – Uplink CCE allocation failure rate = [**N.CCE.UL.AllocReq.Num** – (**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)]/**N.CCE.UL.AllocReq.Num**

Table 4-7 lists the parameters involved in the preceding experience evaluation.

Table 4-7 Parameters related to experience evaluation

Item	Parameter
Switch controlling spectral efficiency prediction for downlink experience evaluation	**DL_EXP_EVAL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter
Switch for evaluation based on multi-frequency experience coefficient	**MULTI_FREQ_EXP_COEFF_EVAL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter
NR cell downlink experience coefficient	**NRCellMobilityConfig**.***NrCellDlExperienceCoeff***
Inter-gNodeB experience evaluation coefficient	**NRCellCaMgmtConfig**.***InterGnbExpEvalCoeff***

NOTE

- The bandwidth capability of a UE can be obtained from the *channelBWs-UL*, *channelBWs-DL*, *channelBWs-UL-v1590*, and *channelBWs-DL-v1590* information elements (IEs) in the *UECapabilityInformation* IE signaled over the Uu interface. For details, see section 4.2 "UE Capability Parameters" in 3GPP TS 38.306 V16.5.0.
- The uplink and downlink bandwidths of a cell are specified by the **NRDUCell**.***UlBandwidth*** and **NRDUCell**.***DlBandwidth*** parameters, respectively. The slot configuration is specified by the **NRDUCell**.***SlotAssignment*** parameter.

The requirements for cells involved in user experience evaluation are as follows:

- UEs cannot measure the SSB beams of independently deployed super uplink cells. Therefore, an independently deployed super uplink cell can be a candidate for user experience evaluation for a UE only if it is the current serving cell of the UE. For details about how to determine whether a cell is an independently deployed cell, see *Super Uplink*.
- The differences in the path loss, interference, and number of uplink receive antennas between the serving and candidate cells affect the uplink service experience of UEs. The impact of the differences in the path loss and interference on user experience can be reduced by setting the **NRDUCellCarrMgmt**.***UlCellExpEvaluationFactor*** parameter on the gNodeB based on the TX/RX modes (such as 4T4R) of the cells involved.
- The gNodeB evaluates user experience based on the measurement reports received from UEs. If the evaluation is affected by factors such as measurement report accuracy, gNodeB configuration differences, and UE compatibility differences, the average uplink and downlink UE throughputs after handovers triggered by experience-based smart carrier selection will not be as expected. If the experience-based smart carrier selection function is

used on live networks, it is recommended that the **NRCellCaMgmtConfig**.***DlExpEvalHyst*** and **NRCellCaMgmtConfig**.***UlExpEvalHyst*** parameters be used for optimization. The larger the values of these parameters, the smaller the impact on the average uplink and downlink UE throughputs after handovers triggered by experience-based smart carrier selection. The smaller the values of these parameters, the larger the impact on the average uplink and downlink UE throughputs after handovers triggered by experience-based smart carrier selection.

- When a cell served by a MetaAAU is a candidate carrier for a UE, virtual grid models cannot be used to predict the downlink spectral efficiency of this candidate carrier. In addition, the actual user experience of the UE on such a candidate carrier is closely related to vertical-direction UE distribution and scheduling. If user experience evaluation is inaccurate, user experience may deteriorate.
- Multi-carrier flow control is supported during user experience evaluation. If the cell corresponding to a candidate carrier or a carrier in a candidate carrier combination for a UE belongs to a gNodeB other than the current serving gNodeB for the UE and the CPU usage of this gNodeB reaches a certain flow control level, the cell will not be engaged in user experience evaluation. The flow control level can be specified using the **gNodeBAlgo**.***MultiCarrierFlowCtrlLevel*** parameter.

4.1.1.1.6 Optimal Carrier or Carrier Combination Selection

The gNodeB selects an optimal carrier or carrier combination based on the user experience evaluation results (including the air interface capability and equivalent cell bandwidths), whether a handover or carrier change is involved, and the number of carriers. Handover, carrier change, and the number of carriers are considered only when an optimal carrier combination needs to be selected for multi-carrier-capable UEs.

Each factor is considered as follows:

- Air interface capability: Candidate carriers or carrier combinations with a higher air interface capability have a higher priority. An air interface capability hysteresis is used to compare the downlink or uplink air interface capability of a candidate carrier or carrier combination with that of the current carrier or carrier combination. If the difference in the uplink or downlink air interface capability between the two carriers or carrier combinations is less than the corresponding hysteresis, the carriers or carrier combinations are considered to have the same uplink or downlink air interface capability. The downlink air interface capability hysteresis and uplink air interface capability hysteresis are specified by the **NRCellCaMgmtConfig**.***DlExpEvalHyst*** and **NRCellCaMgmtConfig**.***UlExpEvalHyst*** parameters, respectively.
- Handover: A handover transiently interrupts data transmission. Therefore, candidate carrier combinations that do not involve a handover have a higher priority.
- Carrier change: A carrier change transiently interrupts data transmission. Therefore, candidate carrier combinations that do not involve an SCC change have a higher priority.
- Equivalent cell bandwidth: Candidate carriers or carrier combinations with a larger equivalent cell bandwidth have a higher priority.

- Number of carriers: In the case of multi-carrier UEs, the smaller the number of carriers for aggregation in a candidate carrier combination, the lower the consumption of channel resources such as physical downlink control channel (PDCCH) and PUCCH resources. Therefore, candidate carrier combinations with a smaller number of carriers have a higher priority.

An optimal carrier or carrier combination is determined based on the preceding factors in sequence.

- For a downlink-preferred UE, the following factors are evaluated in sequence: downlink air interface capability, handover, carrier change, equivalent downlink cell bandwidth, and number of downlink carriers. For example, if the candidates have the same downlink air interface capability, handover will be evaluated. The selection process ends when any factor is evaluated in a candidate's favor. This candidate is then regarded as the optimal carrier or carrier combination.
- For an uplink-preferred UE, the following factors are evaluated in sequence: uplink air interface capability, handover, carrier change, equivalent uplink cell bandwidth, and number of uplink carriers. For example, if the candidates have the same uplink air interface capability, handover will be evaluated. The selection process ends when any factor is evaluated in a candidate's favor. This candidate is then regarded as the optimal carrier or carrier combination.

4.1.1.1.7 Execution of Experience-based Smart Carrier Selection

If the selected optimal carrier or carrier combination is different from the current carrier or carrier combination, a handover or carrier change is required.

- For a single-carrier UE, a handover to the target cell will be initiated directly.
- For a multi-carrier UE for which CA or super uplink has taken effect:
  - If the PCC in the optimal carrier combination is different from the current PCC, a handover to the target cell needs to be initiated. During the handover, the SCCs or supplementary uplink (SUL) carriers in the optimal carrier combination will be configured for the UE involved.
  - If the PCC in the optimal carrier combination is the same as the current PCC:
    - In the case of an SA UE, the gNodeB supports SCC or SUL carrier changes through an RRCReconfiguration message.
    - In the case of an NSA UE, the gNodeB does not support SCC changes through an SgNB Modification Required message.

NOTE

For details about SUL carriers, see *Super Uplink*.

A UE is considered as a single-carrier or multi-carrier UE based on its current status.

If the frequency-priority-based inter-frequency handover function or the CA-UE-specific frequency-priority-based inter-frequency handover function is enabled together with experience-based smart carrier selection, the functions may cause ping-pong handovers between two cells.

An **inter-frequency measurement delay timer** can be set on the gNodeB to prevent the ping-pong handovers. When a UE is handed over (not for coverage reasons) to a cell with frequency-priority-based inter-frequency handover enabled, the gNodeB starts the timer and does not deliver frequency-priority-based inter-frequency measurement configurations to the UE within the timer length. The gNodeB can start **frequency-priority-based inter-frequency handovers** or **CA-UE-specific frequency-priority-based inter-frequency handovers** again based on the specified **retry interval for frequency-priority-based inter-frequency handover**.

Table 4-8 lists the parameters involved in the preceding measures against ping-pong handovers.

Table 4-8 Parameters involved in the measures against ping-pong handovers

Item	Parameter
Frequency-priority-based inter-frequency handover	**FREQ_PRIORITY_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter
CA-UE-specific frequency-priority-based inter-frequency handover	**CA_FREQ_PRI_BASED_INTERF_HO_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter
Inter-frequency measurement delay timer	**NRCellMobilityConfig**.***InterFreqMeasDelay***
Retry interval for frequency-priority-based inter-frequency handover	**NRCellMobilityConfig**.***FreqPriInterFHoRetryIntvl***

NOTE

- For details about frequency-priority-based inter-frequency handover, see *SA Mobility Management in Connected Mode*.
- For details about CA-UE-specific frequency-priority-based inter-frequency handover, see *CA Experience Improvement*.
- It is recommended that the **NRCellMobilityConfig**.***FreqPriInterFHoRetryIntvl*** parameter not be set to **INFINITY**. Otherwise, if frequency-priority-based inter-frequency measurement configurations are no longer delivered after measurement delay, frequency-priority-based inter-frequency handover or CA-UE-specific frequency-priority-based inter-frequency handover cannot take effect.

4.1.1.2 Policy of Experience-based Smart Carrier Selection

The policy of experience-based smart carrier selection is specified by the **NRCellCaMgmtConfig**.***SmartCarrSelPolicy*** parameter.

- If this parameter is set to **EXPERIENCE_FIRST**, an experience-first policy will be used. With this policy, the base station selects optimal carriers or carrier combinations for all UEs based on user experience. For details, see **4.1.1.2.1 Experience-First Policy**.

- If this parameter is set to **NEAR_POINT_FIXED_ANCHOR**, a cell-center fixed anchor policy is used. With this policy, the base station selects optimal carriers or carrier combinations for cell-edge UEs based on user experience, but does not change the PCCs or serving cells for cell-center UEs based on smart carrier selection. For details, see **4.1.1.2.2 Cell-Center Fixed Anchor Policy**.

4.1.1.2.1 Experience-First Policy

If experience-based smart carrier selection is enabled and the **NRCellCaMgmtConfig**.***SmartCarrSelPolicy*** parameter is set to **EXPERIENCE_FIRST**, the experience-first policy takes effect. With this policy, the base station selects an optimal carrier or carrier combination for any UE based on user experience, no matter whether the UE is a cell-center or cell-edge UE. For details about optimal carrier or carrier combination selection in this policy, see the basic process described in **4.1.1.1 Basic Functions of Experience-based Smart Carrier Selection**.

4.1.1.2.2 Cell-Center Fixed Anchor Policy

If experience-based smart carrier selection is enabled and the **NRCellCaMgmtConfig**.***SmartCarrSelPolicy*** parameter is set to **NEAR_POINT_FIXED_ANCHOR**, the cell-center fixed anchor policy takes effect.

It is recommended that this policy be enabled when the following conditions are met: (1) the frequency-priority-based inter-frequency handover function (controlled by the **FREQ_PRIORITY_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter) or the CA-UE-specific frequency-priority-based inter-frequency handover function (controlled by the **CA_FREQ_PRI_BASED_INTERF_HO_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter) is enabled; (2) event A1 is used to trigger frequency-priority-based handovers (**NRCellMobilityConfig**.***FreqPriHoTriggerMode*** set to **A1_BASED**).

When the cell-center fixed anchor policy is used, the base station handles carrier or carrier combination selection differently for cell-center UEs and cell-edge UEs. Therefore, the base station needs to identify whether a UE is a cell-center or cell-edge UE before selecting a carrier or carrier combination for the UE.

Cell-Center UE and Cell-Edge UE Identification

Event A1 and event A2 configurations are delivered to RRC_CONNECTED UEs in order to identify UEs as cell-center or cell-edge UEs. The process is as follows:

1. After a UE accesses a cell, the gNodeB identifies it as a cell-edge UE by default and delivers an event A1 configuration to it.
2. After receiving an event A1 report, the gNodeB identifies the UE as a cell-center UE, deletes the event A1 configuration, and delivers an event A2 configuration.
3. After receiving an event A2 report, the gNodeB identifies the UE as a cell-edge UE, deletes the event A2 configuration, and delivers the event A1 configuration.

Carrier or Carrier Combination Selection for Cell-Center UEs and Cell-Edge UEs

When experience-based smart carrier selection is triggered for a UE, the selection is performed based on its cell-center or cell-edge status:

- If the UE is a cell-center UE:
  - If the UE is a single-carrier UE, an inter-frequency handover for experience-based smart carrier selection will not be performed for it.
  - If the UE is a multi-carrier UE, the PCC of the UE will not be changed and only optimal SCCs will be selected for the UE. For details about the basic process of optimal SCC selection, see **4.1.1.1 Basic Functions of Experience-based Smart Carrier Selection**. An additional step is to filter carriers or carrier combinations. In this step, after receiving measurement reports from a UE, the gNodeB filters carriers or carrier combinations. It retains only the carrier combinations that can keep the PCC of the UE unchanged. Then, the gNodeB selects an optimal carrier combination from the retained ones.
- If the UE is a cell-edge UE, an optimal carrier or carrier combination is selected based on the experience-first policy as described in **4.1.1.1 Basic Functions of Experience-based Smart Carrier Selection**.

4.1.2 Network Analysis

4.1.2.1 Benefits

After experience-based smart carrier selection is enabled, the gNodeB considers service characteristics of UEs along with factors including spectral efficiency, bandwidth, and load to select carriers or carrier combinations that provide optimal user experience for UEs. This increases the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) and average uplink UE throughput (indicated by **User Uplink Average Throughput (DU)**) of the entire network.

This function brings greater benefits to networks with more frequencies, larger coverage difference between frequency bands, larger load difference between cells, or larger UE traffic volumes. The average downlink UE throughput and average uplink UE throughput of individual frequency bands will fluctuate, depending on UE distribution and services.

It is recommended that the experience-based smart carrier selection function be enabled at the network level. If this function is enabled only in a single cell, it is possible that the average UE throughput gains cannot be maximized.

After prohibition of multi-frequency smart selection for NSA UEs whose RRC connections are reestablished takes effect, the SCG service drop rate decreases on NSA networks with UEs whose RRC connections are reestablished on the LTE side. The SCG service drop rate is calculated as follows:

- On the LTE side: (**L.NsaDc.SgNB.Rmv.Att** – **L.NsaDc.MenbTrig.SgNB.NormRel** – **L.NsaDc.SgnbTrig.SgNB.NormRel**)/ **L.NsaDc.SgNB.Rmv.Att** x 100%

- On the NR side: (**N.NsaDc.SgNB.Rel** – **N.NsaDc.SgNB.Rel.SgNBTrigger** – **N.NsaDc.SgNB.Rel.MeNBTrigger.NormalRel** + **N.NsaDc.SgNB.AbnormRel** – **N.NsaDc.SgNB.Rel.InitialAdd.RachFail**)/**N.NsaDc.SgNB.Rel** x 100%

4.1.2.2 Impacts

The impacts between this function and other functions are described in **4.1.3.2.3 Function Impacts**.

Experience-based smart carrier selection has the following impacts on the network:

- Inter-band UE distribution changes. The number of signaling messages over the air interface and the number of handovers increase.
- In light-load scenarios where multiple frequencies are used and the cells working on these frequencies differ greatly in their bandwidths, if the positive gain of experience-based smart carrier selection is less than the negative impact brought by the increase in measurement overheads, the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) and average uplink UE throughput (indicated by **User Uplink Average Throughput (DU)**) decrease.
- Frequent user experience evaluation increases the CPU usage of the main control board. After the **DL_EXP_EVAL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter is selected, the CPU usage of the main control board further increases.
- After the base station takes the ping-pong handover prevention measure that is based on collaboration between experience-based smart carrier selection and frequency-priority-based inter-frequency handover or CA-UE-specific frequency-priority-based inter-frequency handover, the number of frequency-priority-based inter-frequency handovers (indicated by the **N.HO.InterFreq.FreqPri.ExecAttOut** counter) decreases.
- If the information input into a virtual grid model greatly deviates from the information included in the samples used for model training (for example, in abnormal scenarios, the information of a neighboring cell is included in the samples used for model training, but this neighboring cell cannot be measured by UEs when the model is used), the downlink UE-specific spectral efficiency predicted by the virtual grid model is inaccurate, affecting the accuracy of downlink user experience evaluation and further leading to a decrease in the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**).
- The **NRCellCaMgmtConfig**.***DlExpEvalHyst*** and **NRCellCaMgmtConfig**.***UlExpEvalHyst*** parameters must be set based on the traffic and coverage information of the live network. If the parameter settings are inappropriate, the data rates of UEs on the live network will be unstable. The larger the values of these parameters, the lower the probability that experience-based smart carrier selection triggers handovers. The smaller the values of these parameters, the higher the probability that experience-based smart carrier selection triggers handovers.
- If virtual grid models are not built or the gNodeB can obtain the predicted downlink spectral efficiency only of some frequencies through virtual grid models, the downlink air interface capability for user experience evaluation can be calculated only based on the equivalent downlink cell bandwidth and

downlink load factor. If the downlink spectral efficiency of a UE varies greatly between frequencies due to factors such as frequency deployment and UE mobility, the downlink user experience evaluation result may be greatly inaccurate. In this case, the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) decreases.

- After multi-carrier flow control is triggered, the **User Downlink Average Throughput (DU)** and **User Uplink Average Throughput (DU)** on the entire network may decrease. A higher value of the **gNodeBAlgo**.***MultiCarrierFlowCtrlLevel*** parameter may result in a higher **Service Call Drop Rate (CU, Inactive)**.

After prohibition of multi-frequency smart selection for NSA UEs whose RRC connections are reestablished takes effect, the number of handovers triggered by experience-based smart carrier selection and the number of SCC additions decrease in the scenario where NSA networking is used and there are RRC connection reestablishments on the LTE side. The related counters are as follows:

- Counters related to the number of handovers triggered by experience-based smart carrier selection: **N.MultiFreqSmartSel.Exec**, **N.MultiFreqSmartSel.Exec.CA**, **N.MultiFreqSmartSel.DlIntraFreqSEModel.Exec**, **N.MultiFreqSmartSel.DlIntraFreqSEModel.Exec.CA**, **N.MultiFreqSmartSel.Exec.DL**, and **N.MultiFreqSmartSel.Exec.DL.CA**
- Counters related to the number of SCC additions: **N.CA.SCell.Add.Att** and **N.CA.SCell.Add.Succ**

4.1.3 Requirements

4.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-061223	Experience Boosting based on Multi-Band Coordination	NR0S0SCSNR00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL Intra-Freq Spct Eff Mdl Build Allowed	**DL_INTRAF_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	The **DL_EXP_EVAL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter can be selected only when the **DL_INTRAF_SE_MODEL_BUILD_ALLOW** or **DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMulti-Carr**.***ModelBuildingSwitch*** parameter is selected.
Low-frequency TDD	DL Freq Spct Eff Mdl Build Allowed	**DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	The **DL_EXP_EVAL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter can be selected only when the **DL_INTRAF_SE_MODEL_BUILD_ALLOW** or **DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMulti-Carr**.***ModelBuildingSwitch*** parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-frequency spectral efficiency prediction	**DL_INTRA_FREQ_SPCT_EFF_PRED_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	A gNodeB can use the prediction results of virtual grid models and considers UE-specific downlink spectral efficiency for downlink air interface capability evaluation only when all of the following conditions are met: ● The **DL_INTRA_FREQ_SPCT_EFF_PRED_SW** option is selected. ● The **DL_INTRAF_SE_MODEL_BUILD_ALLOW** or **DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter is selected. ● The **DL_EXP_EVAL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Measurement quantity prediction for intra-NR inter-frequency handovers	**NR_IF_HO_MEAS_QTY_PRED_FUN_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	The **MULTI_FREQ_SMART_SEL_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter can be selected for the gNodeB to use the prediction results of virtual grid models only when measurement quantity prediction for intra-NR inter-frequency handovers is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	**MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	• When the **MULTI_FREQ_EXP_COEFF_EVAL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter is selected, at least one of experience-based smart carrier selection and multi-frequency beam coordination must be enabled. • The **DL_EXP_EVAL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter can be selected only when experience-based smart carrier selection is enabled. • Differentiated downlink handling for CA-incapable UEs (controlled by the **CA_INCAP_UE_DL_DIFF_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) can be enabled only when experience-based smart carrier selection is enabled. • The **MULTI_FREQ_SM**

RAT	Function Name	Function Switch	Reference	Description
				ART_SEL_PRED_SW option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter can be selected only when experience-based smart carrier selection is enabled.
Low-frequency TDD	Multi-frequency beam coordination	**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	*Multi-Frequency Smart Selection*	When the **MULTI_FREQ_EXP_COEFF_EVAL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter is selected, at least one of experience-based smart carrier selection and multi-frequency beam coordination must be enabled.

4.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag***	*High Speed Mobility*	High-speed cells (for which the **NRDUCell**.***HighSpeedFlag*** parameter is set to **HIGH_SPEED**) do not support experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Hyper cells do not support experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter).
Low-frequency TDD	Cell Combination	• **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** • **NRDUCellMultiTrp**.***DataTransMode*** set to **BASIC_MODE** or **DPS_MODE**	*Cell Combination*	Combined cells do not support experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter).
Low-frequency TDD	Joint Transmission Cell Combination	• **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** • **NRDUCellMultiTrp**.***DataTransMode*** set to **JOINT_MODE**	*Cell Combination*	Combined cells in joint transmission mode do not support experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	When Fusion Cell is enabled, experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) cannot be enabled.
Low-frequency TDD	Policy on multi-carrier capability combination selection	**NRCellCaMgmtConfig**.***CaBandCombSelectionStrat***	*Carrier Aggregation*	If the **NRCellCaMgmtConfig**.***CaBandCombSelectionStrat*** parameter (specifying the policy on multi-carrier capability combination selection) is set to **DEFAULT**, experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) cannot be enabled.

4.1.3.2.3 Function Impacts

- Impact relationships with functions related to mobility management

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Frequency-priority-based inter-frequency handover	**FREQ_PRIORITY_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	• When the **NRCellCaMgmtConfig**.***SmartCarrSelPolicy*** parameter is set to **EXPERIENCE_FIRST**, you are advised not to enable both experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) and frequency-priority-based inter-frequency handover. Otherwise, ping-pong handovers may occur. • When experience-based smart carrier selection is enabled, it is recommended that event A2 not be used to decide whether to trigger frequency-priority-based inter-frequency handovers (that is, it is recommended that the **NRCellMobilityConfig**.***FreqPriHoTriggerMode***

RAT	Function Name	Function Switch	Reference	Description
				parameter not be set to **A2_BASED**). Otherwise, user experience of non-cell-center UEs may deteriorate.
Low-frequency TDD	UE-number-based connected mode MLB	• **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter • **NRCellMlb**.***MlbTriggerMode*** set to **CELL_USER_NUM**	*Mobility Load Balancing in Connected Mode*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and both experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) and UE-number-based connected mode MLB are enabled. When a handover is triggered for experience-based smart carrier selection, the target cell will not be selected from cells in the MLB state. This prevents ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	• **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter • **NRCellMlb**.***MlbTriggerMode*** set to **SPEC_EFF_BASED_USER_NUM**	*Mobility Load Balancing in Connected Mode*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and both experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) and spectral-efficiency-based connected mode MLB are enabled. When a handover is triggered for experience-based smart carrier selection, the target cell will not be selected from cells in the MLB state. This prevents ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	5QI-specific UE-number-based connected mode MLB	• **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter • **NRCellMlb**.***MlbTriggerMode*** set to **SPECIFIED_5QI_USER_NUM**	*Mobility Load Balancing in Connected Mode*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and both experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) and 5QI-specific UE-number-based connected mode MLB are enabled. When a handover is triggered for experience-based smart carrier selection, the target cell will not be selected from cells in the MLB state. This prevents ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Radio-resource-based connected mode MLB	• **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter • **NRCellMlb**.***MlbTriggerMode*** set to **RADIO_RESOURCE**	*Mobility Load Balancing in Connected Mode*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and both experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) and radio-resource-based connected mode MLB are enabled. When a handover is triggered for experience-based smart carrier selection, the target cell will not be selected from cells in the MLB state. This prevents ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	EPLMN-based intra-RAT inter-PLMN handover	**INTRA_RAT_HO_WITH_GNB_EPLMN_SW** option of the **gNodeBParam**.***EqvPlmnAlgoSwitch*** parameter	*SA Mobility Management in Connected Mode*	When both experience-based smart carrier selection function (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) and EPLMN-based intra-RAT inter-PLMN handover are enabled, the list of PLMNs to which a UE can be handed over contains only the serving PLMN.

- Impact relationships with functions related to carrier management

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Smart carrier selection	**SMART_CARR_SEL_SW** option of the **NRDUCellCollabServ**.***MultiCarrMgmtSw*** parameter	*Carrier Aggregation*	When both experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) and smart carrier selection are enabled, inter-frequency event A5 configurations related to experience-based smart carrier selection are preferentially delivered.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-frequency beam coordination	**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	*Multi-Frequency Smart Selection*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and both multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter) and experience-based smart carrier selection are enabled. When a handover is triggered for experience-based smart carrier selection, the target cell will not be selected from cells in the beam coordination state based on the reported SSB beam measurement information. This prevents ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Service-aware carrier selection	**SERVICE_AWARE_CARR_SEL_SW** option of the **NRDUCellCollabServ**.***MultiCarrMgmtSw*** parameter	*Carrier Aggregation*	Assume that both experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) and service-aware carrier selection are enabled. Experience-based smart carrier selection can take effect to trigger carrier changes for a UE only if the UE has not been considered as a target UE for service-aware carrier selection.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA-UE-specific frequency-priority-based inter-frequency handover	**CA_FREQ_PRI_BASED_INTERF_HO_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*CA Experience Improvement*	● When the smart carrier selection policy (specified by the **NRCellCaMgmtConfig**.***SmartCarrSelPolicy*** parameter) is set to **EXPERIENCE_FIRST**, you are advised not to enable both experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) and CA-UE-specific frequency-priority-based inter-frequency handover (controlled by the **CA_FREQ_PRI_BASED_INTERF_HO_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter). If they are both enabled, ping-pong handovers may occur. ● It is not recommended that event A2 be used for evaluation of

RAT	Function Name	Function Switch	Reference	Description
				CA-UE-specific frequency-priority-based inter-frequency handovers (that is, **NRCellMobilityConfig**.***FreqPriHoTriggerMode*** is set to **A2_BASED**) when experience-based smart carrier selection is enabled. If event A2 is used, user experience of UEs at the cell edge or at a medium distance from the cell center may deteriorate.

- Impact relationships with other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NSA carrier combination selection based on user experience	• NR: **NSA_LTE_SEL_OPT_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter • LTE: **NSA_LTE_FDD_SEL_OPT_SW** or **NSA_LTE_TDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter	*EPC-based NSA Experience Improvement*	Experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) does not take effect for NSA UEs when this function and NSA carrier combination selection based on user experience are both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Networking option optimization flag	**gNBRfspConfig**.***NetworkingOptionOptFlag***	None	If the **gNBRfspConfig**.***NetworkingOptionOptFlag*** parameter corresponding to the RAT/Frequency Selection Priority (RFSP) index of a UE is set to **FALSE**, experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) does not take effect for the UE.
Low-frequency TDD	Operator-specific RB management	**RB_DYNAMIC_SHARING_SW**, **MAX_RB_LIMIT_SW**, or **MIN_RB_GUARANTEE_SW** option of the **NRDUCellAlgoSwitch**.***RanSharingAlgoSwitch*** parameter	*Multi-Operator Sharing*	In RAN sharing with common carrier, it is not recommended that experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) and operator-specific RB management both be enabled. Otherwise, user experience evaluation results may be inaccurate, causing user experience to deteriorate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-frequency MRO under cell-level MRO	• **MRO_SCENARIO_IDENT_SW** option of the **gNBMro**.***ScenarioIdentifySwitch*** parameter • **INTRA_FREQ_MRO_SW** option of the **NRCellMro**.***MroSwitch*** parameter	*MRO*	When both spectral efficiency prediction for downlink experience evaluation and intra-frequency MRO are enabled, event A3 reporting in spectral efficiency prediction for downlink experience evaluation is affected. As a result, the downlink experience evaluation result of the serving cell may be different from the expected result, which may cause the average downlink UE throughput to decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-level MRO	**UE_MRO_SWITCH** option of the **NRCellMro**.***MroSwitch*** parameter	*MRO*	When both spectral efficiency prediction for downlink experience evaluation and UE-level MRO are enabled, event A3 reporting in spectral efficiency prediction for downlink experience evaluation is affected for UEs involved in ping-pong handovers and spectral efficiency prediction for downlink experience evaluation. As a result, the downlink experience evaluation result of the serving cell may be different from the expected result, which may cause the average downlink UE throughput to decrease.
Low-frequency TDD	Energy-efficiency-based inter-frequency handover	**ENERGY_EFFICIENCY_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*Energy Conservation and Emission Reduction*	It is not recommended that experience-based smart carrier selection and energy-efficiency-based inter-frequency handover be both enabled. Otherwise, experience gains and energy saving gains will decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	Experience-based smart carrier selection does not take effect for RedCap UEs.
Low-frequency TDD	Carrier selection based on uplink live streaming services	**UL_LIVE_SERVICE_FREQ_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Ultimate Video Experience*	After carrier selection based on uplink live streaming services is enabled, smart carrier selection is no longer triggered for UEs performing uplink live streaming services.
Low-frequency TDD	UE bandwidth adaptation	**UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter	*Scalable Bandwidth*	After UE bandwidth adaptation is enabled, experience-based smart carrier selection cannot be triggered for a UE identified as an uplink-preferred UE if the effective uplink bandwidth for the UE is not the full uplink bandwidth of the cell.

4.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.1.3.4 Networking

- The network uses at least two frequencies.
- The Xn interface between gNodeBs is normal so that user experience evaluation information can be transmitted.

4.1.3.5 Others

- During key event assurance, it is recommended that the experience-based smart carrier selection function be disabled. This is to prevent network performance deterioration as the function increases CPU usage.
- The switch for experience-based smart carrier selection does not need to be turned on in candidate inter-frequency cells if experience-based smart carrier selection already takes effect in a single cell, as UEs in this cell can be proactively handed over to other cells based on this function. If the gNodeB ID of an inter-frequency cell is different from that of the serving cell of a UE, it is recommended that experience-based smart carrier selection be enabled for at least one cell under the gNodeB serving the inter-frequency cell. Otherwise, this inter-frequency cell cannot serve as a candidate carrier or as a PCC in a candidate carrier combination.
- If a cell with experience-based smart carrier selection disabled is activated first, for example, when its serving base station is reset, the cell cannot serve as a candidate carrier or as a PCC in a candidate carrier combination for user experience evaluation within 20 minutes after the activation.
- It is recommended that the number of frequencies configured on the gNodeB be greater than the maximum number of carriers supported by multi-carrier UEs for CA. This will ensure that the gNodeB selects an optimal carrier combination for such UEs. The gNodeB can select an optimal carrier or carrier combination for a UE when the number of frequencies configured on the gNodeB is equal to the maximum number of carriers supported by the UE for CA in the following situations:

  The UE does not support SRS carrier switching, and the bandwidths of carriers in the candidate carrier combination are different.
- If experience-based smart carrier selection takes effect and there are multi-carrier UEs supporting CA on the network, ensure that the basic functions of CA normally take effect.

4.1.4 Operation and Maintenance

4.1.4.1 Data Configuration

4.1.4.1.1 Data Preparation

Table 4-9 describes the parameters used for function activation. For details about other related parameters, see Table 4-10. This section does not describe parameters related to cell establishment. The following parameters do not take effect for SUL cells, for which the **NRDUCell**.***DuplexMode*** parameter is set to **CELL_SUL**.

Table 4-9 Parameters used for activation

NE	Parameter Name	Parameter ID	Option	Setting Notes
gNodeB	Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	MULTI_FREQ_SMART_SEL_SW	You are advised to select this option.
gNodeB	Smart Carrier Selection Policy	**NRCellCaMgmtConfig**.***SmartCarrSelPolicy***	N/A	Set this parameter based on operators' requirements.
gNodeB	Downlink Experience Evaluation Hysteresis	**NRCellCaMgmtConfig**.***DlExpEvalHyst***	N/A	Use the recommended value.
gNodeB	Uplink Experience Evaluation Hysteresis	**NRCellCaMgmtConfig**.***UlExpEvalHyst***	N/A	Use the recommended value.
gNodeB	Cell Uplink Experience Evaluation Factor	**NRDUCellCarrMgmt**.***UlCellExpEvaluationFactor***	N/A	It is recommended that operators consult Huawei engineers before setting this parameter.
gNodeB	Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	DL_EXP_EVAL_SPCT_EFF_PRED_SW	You are advised to select this option.
gNodeB	Inter-Freq Measurement Delay	**NRCellMobilityConfig**.***InterFreqMeasDelay***	N/A	Use the recommended value.
gNodeB	Carrier Selection Minimum Interval	**NRCellCaMgmtConfig**.***CarrSelMinInterval***	N/A	Use the recommended value.

NE	Parameter Name	Parameter ID	Option	Setting Notes
gNodeB	Intrf-based Inter-Freq HO RSRP Thld	**NRCellInterFHoMeaGrp**.***IntrfInterFreqHoRsrpThld***	N/A	Use the recommended value.
gNodeB	Multi-Freq Smart Sel A5 RSRP Thld2	**NRCellInterFHoMeaGrp**.***MultiFreqSmtSelA5RsrpThld2***	N/A	Set this parameter based on operators' requirements.
gNodeB	Exp Based PCC Freq Sel RSRP Offset	**NRCellFreqRelation**.***ExpBasedPCCFreqSelRsrpOfs***	N/A	Set this parameter based on operators' requirements.
gNodeB	Inter-gNodeB Exp Evaluation Coefficient	**NRCellCaMgmtConfig**.***InterGnbExpEvalCoeff***	N/A	Use the recommended value.
gNodeB	Networking Option Opt Flag	**gNBQciBearer**.***NetworkingOptionOptFlag***	N/A	Set this parameter based on operators' requirements.
gNodeB	Networking Option Opt Flag	**gNBRfspConfig**.***NetworkingOptionOptFlag***	N/A	Set this parameter based on operators' requirements.
gNodeB	Multi-Frequency Smart Selection Mode	**NRCellAlgoSwitch**.***MultiFreqSmartSelMode***	N/A	Set this parameter based on operators' requirements.
gNodeB	Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	MULTI_FREQ_SMART_SEL_PRED_SW	Set this option based on operators' requirements. Before this option is selected, ensure that the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter has been selected.

NE	Parameter Name	Parameter ID	Option	Setting Notes
gNodeB	Multi-Carrier Capb Comb Select Strat	**NRCellCaMgmtConfig**.***CaBandCombSelectionStrat***	N/A	If this parameter is set to **DEFAULT**, experience-based smart carrier selection cannot be enabled. This parameter can be set to **BANDWIDTH_FIRST**, **PEAK_RATE_FIRST**, or **UL_PEAK_RATE_FIRST** based on operators' requirements.
gNodeB	Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	CA_INCAP_UE_DL_DIFF_SW	Set this option based on operators' requirements.
gNodeB	Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	MULTI_FREQ_EXP_COEFF_EVAL_SW	Set this option based on operators' requirements.
gNodeB	Multi-carrier Flow Control Level	**gNodeBAlgo**.***MultiCarrierFlowCtrlLevel***	N/A	Use the recommended value.

Table 4-10 Other related parameters

NE	Parameter Name	Parameter ID	Option	Setting Notes
gNodeB	NR Move to E-UTRAN UL SINR Low Threshold	**NRDUCellSrsMeas**.***NrToEutranSinrLowThld***	N/A	Use the recommended value.
gNodeB	Measurement Algorithm Switch	**gNBMobilityCommParam**.***MeasAlgoSwitch***	GAP_AVOID_NR_RES_SW	You are advised to select this option.
gNodeB	Networking Option Opt Sw	**gNodeBParam**.***NetworkingOptionOptSw***	NSA_REEST_FORB_MUL_FREQ_SEL_SW	You are advised to select this option when NSA networking is used and there are RRC connection reestablishments on the LTE side.

NE	Parameter Name	Parameter ID	Option	Setting Notes
eNodeB	NSA DC Algorithm Ext Switch	**EnodebAlgoExtSwitch**.***NsaDcAlgoExtSwitch***	NSA_REEST_NOTIFY_TO_NR_SW	You are advised to select this option when NSA networking is used and there are RRC connection reestablishments on the LTE side.

4.1.4.1.2 Using MML Commands

Before using MML commands, refer to **4.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Configuration on the eNodeB

```
//(Optional) Enabling "prohibition of multi-frequency smart selection for NSA UEs whose RRC connections are reestablished" when NSA networking is used and there are RRC connection reestablishments on the LTE side
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=NSA_REEST_NOTIFY_TO_NR_SW-1;
```

Configuration on the gNodeB

```
//Setting the CaBandCombSelectionStrat parameter based on operators' requirements (In the following example, this parameter is set to PEAK_RATE_FIRST.)
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaBandCombSelectionStrat=PEAK_RATE_FIRST;
//Selecting the MULTI_FREQ_SMART_SEL_SW option
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=MULTI_FREQ_SMART_SEL_SW-1;
//Setting the experience-first policy for smart carrier selection
MOD NRCELLCAMGMTCONFIG: NrCellId=0, SmartCarrSelPolicy=EXPERIENCE_FIRST;
//Setting the downlink experience evaluation hysteresis for smart carrier selection
MOD NRCELLCAMGMTCONFIG: NrCellId=0, DlExpEvalHyst=20;
//Setting the uplink experience evaluation hysteresis for smart carrier selection
MOD NRCELLCAMGMTCONFIG: NrCellId=0, UlExpEvalHyst=20;
//Setting the uplink experience evaluation factor for the cell
MOD NRDUCELLCARRMGMT: NrDuCellId=0, UlCellExpEvaluationFactor=0;
//(Recommended when the DL_INTRAF_SE_MODEL_BUILD_ALLOW option is selected) Selecting the DL_EXP_EVAL_SPCT_EFF_PRED_SW option to improve the spectral efficiency prediction accuracy
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=DL_EXP_EVAL_SPCT_EFF_PRED_SW-1;
//Modifying the inter-frequency measurement delay as required. The value 6 is used as an example in the following command.
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqMeasDelay=6;
//Setting the minimum interval for carrier selection
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CarrSelMinInterval=30;
//Setting the lower uplink SINR threshold used for evaluation of UE transfer from NR to E-UTRAN
MOD NRDUCELLSRSMEAS: NrDuCellId=0, NrToEutranSinrLowThld=-30;
//(Recommended for inter-frequency measurements) Selecting the GAP_AVOID_NR_RES_SW option
MOD GNBMOBILITYCOMMPARAM: MeasAlgoSwitch=GAP_AVOID_NR_RES_SW-1;
//Setting RSRP threshold 2 for event A5 related to multi-frequency smart selection to, for example, –104
```

```
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
MultiFreqSmtSelA5RsrpThld2=-104;
//(Required only when RSRP threshold 2 for event A5 related to multi-frequency smart selection is set to
255) Setting the RSRP threshold for interference-based inter-frequency handovers
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, IntrfInterFreqHoRsrpThld=-104;
//Setting the RSRP offset for experience-based PCC frequency selection to –3
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=8000, ExpBasedPCCFreqSelRsrpOfs=-3;
//Setting the inter-gNodeB user experience evaluation coefficient
MOD NRCELLCAMGMTCONFIG: NrCellId=0, InterGnbExpEvalCoeff=70;
//Setting network architecture selection flags for bearers of specific QCIs based on the network plan (The
following settings indicate that experience-based smart carrier selection is not allowed.)
MOD GNBQCIBEARER: Qci=1, NetworkingOptionOptFlag=FORBID;
MOD GNBQCIBEARER: Qci=9, NetworkingOptionOptFlag=PERMIT;
//Setting the NetworkingOptionOptFlag parameter to allow experience-based smart carrier selection for
UEs with a specific RFSP ID based on the network plan if the RFSP ID is configured on the core network
MOD GNBRFSPCONFIG: OperatorId=1, RfspIndex=1, NetworkingOptionOptFlag=TRUE;
//Setting the multi-frequency smart selection mode (The following uses UL_DL_MODE as an example.)
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqSmartSelMode=UL_DL_MODE;
//(Recommended when the VG_MODEL_ALLOW_BUILD_FLAG option is selected) Selecting the
MULTI_FREQ_SMART_SEL_PRED_SW option to reduce the number of invalid inter-frequency measurements
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=MULTI_FREQ_SMART_SEL_PRED_SW-1;
//Configuring differentiated downlink handling for CA-incapable UEs based on operators' requirements. In
the following example, this function is enabled.
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=CA_INCAP_UE_DL_DIFF_SW-1;
//Selecting the MULTI_FREQ_EXP_COEFF_EVAL_SW option if the downlink experience coefficient factor
needs to be considered for downlink air interface capability evaluation
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=MULTI_FREQ_EXP_COEFF_EVAL_SW-1;
//Setting the multi-carrier flow control level
MOD GNODEBALGO: MultiCarrierFlowCtrlLevel=MEDIUM;
//(Optional) Enabling "prohibition of multi-frequency smart selection for NSA UEs whose RRC connections
are reestablished" when NSA networking is used and there are RRC connection reestablishments on the LTE
side
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_REEST_FORB_MUL_FREQ_SEL_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Configuration on the eNodeB

```
//Disabling "prohibition of multi-frequency smart selection for NSA UEs whose RRC connections are
reestablished"
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=NSA_REEST_NOTIFY_TO_NR_SW-0;
```

Configuration on the gNodeB

```
//Deselecting the MULTI_FREQ_SMART_SEL_PRED_SW option
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=MULTI_FREQ_SMART_SEL_PRED_SW-0;
//Deselecting the MULTI_FREQ_SMART_SEL_SW option
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=MULTI_FREQ_SMART_SEL_SW-0;
//Disabling "prohibition of multi-frequency smart selection for NSA UEs whose RRC connections are
reestablished"
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_REEST_FORB_MUL_FREQ_SEL_SW-0;
```

4.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.1.4.2 Activation Verification

If any of the counters listed in **Table 4-11** returns a non-zero value, experience-based smart carrier selection has taken effect.

Table 4-11 Counters related to experience-based smart carrier selection

Counter ID	Counter Name
1911833645	**N.MultiFreqSmartSel.Exec**
1911833646	**N.MultiFreqSmartSel.Exec.CA**
1911833846	**N.MultiFreqSmartSel.DlIntraFreqSEModel.Exec**
1911833847	**N.MultiFreqSmartSel.DlIntraFreqSEModel.Exec.CA**
1911842641	**N.MultiFreqSmartSel.Exec.DL**
1911842642	**N.MultiFreqSmartSel.Exec.DL.CA**

4.1.4.3 Network Monitoring

Observe the average downlink and uplink UE throughputs to check whether UE data rates increase after this function is enabled.

- The average downlink UE throughput is measured by **User Downlink Average Throughput (DU)**.
- The average uplink UE throughput is measured by **User Uplink Average Throughput (DU)**.

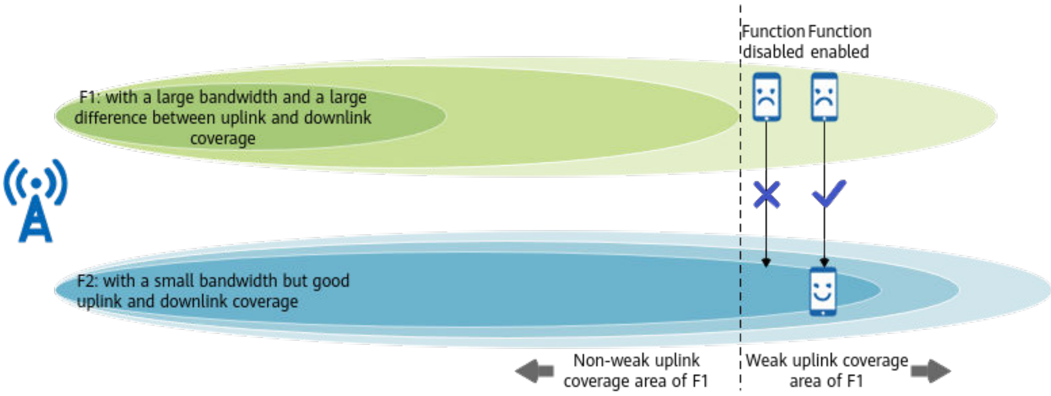
On the MAE-Access, start a signaling trace management task with the parameters for **Smart Carrier Selection Monitoring** under **User Performance Monitoring** specified to monitor experience-based smart carrier selection.

4.2 Protection of UEs Under Weak Uplink Coverage

4.2.1 Principles

Uplink coverage differs greatly from downlink coverage for cells operating on certain frequencies. If uplink coverage becomes limited first during UE movement, uplink and downlink user experience will be affected. To ensure service continuity in multi-carrier scenarios, protection of UEs under weak uplink coverage is introduced. This function identifies UEs under weak uplink coverage and hands over these UEs to inter-frequency cells, as shown in **Figure 4-4**.

Figure 4-4 Schematic diagram of protection of UEs under weak uplink coverage



The protection of UEs under weak uplink coverage function involves two phases:

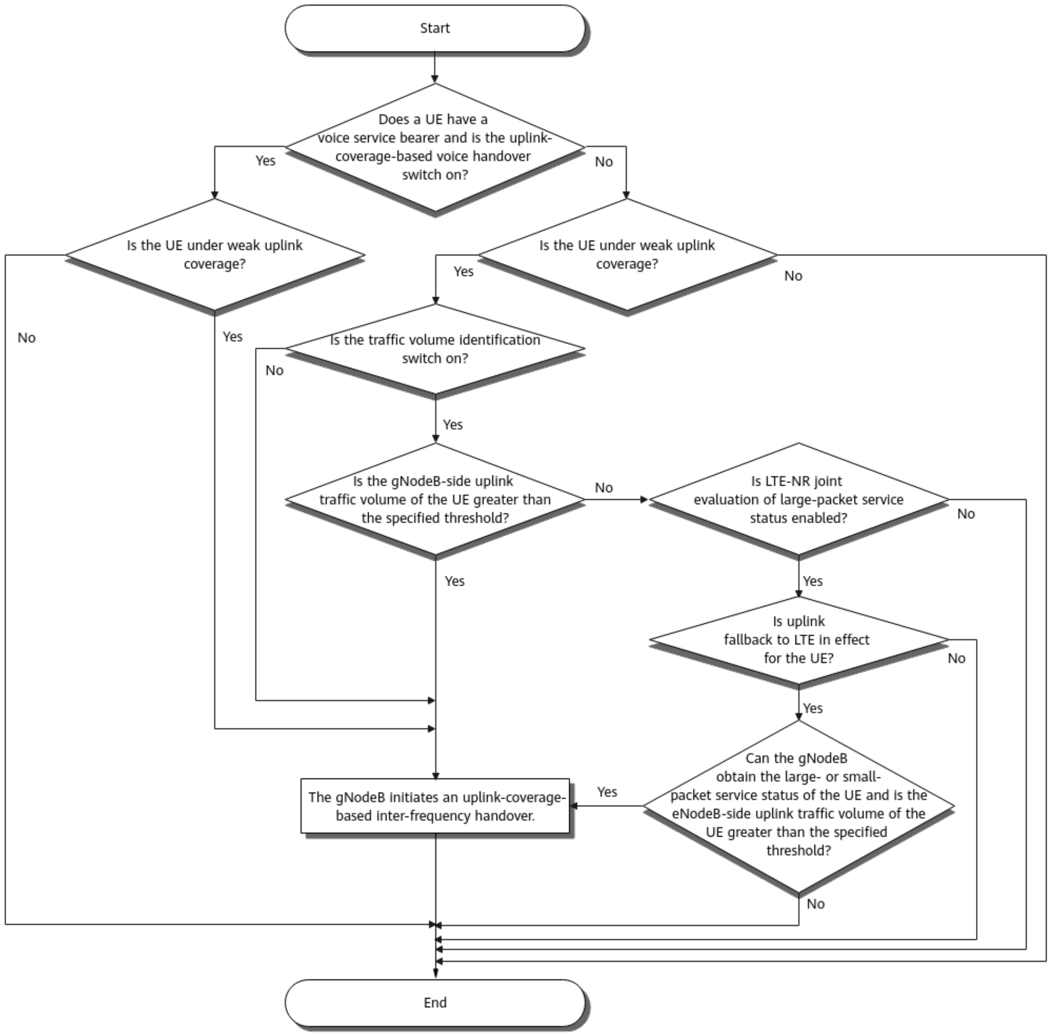
1. The gNodeB identifies target UEs under weak uplink coverage and determines whether to initiate uplink-coverage-based inter-frequency handovers for the identified UEs. For details, see **4.2.1.1 Decision on Whether to Initiate an Uplink-Coverage-based Inter-Frequency Handover**.
2. The gNodeB instructs target UEs to perform uplink-coverage-based inter-frequency handovers. For details, see **4.2.1.2 Uplink-Coverage-based Inter-Frequency Handover Execution**.

This function is controlled by the **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter.

4.2.1.1 Decision on Whether to Initiate an Uplink-Coverage-based Inter-Frequency Handover

The gNodeB identifies target UEs under weak uplink coverage periodically and determines whether to initiate uplink-coverage-based inter-frequency handovers for identified UEs. **Figure 4-5** shows the process of determining whether to initiate an uplink-coverage-based inter-frequency handover for a UE within a **period**.

Figure 4-5 Flowchart of deciding whether to initiate an uplink-coverage-based inter-frequency handover



The process is as follows:

1. The gNodeB checks the real-time uplink coverage of all UEs in the cell to identify target UEs under weak uplink coverage.

   It measures the SINR of each UE for five consecutive times at intervals specified by the **NSA uplink path selection SINR time-to-trigger**. The initial coverage status is considered to be strong coverage for all UEs by default.

   – If a UE has a voice service bearer (that is, a 5QI 1 bearer) but the **uplink-coverage-based voice handover switch** is off, or if a UE has no voice service bearer (that is, no 5QI 1 bearer), the gNodeB determines the coverage status of the UE based on the following rules:

     ■ If the UE's SINRs measured for five consecutive times are less than the **lower uplink SINR threshold used for evaluation of UE transfer from NR to E-UTRAN** minus 2 dB, the coverage status is recorded as weak coverage. In this case, the process proceeds to step 2.

  - If the UE's SINRs measured for five consecutive times are greater than the **lower uplink SINR threshold used for evaluation of UE transfer from NR to E-UTRAN** plus 2 dB, the coverage status is recorded as strong coverage. In this scenario, the process ends, and the next identification period starts.
  - Otherwise, the status remains unchanged.
- If a UE has a voice service bearer (that is, a 5QI 1 bearer) and the **uplink-coverage-based voice handover switch** is on, the gNodeB determines the coverage status of the UE based on the following rules:
  - If the UE's SINRs measured for five consecutive times are less than the **SINR threshold for uplink-coverage-based handover of voice-service UEs** minus 2 dB, the coverage status is recorded as weak coverage. For this UE, the gNodeB can directly perform an uplink-coverage-based inter-frequency handover. For details, see **4.2.1.2 Uplink-Coverage-based Inter-Frequency Handover Execution**.
  - If the UE's SINRs measured for five consecutive times are greater than the **SINR threshold for uplink-coverage-based handover of voice-service UEs** plus 2 dB, the coverage status is recorded as strong coverage. In this scenario, the process ends, and the next identification period starts.
  - Otherwise, the status remains unchanged.

2. If the UE is under weak coverage, the gNodeB further determines whether to initiate an uplink-coverage-based inter-frequency handover based on the status of the **traffic volume identification switch**.
   - If the **traffic volume identification switch** is on, the gNodeB checks whether the uplink traffic volume to be transmitted by the UE is greater than the **threshold for identifying common large-packet UEs in the uplink**.
     - If the volume is greater than the threshold, the gNodeB can directly initiate an uplink-coverage-based inter-frequency handover. For details, see **4.2.1.2 Uplink-Coverage-based Inter-Frequency Handover Execution**.
     - If the volume is less than or equal to the threshold and the **LTE-NR joint evaluation of large-packet service status function** is disabled, the procedure ends and the next identification period starts.
     - If the volume is less than or equal to the threshold and the **LTE-NR joint evaluation of large-packet service status function** is enabled, the processing varies as follows:
       - If uplink fallback to LTE is in effect for the UE and:

         If the gNodeB can obtain the service status (large- or small-packet service) of the UE from the eNodeB and the eNodeB-side uplink traffic volume to be transmitted by the UE is greater than the **uplink large-packet traffic volume threshold**, the gNodeB can initiate an uplink-coverage-based inter-frequency handover. For details, see **4.2.1.2 Uplink-Coverage-based Inter-Frequency Handover Execution**.

If the gNodeB can obtain the service status (large- or small-packet service) of the UE from the eNodeB and the eNodeB-side uplink traffic volume to be transmitted by the UE is less than or equal to the **uplink large-packet traffic volume threshold**, the procedure ends and the next identification period starts.

If the gNodeB cannot obtain the service status (large- or small-packet service) of the UE from the eNodeB, the procedure ends and the next identification period starts.

- If uplink fallback to LTE is not in effect for the UE, the procedure ends and the next identification period starts.

For details about uplink fallback to LTE and whether the gNodeB can obtain the service status (large- or small-packet service) of UEs with uplink fallback to LTE in effect from the eNodeB, see *EPC-based NSA Basics* .

– If the **traffic volume identification switch** is off, the gNodeB can directly initiate an uplink-coverage-based inter-frequency handover. For details, see **4.2.1.2 Uplink-Coverage-based Inter-Frequency Handover Execution**.

The parameters involved in deciding whether to initiate an uplink-coverage-based inter-frequency handover are listed in **Table 4-12**.

Table 4-12 Parameters involved in deciding whether to initiate an uplink-coverage-based inter-frequency handover

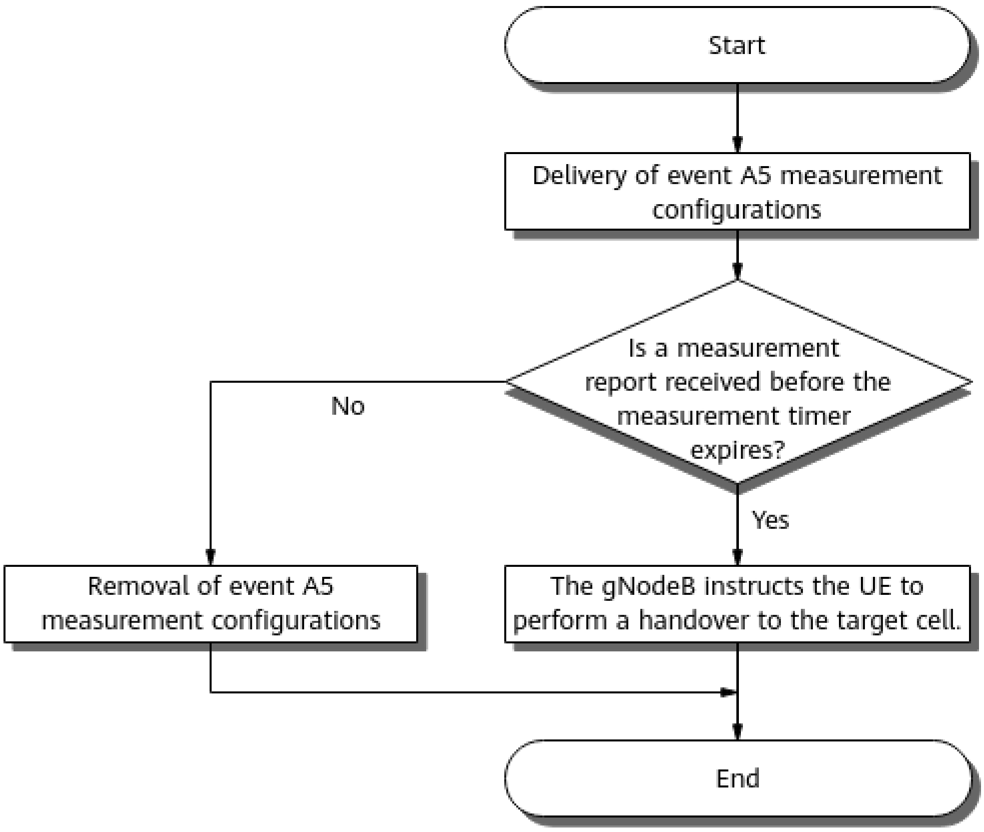
Item	Parameter
Period	**gNodeBParam**.***NsaSaSelOptPeriod***
NSA uplink path selection SINR time-to-trigger	**NRDUCellSrsMeas**.***NsaUlPathSelSinrTimeToTrig***
Uplink-coverage-based voice handover switch	**VOICE_UL_COV_BASED_HO_SW** option of the **NRCellAlgoSwitch**.***VoiceStrategySwitch*** parameter
Lower uplink SINR threshold used for evaluation of UE transfer from NR to E-UTRAN	**NRDUCellSrsMeas**.***NrToEutranSinrLowThld***
SINR threshold for uplink-coverage-based voice handover	**NRCellMobilityConfig**.***VoiceUlCovBasedHoSinrThld***
Traffic volume identification switch	**VOLUME_IDENTIFY_SW** option of the **NRCellAlgoSwitch**.***ServiceFunctionSwitch*** parameter

Item	Parameter
Threshold for identifying common large-packet UEs in the uplink	**NRDUCellServExp**.***CommonUlLargePktIdentThld***
LTE-NR joint evaluation of large-packet service status	**LNR_LRG_PKT_UL_WEAK_COV_SW** option of the **NRCellMobilityConfig**.***InterFreqHoExtSwitch*** parameter
Uplink large-packet traffic volume threshold	**CellServiceDiffCfg**.***UlLargePacketDataVoThd***

4.2.1.2 Uplink-Coverage-based Inter-Frequency Handover Execution

The base station initiates uplink-coverage-based inter-frequency handovers for the identified target UEs. This function allows measurement-based handover but does not allow blind handover. **Figure 4-6** shows the process of executing an uplink-coverage-based inter-frequency handover.

Figure 4-6 Process of executing an uplink-coverage-based inter-frequency handover



The process is as follows:

1. The gNodeB delivers event A5 measurement configurations to the identified target UEs for uplink-coverage-based inter-frequency handovers.

   The delivered event A5 measurement configurations include the target frequencies and event A5 thresholds.

   – Target frequencies refer to frequencies whose **Frequency Priority for Connected Mode** have been specified. For details about the maximum number of neighboring NG-RAN frequencies that can be delivered by the gNodeB, see *SA Mobility Management in Connected Mode*.
   – There are two thresholds for event A5: **RSRP threshold 1 for event A5 related to uplink-coverage-based inter-frequency handover** and **RSRP threshold 2 for event A5 related to coverage-based inter-frequency handover**.

2. The gNodeB waits for event A5 measurement reports before the **measurement timer for protection against weak uplink coverage** expires.

   – If the gNodeB receives an event A5 measurement report before the timer expires, it performs step **3**.
   – If the gNodeB does not receive any event A5 measurement report on any frequency before the timer expires, it removes event A5 measurement configurations and ends the procedure.

3. The gNodeB selects a target cell.

   The gNodeB selects a cell with the highest RSRP as the target cell based on the event A5 measurement report from the UE.

4. The gNodeB instructs the UE to perform a handover to the target cell.

   – If the PCI of the target cell is in the neighboring cell list configured on the gNodeB and PCI confusion does not occur, then:
     - If the UE supports handover, the gNodeB instructs the UE to perform a handover to the target cell.
     - If the UE does not support handover, the gNodeB instructs the UE to perform a redirection to the target cell.
   – If the PCI of the target cell is not in the neighboring cell list configured on the gNodeB or PCI confusion occurs, then:
     - If the intra-NR ANR switch is on and the UE supports CGI measurement and handover, the gNodeB instructs the UE to perform a handover to a neighboring NR cell with the best signal quality based on an ANR-maintained neighboring cell list.
     - If the intra-NR ANR switch is on and the UE supports CGI measurement but does not support handover, the gNodeB instructs the UE to perform a redirection to a neighboring NR cell with the best signal quality based on an ANR-maintained neighboring cell list.
     - If the intra-NR ANR switch is off or the UE does not support CGI measurement, the gNodeB stops the procedure.

The parameters involved in the execution of an uplink-coverage-based inter-frequency handover are listed in **Table 4-13**.

Table 4-13 Parameters involved in the execution of an uplink-coverage-based inter-frequency handover

Item	Parameter
Frequency Priority for Connected Mode	**NRCellFreqRelation**.***ConnFreqPriority***
RSRP threshold 1 for event A5 related to uplink-coverage-based inter-frequency handover	**NRCellInterFHoMeaGrp**.***UlCovInterFreqA5RsrpThld1***
RSRP Threshold 2 for event A5 related to coverage-based inter-frequency handover	**NRCellInterFHOMeaGrp**.***CovInterFreqA5RsrpThld2***
Measurement timer for protection against weak uplink coverage	**NRCellMobilityConfig**.***UlWeakCovPrtMeasTimer***
Intra-NR ANR switch	**NR_NR_ANR_SW** option of the **NRCellAlgoSwitch**.***AnrSwitch*** parameter

NOTE

- For details about PCI confusion, see *PCI Conflict Detection and Self-Optimization*.
- For details about ANR, see *ANR*.

4.2.2 Network Analysis

4.2.2.1 Benefits

This function applies to multi-carrier scenarios where uplink coverage differs greatly from downlink coverage for cells on certain frequencies and uplink coverage of inter-frequency neighboring cells differs from each other greatly. It is recommended that this function be enabled in cells on the frequencies with a large difference between uplink coverage and downlink coverage.

This function increases the average uplink UE throughput (indicated by **User Uplink Average Throughput (DU)**) in weak uplink coverage scenarios. The gain varies depending on the following factors: ratio between inter-frequency sites, distribution of UEs (in the cell center, between the cell center and the cell edge, and at the cell edge), bandwidth of inter-frequency cells, uplink load, number of base station receive antennas, and interference. The gain cannot be ensured in the following conditions: the ratio between inter-frequency sites is not 1:1, the proportion of cell edge UEs is low, the bandwidth of inter-frequency cells is small, uplink load is high, the number of base station receive antennas is small, and interference is high.

The service drop rate (**Service Call Drop Rate (CU, Inactive)**) decreases when all of the following conditions are met:

- The **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter is selected.
- The **VOLUME_IDENTIFY_SW** option of the **NRCellAlgoSwitch**.***ServiceFunctionSwitch*** parameter is selected.
- The **NRDUCellSrsMeas**.***NrToEutranSinrLowThld*** parameter is set to an appropriate value based on site requirements.
- The **NRDUCellServExp**.***CommonUlLargePktIdentThld*** parameter is set to an appropriate value based on site requirements.

4.2.2.2 Impacts

The impacts between this function and other functions are described in **4.2.3.2.3 Function Impacts**.

Protection of UEs under weak uplink coverage has the following network impacts:

- Service interruption occurs when a UE synchronizes with the target cell and initiates random access during a handover, causing delay and affecting throughput.
- Measurement gaps are used for inter-frequency measurement, leading to throughput decrease.
- Current networking and cooperation with other types of handover (such as frequency-priority-based inter-frequency handover) must be considered during function deployment. Otherwise, ping-pong handovers may occur, affecting user experience. To prevent ping-pong handovers, there are requirements for feature cooperation, network planning, and data configuration. For details, see **4.2.3.2.3 Function Impacts**, **4.2.3.4 Networking**, and **4.2.4.1 Data Configuration**, respectively.
- If the **VOICE_UL_COV_BASED_HO_SW** option of the **NRCellAlgoSwitch**.***VoiceStrategySwitch*** parameter is selected, the **Service Call Drop Rate (CU)** may increase.

4.2.3 Requirements

4.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-061223	Experience Boosting based on Multi-Band Coordination	NR0S0SCSNR00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Protection of UEs under weak uplink coverage	**UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	LTE-NR joint evaluation of large-packet service status (controlled by the **LNR_LRG_PKT_UL_WEAK_COV_SW** option of the **NRCellMobilityConfig**.***InterFreqHoExtSwitch*** parameter) can be enabled only when protection of UEs under weak uplink coverage has been enabled.
Low-frequency TDD	Function of requesting LTE-side large-packet status of ULFB UEs	**ULFB_LTE_UL_LRG_PKT_REQ_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter	*EPC-based NSA Basics*	LTE-NR joint evaluation of large-packet service status (controlled by the **LNR_LRG_PKT_UL_WEAK_COV_SW** option of the **NRCellMobilityConfig**.***InterFreqHoExtSwitch*** parameter) can be enabled only when the function of requesting LTE-side large-packet status of ULFB UEs has been enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Traffic volume identification	**VOLUME_IDENTIFY_SW** option of the **NRCellAlgoSwitch**.***ServiceFunctionSwitch*** parameter	*Multi-Frequency Smart Selection*	LTE-NR joint evaluation of large-packet service status (controlled by the **LNR_LRG_PKT_UL_WEAK_COV_SW** option of the **NRCellMobilityConfig**.***InterFreqHoExtSwitch*** parameter) can take effect only when the traffic volume identification switch is turned on.

4.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User-experience-based coverage expansion	**COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	When user-experience-based coverage expansion is enabled, protection of UEs under weak uplink coverage (controlled by the **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) is not supported.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User-experience-based NSA SCG coverage expansion	**NSA_COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	When user-experience-based NSA SCG coverage expansion is enabled, protection of UEs under weak uplink coverage (controlled by the **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) is not supported.

4.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink-interference-based inter-frequency handover	**UL_INTRF_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	In multi-carrier scenarios, ping-pong handovers may easily occur between a cell with uplink-interference-based inter-frequency handover enabled and an inter-frequency cell with protection of UEs under weak uplink coverage (controlled by the **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) enabled. As such, you are advised to further select the **INTER_FREQ_UL_INTRF_SW** option of the **NRCellAlgoSwitch**.***NCellInfoCtrlSwitch*** parameter for this inter-frequency cell, so that handovers to high-interference cells are not triggered during handover decision-making.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-number-based connected mode MLB	• **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter • **NRCellMlb**.***MlbTriggerMode*** set to **CELL_USER_NUM**	*Mobility Load Balancing in Connected Mode*	In multi-carrier scenarios, ping-pong handovers may easily occur between a cell with UE-number-based connected mode MLB enabled and an inter-frequency cell with protection of UEs under weak uplink coverage (controlled by the **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) enabled. As such, you are advised to further select the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter for this inter-frequency cell. In this way, ping-pong handovers are prevented as cells in the MLB state will not be selected as the target cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	• **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter • **NRCellMlb**.***MlbTriggerMode*** set to **SPEC_EFF_BASED_USER_NUM**	*Mobility Load Balancing in Connected Mode*	In multi-carrier scenarios, ping-pong handovers may easily occur between a cell with spectral-efficiency-based connected mode MLB enabled and an inter-frequency cell with protection of UEs under weak uplink coverage (controlled by the **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) enabled. As such, you are advised to further select the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter for this inter-frequency cell. In this way, ping-pong handovers are prevented as cells in the MLB state will not be selected as the target cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	5QI-specific UE-number-based connected mode MLB	• **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter • **NRCellMlb**.***MlbTriggerMode*** set to **SPECIFIED_5QI_USER_NUM**	*Mobility Load Balancing in Connected Mode*	In multi-carrier scenarios, ping-pong handovers may easily occur between a cell with 5QI-specific UE-number-based connected mode MLB enabled and an inter-frequency cell with protection of UEs under weak uplink coverage (controlled by the **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) enabled. As such, you are advised to further select the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter for this inter-frequency cell. In this way, ping-pong handovers are prevented as cells in the MLB state will not be selected as the target cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Radio-resource-based connected mode MLB	• **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter • **NRCellMlb**.***MlbTriggerMode*** set to **RADIO_RESOURCE**	*Mobility Load Balancing in Connected Mode*	In multi-carrier scenarios, ping-pong handovers may easily occur between a cell with radio-resource-based connected mode MLB enabled and an inter-frequency cell with protection of UEs under weak uplink coverage (controlled by the **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) enabled. As such, you are advised to further select the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter for this inter-frequency cell. In this way, ping-pong handovers are prevented as cells in the MLB state will not be selected as the target cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	• **UL_SINR_MOBILITY_TO_EUTRAN_SW** option of the **NRCellAlgoSwitch**.***InterRatServiceMobilitySw*** parameter • **MOBILITY_TO_EUTRAN_SW** option of the **NRCellAlgoSwitch**.***InterRatServiceMobilitySw*** parameter	*Interoperability Between E-UTRAN and NG-RAN in Connected Mode*	When protection of UEs under weak uplink coverage (controlled by the **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) and uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN are both enabled, which of the functions takes effect for a UE depends on the measurement report received first from the UE. If the base station receives an inter-frequency event A5 measurement report first, protection of UEs under weak uplink coverage takes effect. If the base station receives an inter-RAT event B2 measurement report first, uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SA-to-NSA switching based on weak uplink coverage	• **NSA_SA_UL_SEL_OPT_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter • **NSA_SA_UL_SEL_OPT_SW** option of the **gNBOperator**.***OperatorInterRatPolicySw*** parameter	*NSA-SA Selection Based on User Experience*	When protection of UEs under weak uplink coverage (controlled by the **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) and SA-to-NSA switching based on weak uplink coverage are both enabled, which of the functions takes effect for a UE depends on whether the conditions for the latter function are met. If the conditions are met, the SA-to-NSA switching is preferentially performed. If the conditions are not met, protection of UEs under weak uplink coverage takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Voice-capability-based inter-frequency handover back to the HPLMN	**VOICE_CAPB_MOB_TO_HPLMN_SW** option of the **gNBOperator**.***HplmnAlgoSwitch*** parameter	*SA Mobility Management in Connected Mode*	In the handover decision-making phase for protection of UEs under weak uplink coverage, if voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE and the **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter is selected for the serving cell, then: ● When the **gNBFreqPriorityGroup**.***RoamingUeHplmnInterFHoFlag*** parameter is set to **TRUE** for the target cell's frequency in the frequency list group bound to the roaming UE, the following mechanism is applied: – If the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. – If the roaming UE does not support

RAT	Function Name	Function Switch	Reference	Description
				handovers, the gNodeB instructs the roaming UE to immediately perform a redirection to the frequency. ● When the **gNBFreqPriorityGroup**.***RoamingUeHplmnInterFHoFlag*** parameter is set to **FALSE** for the target cell's frequency in the frequency list group bound to the roaming UE, the gNodeB instructs the roaming UE to immediately perform a redirection to the target cell's frequency.

4.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.2.3.4 Networking

- The network uses at least two frequencies.
- Network planning should avoid handovers back to source cells after several unnecessary handovers of the same or different types. Otherwise, ping-pong handovers occur, affecting user experience. Unnecessary handovers refer to handovers other than coverage-based intra-frequency handover and coverage-based inter-frequency handover.

  For example, network planning should avoid the following situation: a UE is handed over from cell 1 to cell 2 through an uplink-interference-based inter-frequency handover, then from cell 2 to cell 3 through a service-based inter-frequency handover, and finally from cell 3 back to cell 1 through a frequency-priority-based handover.

4.2.3.5 Cells

When the **NRCell**.***DuplexMode*** parameter is set to **CELL_SDL** for a cell, protection of UEs under weak uplink coverage cannot be enabled.

4.2.3.6 Others

UEs need to support inter-frequency intra-FR1 TDD handover. The inter-frequency handover capability is indicated by the *handoverInterF* IE in the UECapabilityInformation message on the Uu interface.

The *handoverInterF* IE can be carried in the following IEs: *measAndMobParametersXDD-Diff*, *measAndMobParametersFRX-Diff*, *tdd-Add-UE-NR-Capabilities*, and *fr1-Add-UE-NR-Capabilities*. Therefore, the inter-frequency handover capability of a UE needs to be determined based on the values of *handoverInterF* in these IEs. For details about how to determine the capability based on the values of *handoverInterF*, see section 4.2 "UE Capability Parameters" in 3GPP TS 38.306 V15.9.0.

When the values of *handoverInterF* in the aforementioned IEs differ from each other, the methods to evaluate whether UEs support inter-frequency handover vary among vendors. Therefore, 3GPP TS 38.306 V16.1.0 has updated the definition of the inter-frequency handover capability and describes how to determine the capability based on different values of *handoverInterF* in the IEs. For details, see "Annex B: UE capability indication for UE capabilities with both FDD/TDD and FR1/FR2 differentiations" in 3GPP TS 38.306 V16.1.0.

To ensure UE compatibility and improve UE capability determination accuracy, Huawei has made compatibility optimization in accordance with protocols. This optimization is controlled by the **INTER_FREQ_HO_CAPB_IND_OPT_SW** option of the **gNBMobilityCommParam**.***ProtocolCompatibilitySw*** parameter. It is recommended that this option be selected. Selecting this option can reduce the number of NR inter-frequency handovers caused by UE compatibility issues, that is, the number of invalid inter-frequency handover attempts.

- If this option is selected:

  The gNodeB refers to the inter-frequency handover capability definition updated in 3GPP TS 38.306 V16.1.0 to determine the UE capability based on *handoverInterF*.
- If this option is deselected:

  The gNodeB refers to the inter-frequency handover capability definition provided in versions earlier than 3GPP TS 38.306 V16.1.0 to determine the UE capability based on *handoverInterF*.

4.2.4 Operation and Maintenance

4.2.4.1 Data Configuration

4.2.4.1.1 Data Preparation

Before activating this function, configure NR cell frequency relationships and neighbor relationships. For details about related parameters, see **Table 4-14** and the table with the parameters used to configure neighboring cells in *SA Mobility Management in Connected Mode*, respectively. In RAN sharing with common carrier, the parameters related to the NR external-cell PLMN list (described in **Table 4-15**) must also be configured.

For details about the function activation switch, see **Table 4-16**.

For details about the parameters used for NR inter-frequency handover measurement configuration, see **Table 4-17**.

For details about other related parameters, see **Table 4-18**. This section does not describe parameters related to cell establishment.

Table 4-14 Parameters used for NR cell frequency relationship configuration

Parameter Name	Parameter ID	Option	Setting Notes
NR Cell ID	**NRCellFreqRelation.*NrCellId***	N/A	Set this parameter based on the network plan.
SSB Frequency Position	**NRCellFreqRelation.*SsbFreqPos***	N/A	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	**NRCellFreqRelation.*SsbDescMethod***	N/A	Set this parameter to its recommended value.
Frequency Band	**NRCellFreqRelation.*FrequencyBand***	N/A	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Subcarrier Spacing	**NRCellFreqRelation**.***SubcarrierSpacing***	N/A	Set this parameter based on the network plan.
Frequency Priority for Connected Mode	**NRCellFreqRelation**.***ConnFreqPriority***	N/A	Set this parameter based on the network plan.

Table 4-15 Parameters related to the NR external-cell PLMN list

Parameter Name	Parameter ID	Option	Setting Notes
NR Networking Option	**NRExternalNCellPlmn**.***NrNetworkingOption***	N/A	Set this parameter based on the network plan.

Table 4-16 Parameters used for function activation in an NR cell

Parameter Name	Parameter ID	Option	Setting Notes
Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	UL_WEAK_COV_UE_PROTECT_SW	You are advised to select this option.

Table 4-17 Parameters used for measurement configuration for NR inter-frequency handover

Parameter Name	Parameter ID	Option	Setting Notes
NR Cell ID	**NRCellInterFHoMeaGrp**.***NrCellId***	N/A	Set this parameter based on the network plan.
Inter-freq Handover Measurement Group ID	**NRCellInterFHoMeaGrp**.***InterFreqHoMeasGroupId***	N/A	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Inter-freq Measurement Event Time to Trigger	**NRCellInterFHoMeaGrp**.***InterFreqA4A5TimeToTrig***	N/A	Set this parameter to its recommended value.
Inter-freq Measurement Event Hysteresis	**NRCellInterFHoMeaGrp**.***InterFreqA4A5Hyst***	N/A	Set this parameter to its recommended value.
UL Cov Inter-freq A5 RSRP Threshold1	**NRCellInterFHoMeaGrp**.***UlCovInterFreqA5RsrpThld1***	N/A	To prevent ping-pong handovers between cells, it is recommended that RSRP threshold 1 for event A5 related to uplink-coverage-based inter-frequency handovers be less than or equal to the RSRP threshold for event A4 and RSRP threshold 2 for event A5 related to other types of inter-frequency handovers, including but not limited to protection of UEs under weak uplink coverage, coverage-based inter-frequency handover, frequency-priority-based inter-frequency handover, operator-specific-priority-based inter-frequency handover, service-based inter-frequency handover, and voice-quality-based inter-frequency handover.
Cov Inter-freq Handover A5 RSRP Threshold2	**NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld2***	N/A	Set this parameter to its recommended value.
QoS Class Identifier	**NRCellQciBearer**.***Qci***	N/A	Set this parameter based on the network plan.
Inter-freq HO Measurement Group ID	**NRCellQciBearer**.***InterFreqHoMeasGroupId***	N/A	Set this parameter based on the network plan.
QoS Class Identifier	**gNBQciBearer**.***Qci***	N/A	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
QCI Priority for Handover	**gNBQciBearer**.***QciPriorityForHo***	N/A	Set this parameter based on the network plan.
Protocol Compatibility Switch	**gNBMobilityCommParam**.***ProtocolCompatibilitySw***	INTER_FREQ_HO_CAPB_IND_OPT_SW	You are advised to select this option.

Table 4-18 Other related parameters

NE	Parameter Name	Parameter ID	Option	Setting Notes
gNodeB	NR Move to E-UTRAN UL SINR Low Threshold	**NRDUCellSrsMeas**.***NrToEutranSinrLowThld***	N/A	Set this parameter to its recommended value. In NSA networking, if both protection of UEs under weak uplink coverage and fallback to LTE are enabled, it is recommended that related parameter settings meet the following criterion so that protection of UEs under weak uplink coverage preferentially takes effect: **NRDUCellSrsMeas**.***NrToEutranSinrLowThld*** – 2 dB > **NRDUCellSrsMeas**.***NsaUlFackToLteSinrThld*** + **NRDUCellSrsMeas**.***NsaUlFackToLteSinrHyst***.

NE	Parameter Name	Parameter ID	Option	Setting Notes
gNodeB	Voice Strategy Switch	**NRCellAlgoSwitch**.***VoiceStrategySwitch***	VOICE_UL_COV_BASED_HO_SW	It is recommended that this option be selected for an NR cell with VoNR enabled if the SINR threshold for uplink-coverage-based handover of voice-service UEs and the lower uplink SINR threshold used for evaluation of UE transfer from NR to E-UTRAN need to be set to different values.
gNodeB	Voice UL Coverage-based HO SINR Threshold	**NRCellMobilityConfig**.***VoiceUlCovBasedHoSinrThld***	N/A	Set this parameter to its recommended value.
gNodeB	Service Function Switch	**NRCellAlgoSwitch**.***ServiceFunctionSwitch***	VOLUME_IDENTIFY_SW	You are advised to select this option.
gNodeB	Inter-Frequency Handover Extend Switch	**NRCellMobilityConfig**.***InterFreqHoExtSwitch***	LNR_LRG_PKT_UL_WEAK_COV_SW	You are advised to select this option.
gNodeB	Common UL Large Pkt Ident Thld	**NRDUCellServExp**.***CommonUlLargePktIdentThld***	N/A	Set this parameter to its recommended value.
gNodeB	NSA UL Path Select SINR Time-to-Trigger	**NRDUCellSrsMeas**.***NsaUlPathSelSinrTimeToTrig***	N/A	Set this parameter to its recommended value.
gNodeB	NSA/SA Selection Opt Period	**gNodeBParam**.***NsaSaSelOptPeriod***	N/A	Set this parameter to its recommended value.
gNodeB	UL Weak Coverage Protection Measurement Timer	**NRCellMobilityConfig**.***UlWeakCovPrtMeasTimer***	N/A	Set this parameter to its recommended value.

NE	Parameter Name	Parameter ID	Option	Setting Notes
eNodeB	UL Large Packet Data Volume Threshold	**CellServiceDiffCfg**.***UlLargePacketDataVoThd***	N/A	Set this parameter to its recommended value.

4.2.4.1.2 Using MML Commands

Before using MML commands, refer to **4.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Configurations on the eNodeB side:

```
//Setting the uplink large-packet traffic volume threshold (local cell ID 0 is used as an example)
MOD CELLSERVICEDIFFCFG: LocalCellId=0, UlLargePacketDataVoThd=25;
```

Configurations on the gNodeB side:

For low-frequency TDD cells

```
//Adding an NR cell frequency relationship
ADD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=8000, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N77, SubcarrierSpacing=30KHZ, ConnFreqPriority=16;
//Adding an external cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, CellName="A", PhysicalCellId=0,
Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8000, FrequencyBand=N77,
NrNetworkingOption=UNLIMITED;
//Adding a PLMN list for the external NR cell in RAN sharing with common carrier mode
ADD NREXTERNALNCELLPLMN: Mcc="302", Mnc="220", gNBId=1, CellId=0, SharedMcc="460",
SharedMnc="03", Tac=0, NrNetworkingOption=SA;
//Adding an NR cell relationship
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0, CellIndividualOffset=DB0;
//Configuring measurement parameter groups for inter-frequency handovers
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, UlCovInterFreqA5RsrpThld1=-110,
CovInterFreqA5RsrpThld2=-106;
MOD NRCELLINTERFHOMEAGRP: NrCellId=0,
InterFreqHoMeasGroupId=1,InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2,
UlCovInterFreqA5RsrpThld1=-110, CovInterFreqA5RsrpThld2=-106;
//Setting QCI-specific measurement parameters
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, InterFreqHoMeasGroupId=0;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, InterFreqHoMeasGroupId=1;
//Configuring QCI priorities for handovers
MOD GNBQCIBEARER: Qci=1, QciPriorityForHo=1;
MOD GNBQCIBEARER: Qci=9, QciPriorityForHo=7;
//Selecting the INTER_FREQ_HO_CAPB_IND_OPT_SW option
MOD GNBMOBILITYCOMMPARAM: ProtocolCompatibilitySw=INTER_FREQ_HO_CAPB_IND_OPT_SW-1;
//Changing the uplink SINR threshold for NR inter-frequency handovers
```

```
MOD NRDUCELLSRSMEAS: NrDuCellId=0, NrToEutranSinrLowThld=-30;
//Enabling protection of UEs under weak uplink coverage to take uplink traffic volume into consideration
MOD NRCELLALGOSWITCH: NrCellId=0, ServiceFunctionSwitch=VOLUME_IDENTIFY_SW-1;
//Changing the threshold for identifying common large-packet UEs in the uplink as required (It is
recommended that the default value be used. The value 80 is used as an example.)
MOD NRDUCELLSERVEXP: NrDuCellId=0, CommonUlLargePktIdentThld=80;
//Changing the SINR-based time-to-trigger for NSA uplink path selection as required (It is recommended
that the default value be used. The value 10 is used as an example.)
MOD NRDUCELLSRSMEAS: NrDuCellId=0, NsaUlPathSelSinrTimeToTrig=10;
//Changing the NSA/SA selection optimization period as required (It is recommended that the default value
be used. The value 60 is used as an example.)
MOD GNODEBPARAM: NsaSaSelOptPeriod=60;
//Selecting the VOICE_UL_COV_BASED_HO_SW option and setting the VoiceUlCovBasedHoSinrThld
parameter if weak uplink coverage determination for voice services is required (as the uplink weak
coverage determination procedure for voice services is decoupled from that for data services)
MOD NRCELLALGOSWITCH: NrCellId=0, VoiceStrategySwitch=VOICE_UL_COV_BASED_HO_SW-1;
MOD NRCELLMOBILITYCONFIG: NrCellId=0, VoiceUlCovBasedHoSinrThld=-30;
//Activating protection of UEs under weak uplink coverage
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=UL_WEAK_COV_UE_PROTECT_SW-1;
//(Optional) Enabling LTE-NR joint evaluation of large-packet service status in NSA networking, to support
uplink-coverage-based inter-frequency handover for UEs with uplink fallback to LTE in effect
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqHoExtSwitch=LNR_LRG_PKT_UL_WEAK_COV_SW-1;
//Setting the measurement timer for protection against weak uplink coverage
MOD NRCELLMOBILITYCONFIG: NrCellId=0, UlWeakCovPrtMeasTimer=4;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Deactivating protection of UEs under weak uplink coverage
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=UL_WEAK_COV_UE_PROTECT_SW-0;
```

4.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.2.4.2 Activation Verification

If any of the counters listed in **Table 4-19** has a non-zero value, protection of UEs under weak uplink coverage has taken effect.

Table 4-19 Counters related to the number of uplink-coverage-based inter-frequency handovers

Counter ID	Counter Name
1911833798	**N.HO.InterFreq.ULCoverage.PrepAttOut**
1911833799	**N.HO.InterFreq.ULCoverage.ExecAttOut**
1911833800	**N.HO.InterFreq.ULCoverage.ExecSuccOut**
1911833801	**N.NsaDc.InterFreq.ULCoverage.PSCell.Change.Succ**

Counter ID	Counter Name
1911833802	**N.NsaDc.InterFreq.ULCoverage.PSCell.Change.Att**

4.2.4.3 Network Monitoring

The **User Uplink Average Throughput (DU)** can be used to monitor throughput increase.

The **Service Call Drop Rate (CU, Inactive)** can be used to monitor service drop decrease.

The **Inter-Frequency Handover Out Success Rate (CU)** can be used to monitor handover performance.

4.3 User-Experience-based Coverage Expansion

4.3.1 Principles

On a network deployed with carriers that provide different coverage, there may be coverage capability and bandwidth differences between frequency bands. If a frequency band with good uplink and downlink coverage but a small bandwidth becomes congested because a large number of UEs access this frequency band, the experience of some UEs will be affected. The causes of congestion on the frequency band that has good uplink and downlink coverage but a small bandwidth are as follows:

- UEs are handed over from the cell working on a frequency band with a large bandwidth but a large difference between uplink and downlink coverage to the cell working on a frequency band with a small bandwidth but good uplink and downlink coverage through coverage-based inter-frequency handovers.
- Cell-edge UEs in the cell working on a frequency band with a small bandwidth but good uplink and downlink coverage cannot be handed over to the cell working on a frequency band with a large bandwidth but a large difference between uplink and downlink coverage. Therefore, these UEs always camp on the former cell.

To reduce the load of cells on frequency bands with good coverage but small bandwidths and improve the traffic absorption capability of cells on frequency bands with large bandwidths but a large difference between uplink and downlink coverage, the user-experience-based coverage expansion function is introduced. This function expands the overlapping coverage between frequency bands and selects optimal carriers for UEs.

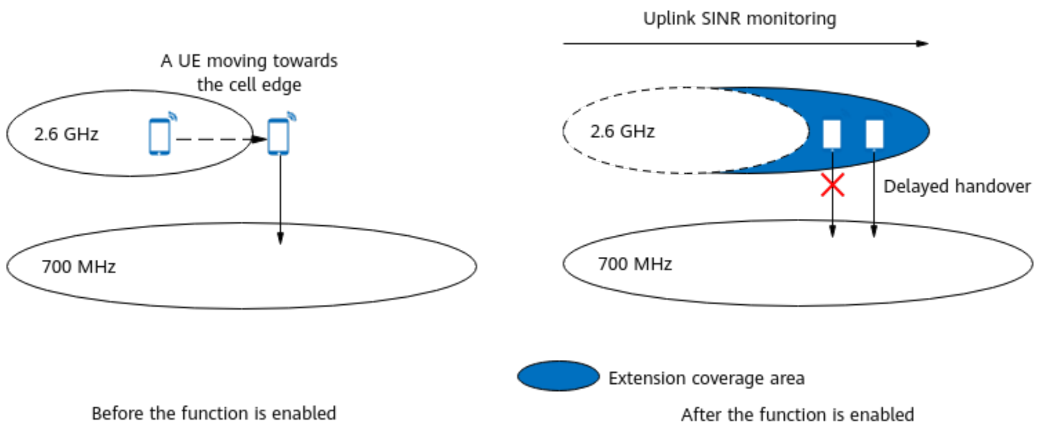
Based on user experience estimation and uplink SINR monitoring, user-experience-based coverage expansion optimizes the load distribution between cells on different frequency bands. Therefore, the load of cells on frequency bands with good uplink and downlink coverage but small bandwidths is reduced, increasing the average downlink throughput of cell-edge UEs in these cells and further increasing the average downlink UE throughput on the entire network. In this document, the sub-1 GHz and non-sub-1 GHz frequency bands are used as examples. The principles for other frequency bands are similar.

User-experience-based coverage expansion is controlled by the **COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter. Currently, it is supported only in SA networking. This function includes the following sub-functions:

- Threshold adaptation for coverage-based inter-frequency handovers

  When this sub-function takes effect, the gNodeB expands the coverage area of cells on non-sub-1 GHz frequency bands to delay handovers of UEs from these cells to those on sub-1 GHz frequency bands, reducing the load of the latter cells. **Figure 4-7** shows an example where 2.6 GHz and 700 MHz cells are deployed. In the example, the gNodeB expands the coverage area of the 2.6 GHz NR cell to delay handovers of UEs in the expansion area to the 700 MHz NR cell. For details about threshold adaptation for coverage-based inter-frequency handovers, see **4.3.1.1 Threshold Adaptation for Coverage-based Inter-Frequency Handovers**.

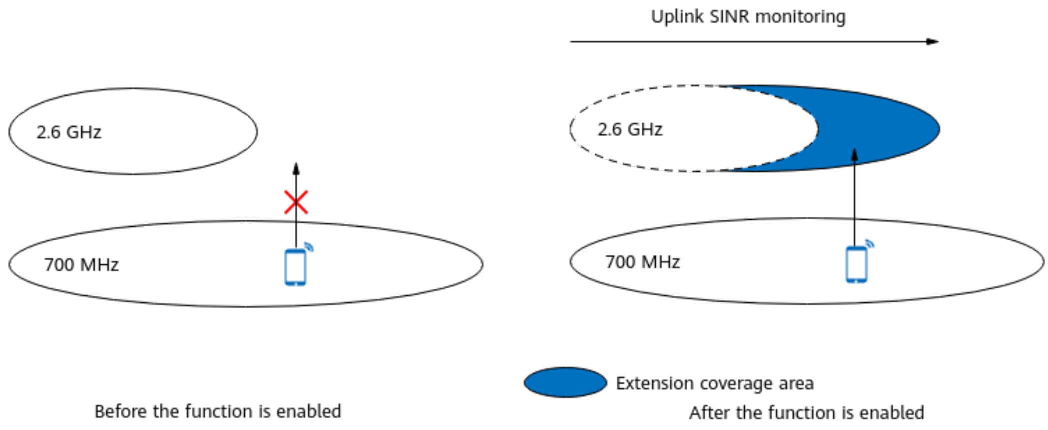
  Figure 4-7 Threshold adaptation for coverage-based inter-frequency handovers



- Threshold adaptation for inter-frequency handovers triggered by experience-based smart carrier selection

  When this sub-function takes effect, the gNodeB expands the coverage area of cells on non-sub-1 GHz frequency bands and increases the probability of handovers from cells on sub-1 GHz frequency bands to those on non-sub-1 GHz frequency bands, reducing the load of cells on sub-1 GHz frequency bands. **Figure 4-8** shows an example where 2.6 GHz and 700 MHz cells are deployed. In the example, the gNodeB expands the coverage area of the 2.6 GHz NR cell to increase the probability of UE handovers from the 700 MHz NR cell to the 2.6 GHz NR cell. For details about threshold adaptation for inter-frequency handovers triggered by experience-based smart carrier selection, see **4.3.1.2 Threshold Adaptation for Inter-Frequency Handovers Triggered by Experience-based Smart Carrier Selection**.

Figure 4-8 Threshold adaptation for inter-frequency handovers triggered by experience-based smart carrier selection



NOTE

In this document, frequency bands within FR1 are classified into sub-1 GHz frequency bands and non-sub-1 GHz frequency bands. The former are those with a downlink center frequency less than 1 GHz, and the latter are those other than the sub-1 GHz frequency bands.

4.3.1.1 Threshold Adaptation for Coverage-based Inter-Frequency Handovers

After threshold adaptation for coverage-based inter-frequency handovers is enabled, the gNodeB selects target UEs for which coverage can be expanded and then determines whether to initiate inter-frequency handovers for the UEs based on the following information: whether the UEs have reported event A2 related to coverage-based inter-frequency handovers or whether the UEs are identified as UEs under weak uplink coverage.

1. The gNodeB selects target UEs for which coverage can be expanded. For details, see **gNodeB Selecting Target UEs of Coverage Expansion**.
2. The gNodeB determines whether inter-frequency handovers can be initiated using either of the following methods:
   - If a UE has reported event A2 related to coverage-based inter-frequency handovers, the gNodeB further determines whether an inter-frequency handover can be initiated for the UE. For details, see **Coverage-based Inter-Frequency Handover**.
   - If a UE is identified as a UE under weak uplink coverage, the gNodeB further determines whether an inter-frequency handover can be initiated for the UE. For details, see **Uplink-Coverage-based Inter-Frequency Handover**. For details about how the base station determines UEs under weak uplink coverage, see **4.2.1.1 Decision on Whether to Initiate an Uplink-Coverage-based Inter-Frequency Handover**.

gNodeB Selecting Target UEs of Coverage Expansion

The following requirements must be met for a target UE of coverage expansion:

- The UE is an SA UE.

- The UE is not a VoNR UE. For details about VoNR, see *VoNR*.
- The UE is not running an emergency service.
- The UE has at least one bearer with a QCI for which the **gNBQciBearer**.***NetworkingOptionOptFlag*** parameter is set to **PERMIT** and does not have a bearer with a QCI for which this parameter is set to **FORBID**.
- The **gNBRfspConfig**.***NetworkingOptionOptFlag*** parameter corresponding to the RAT/Frequency Selection Priority (RFSP) index of the UE is set to **TRUE**.
- The currently effective threshold for event A2 related to coverage-based inter-frequency handovers is greater than the specified **RSRP threshold used in coverage threshold adaptation** for the UE.

  For non-super-uplink UEs: Currently effective threshold for event A2 related to coverage-based inter-frequency handovers = **Threshold for event A2 related to coverage-based inter-frequency handovers** + **Offset to the threshold for event A2 related to inter-frequency handovers**

  For super uplink UEs: Currently effective threshold for event A2 related to coverage-based inter-frequency handovers = **Threshold for event A2 related to coverage-based inter-frequency handovers** + **Offset to the threshold for event A2 related to inter-frequency handovers** + **Offset to the RSRP threshold for SUL coverage-based handovers**
- The UE is being served by a cell that has at least one neighboring frequency for which the **event for triggering coverage-based inter-frequency handovers is event A5**.

Table 4-20 lists the parameters involved in the preceding requirements for target UEs of coverage expansion.

Table 4-20 Parameters related to target UE selection for coverage expansion

Item	Parameter
RSRP threshold used in coverage threshold adaptation	**gNBMobilityCommParam**.***AdaptCovRsrpThld***
Threshold for event A2 related to coverage-based inter-frequency handovers	**NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrpThld*** (If this threshold needs to be frequency-specific, operators can set the **NRCellFreqRelation**.***InterFreqHoA1A2A51FreqOfs*** parameter to different values for different frequencies. The frequency-specific threshold will be equal to **NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrpThld*** plus **NRCellFreqRelation**.***InterFreqHoA1A2A51FreqOfs***.)
Offset to the threshold for event A2 related to inter-frequency handovers	**gNBOperatorQciParam**.***InterFreqA2RsrpThldOfs***
Offset to the RSRP threshold for SUL coverage-based handovers	**NRCellMeasConfig**.***SulCovHoRsrpThldOffset***

Item	Parameter
Event A5 for triggering coverage-based inter-frequency handovers	**NRCellFreqRelation**.***InterFreqHoEventType*** set to **EVENT_A5**

NOTE

- For details about RFSP, see *Flexible User Steering*.
- For details about the coverage-based inter-frequency handover function, see *SA Mobility Management in Connected Mode*.

Coverage-based Inter-Frequency Handover

When threshold adaptation for coverage-based inter-frequency handovers is enabled, the differences in coverage-based inter-frequency handovers between target UEs of coverage expansion and common UEs are as follows:

- The gNodeB updates the following thresholds related to coverage-based inter-frequency handovers for target UEs of coverage expansion. For these UEs:
  - Threshold for event A2 related to coverage-based inter-frequency handovers = RSRP threshold used in coverage threshold adaptation (**gNBMobilityCommParam**.***AdaptCovRsrpThld***)
  - Threshold for event A1 related to coverage-based inter-frequency handovers = Value of the **gNBMobilityCommParam**.***AdaptCovRsrpThld*** parameter + 4 dB
  - Threshold 2 for event A5 related to coverage-based inter-frequency handovers = Value of the **gNBMobilityCommParam**.***AdaptCovRsrpThld*** parameter + 4 dB
- After receiving A5 measurement reports from target UEs of coverage expansion, the gNodeB performs the following operations:
  - If the reported inter-frequency cell with the largest RSRP is served by the same gNodeB (the intra-base-station scenario), a handover to this cell is allowed.
  - If the reported inter-frequency cell with the largest RSRP is served by another gNodeB (the inter-base-station scenario), a further check on the setting of the **COV_THLD_ADAPT_SW** option of the **NRExternalNCell**.***MobilityFunInd*** parameter for this inter-frequency cell is required to determine whether a handover will be initiated:
    - If this option is deselected, the gNodeB checks whether the RSRP of this inter-frequency cell is greater than threshold 2 for event A5 related to coverage-based inter-frequency handovers. This threshold equals the sum of the values of the **NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld2*** and **gNBOperatorQciParam**.***InterFreqA5RsrpThld2Ofs*** parameters.
      - If the RSRP of this inter-frequency cell is greater than this threshold, handovers to this cell are allowed.

- If the RSRP of this inter-frequency cell is less than or equal to the threshold, handovers to this cell are prohibited.

- If this option is selected, handovers to this inter-frequency cell are allowed.

Uplink-Coverage-based Inter-Frequency Handover

The basic process of uplink-coverage-based inter-frequency handover triggered for user-experience-based coverage expansion is similar to that of uplink-coverage-based inter-frequency handover (see **4.2.1.2 Uplink-Coverage-based Inter-Frequency Handover Execution**) triggered for protection of UEs under weak uplink coverage. The differences are as follows:

Threshold 1 for event A5 is specified by different parameters depending on whether the UE involved is a target UE of coverage expansion as follows:

- If the UE is a target UE of coverage expansion, the threshold takes the value of the **NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld1*** parameter.
- If the UE is not a target UE of coverage expansion, the threshold takes the value of the **NRCellInterFHoMeaGrp**.***UlCovInterFreqA5RsrpThld1*** parameter.

4.3.1.2 Threshold Adaptation for Inter-Frequency Handovers Triggered by Experience-based Smart Carrier Selection

After threshold adaptation for inter-frequency handovers triggered by experience-based smart carrier selection is enabled, the gNodeB imposes certain requirements when selecting target cells for inter-frequency handovers of non-VoNR SA UEs. These requirements are different from those involved in inter-frequency handovers triggered by the function described in **4.1 Experience-based Smart Carrier Selection** as follows:

- Additional uplink SINR requirements for the target cell: The predicted uplink SINR of a UE in the target cell must be greater than or equal to the **lower uplink SINR threshold used for evaluation of UE transfer from NR to E-UTRAN** minus 2 dB.
- Additional RSRP requirement for the target cell in intra-base-station scenarios: RSRP threshold for the target cell must be equal to the **RSRP threshold used in coverage threshold adaptation** plus the **RSRP offset for experience-based PCC frequency selection** plus 4 dB.
- Additional RSRP requirement for the target cell in inter-base-station scenarios:
  - If the **coverage threshold adaptation switch** is on for the target cell, the RSRP threshold for the target cell must be equal to the **RSRP threshold used in coverage threshold adaptation** plus the **RSRP offset for experience-based PCC frequency selection** plus 4 dB.
  - If the **coverage threshold adaptation switch** is off for the target cell:
    - If **RSRP threshold 2 for event A5 related to multi-frequency smart selection** is set to **255**, the RSRP threshold for the target cell must be equal to the **RSRP threshold for interference-based inter-frequency handovers** plus **RSRP offset for experience-based PCC frequency selection**.

- If **RSRP threshold 2 for event A5 related to multi-frequency smart selection** is set to a value other than **255**, the RSRP threshold for the target cell must be equal to **RSRP threshold 2 for event A5 related to multi-frequency smart selection** plus **RSRP offset for experience-based PCC frequency selection**.

Table 4-21 lists the parameters involved in "threshold adaptation for inter-frequency handovers triggered by experience-based smart carrier selection".

Table 4-21 Parameters related to "threshold adaptation for inter-frequency handovers triggered by experience-based smart carrier selection"

Item	Parameter
Lower uplink SINR threshold used for evaluation of UE transfer from NR to E-UTRAN	**NRDUCellSrsMeas**.*NrToEutranSinrLowThld*
RSRP threshold used in coverage threshold adaptation	**gNBMobilityCommParam**.*AdaptCovRsrpThld*
RSRP offset for experience-based PCC frequency selection	**NRCellFreqRelation**.*ExpBasedPCCFreqSelRsrpOfs*
Coverage threshold adaptation switch	**COV_THLD_ADAPT_SW** option of the **NRExternalNCell**.*MobilityFunInd* parameter
RSRP threshold 2 for event A5 related to multi-frequency smart selection	**NRCellInterFHoMeaGrp**.*MultiFreqSmtSelA5RsrpThld2*
RSRP threshold for interference-based inter-frequency handovers	**NRCellInterFHoMeaGrp**.*IntrfInterFreqHoRsrpThld*

4.3.2 Network Analysis

4.3.2.1 Benefits

User-experience-based coverage expansion reduces the load of sub-1 GHz frequency bands and increases the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) on the entire network.

The benefits are related to the following factors:

- The higher the proportion of transmission duration on sub-1 GHz frequency bands, the greater the benefits.

  Proportion of transmission duration on sub-1 GHz frequency bands = Transmission duration on sub-1 GHz frequency bands/Transmission duration on all frequency bands

  The transmission duration is measured using the **N.ThpTime.DL.RmvLastSlot** counter.
- The heavier the load on sub-1 GHz frequency bands relative to that on non-sub-1 GHz frequency bands, the greater the benefits.
- If both sub-1 GHz and non-sub-1 GHz frequency bands exist, the benefits are greater.
- If sub-1 GHz and non-sub-1 GHz frequency bands overlap, the benefits are greater.

4.3.2.2 Impacts

The impacts between this function and other functions are described in **4.3.3.2.3 Function Impacts**.

User-experience-based coverage expansion has the following impacts on the network:

- The traffic volume and UE distribution on each frequency band change.
- The inter-frequency handover success rate (indicated by **Inter-Frequency Handover Out Success Rate (CU)**) decreases and the service drop rate (indicated by **Service Call Drop Rate (CU, Inactive)**) increases due to the extended cell edge.

4.3.3 Requirements

4.3.3.1 Licenses

None

4.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User-experience-based coverage expansion	**COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	• Threshold adaptation for coverage-based inter-frequency handovers takes effect only when all of the following conditions are met: – User-experience-based coverage expansion is enabled. – The **NRCellMobilityConfig**.***MeasTrigQuantity*** parameter (specifying the triggering quantity for event A1/A2 related to coverage-based inter-frequency measurements) is set to **RSRP**. – The **NRCellFreqRelation**.***InterFreqHoEventType*** parameter (specifying the event for triggering coverage-based inter-frequency handovers) is set to **EVENT_A5**.

RAT	Function Name	Function Switch	Reference	Description
				– Coverage-based inter-frequency handover (controlled by the **COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter) or inter-frequency redirection (controlled by the **REDIRECTION** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter) is enabled. ● "Threshold adaptation for inter-frequency handovers triggered by experience-based smart carrier selection" takes effect only when both user-experience-based coverage expansion and experience-based smart carrier selection (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***) are enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Triggering quantity for events A1 and A2 related to coverage-based inter-frequency measurements	**NRCellMobilityConfig**.***MeasTrigQuantity***	*SA Mobility Management in Connected Mode*	Threshold adaptation for coverage-based inter-frequency handovers takes effect only when all of the following conditions are met: ● User-experience-based coverage expansion (controlled by the **COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter) is enabled. ● The **NRCellMobilityConfig**.***MeasTrigQuantity*** parameter (specifying the triggering quantity for event A1/A2 related to coverage-based inter-frequency measurements) is set to **RSRP**. ● The **NRCellFreqRelation**.***InterFreqHoEventType*** parameter (specifying the event for triggering coverage-based inter-frequency handovers) is set to **EVENT_A5**. ● Coverage-based inter-frequency handover
Low-frequency TDD	Event for triggering inter-frequency handovers	**NRCellFreqRelation**.***InterFreqHoEventType***	*SA Mobility Management in Connected Mode*	
Low-frequency TDD	Coverage-based inter-frequency handover	**COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	
Low-frequency TDD	Inter-frequency redirection	**REDIRECTION** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	

RAT	Function Name	Function Switch	Reference	Description
				(controlled by the **COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.*InterFreqHoSwitch* parameter) or inter-frequency redirection (controlled by the **REDIRECTION** option of the **NRCellAlgoSwitch**.*InterFreqHoSwitch* parameter) is enabled.
Low-frequency TDD	Experience-based smart carrier selection	**MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.*MultiFreqAlgoSwitch* parameter	*Multi-Frequency Smart Selection*	Threshold adaptation for inter-frequency handovers triggered by experience-based smart carrier selection takes effect only when both user-experience-based coverage expansion (controlled by the **COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.*MobilityAlgoSwitch* parameter) and experience-based smart carrier selection are enabled.

4.3.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Protection of UEs under weak uplink coverage	**UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	When user-experience-based coverage expansion is enabled (with the **COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter selected), protection of UEs under weak uplink coverage cannot be enabled.
Low-frequency TDD	Triggering quantity for events A1 and A2 related to coverage-based inter-frequency measurements	**NRCellMobilityConfig**.***MeasTrigQuantity***	*SA Mobility Management in Connected Mode*	If the **NRCellMobilityConfig**.***MeasTrigQuantity*** parameter is set to **RSRP_OR_RSRQ** or **RSRP_OR_SINR**, user-experience-based coverage expansion cannot be enabled.

4.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	**MOBILITY_TO_EUTRAN_SW** option of the **NRCellAlgoSwitch**.***InterRatServiceMobilitySw*** parameter	*Interoperability Between E-UTRAN and NG-RAN in Connected Mode*	If user-experience-based coverage expansion (controlled by the **COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter) is enabled together with downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, the RSRP threshold used in coverage threshold adaptation (specified by the **gNBMobilityCommParam**.***AdaptCovRsrpThld*** parameter) must be greater than or equal to the inter-RAT event A2 threshold. This prevents an increase in the number of outgoing handover executions from NG-RAN to E-UTRAN (indicated by **N.HO.InterRAT.N2E.ExecAttOut**). The inter-RAT event A2 threshold is the sum of the values of the **NRCellInterRHoMeaGrp**.***InterRatHoA2RsrpThld***, **gNBOperatorQciParam**.***EutranA2B2RsrpThldOffset***, **NRCellMeasConfig**.***SulCovHoRsrpThldOffset***, **gNBRfspConfig**.***InterRatA1A2B2RsrpThl***

RAT	Function Name	Function Switch	Reference	Description
				dOfs, **NRCellOpPolicy**.***EutranA1A2B2RsrpThldOfs***, and **NRCellInterRHoMeaGrp**.***InterRatHoA1A2Hyst*** parameters.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	• **UL_SINR_MOBILITY_TO_EUTRAN_SW** option of the **NRCellAlgoSwitch**.***InterRatServiceMobilitySw*** parameter • **MOBILITY_TO_EUTRAN_SW** option of the **NRCellAlgoSwitch**.***InterRatServiceMobilitySw*** parameter	*Interoperability Between E-UTRAN and NG-RAN in Connected Mode*	Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN uses the same weak coverage evaluation mechanism as user-experience-based coverage expansion does, and the weak-uplink-coverage-based handover mechanism is triggered on a first-come, first-served basis. • If uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN has been enabled and then user-experience-based coverage expansion is enabled (with the **COV_THLD_ADAPT_SW** of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter selected), the number of outgoing handover executions from NG-RAN to E-UTRAN (indicated by **N.HO.InterRAT.N2E.ExecAttOut**) decreases. • If uplink coverage-based inter-RAT mobility from NG-RAN to

RAT	Function Name	Function Switch	Reference	Description
				E-UTRAN is enabled after user-experience-based coverage expansion is enabled (with the **COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter selected), the number of weak-coverage-based inter-frequency handovers (indicated by **N.HO.InterFreq.ULCoverage.ExecAttOut**) decreases.
Low-frequency TDD	Flexible user steering	**gNBRfspConfig**.***NetworkingOptionOptFlag***	None	If the **gNBRfspConfig**.***NetworkingOptionOpt-Flag*** parameter corresponding to the RFSP index of a UE is set to **FALSE**, user-experience-based coverage expansion (controlled by the **COV_THLD_ADAPT_SW** option of the **gNBMobilityComm-Param**.***MobilityAlgoSwitch*** parameter) does not take effect for the UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	VoNR	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	If the **VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter is selected, user-experience-based coverage expansion (controlled by the **COV_THLD_ADAPT_SW** option of the **gNBMobilityComm-Param**.***MobilityAlgoSwitch*** parameter) does not take effect for UEs with voice-dedicated bearers set up.
Low-frequency TDD	Experience-based smart carrier selection	**MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	Cell edge extension alone cannot ensure user experience. Therefore, experience-based smart carrier selection must be enabled before user-experience-based coverage expansion (controlled by the **COV_THLD_ADAPT_SW** option of the **gNBMobilityComm-Param**.***MobilityAlgoSwitch*** parameter) is enabled. This way, experience-based smart carrier selection is more likely to take effect, further improving user experience.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Measurement-based inter-frequency handover	**COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	User-experience-based coverage expansion modifies the thresholds for events A1 and A2 and threshold 2 for event A5 related to coverage-based inter-frequency handover only when measurement-based inter-frequency handover (controlled by the **COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter) or measurement-based inter-frequency redirection (controlled by the **REDIRECTION** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter) is enabled in coverage scenarios.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Measurement-based inter-frequency redirection	**REDIRECTION** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	User-experience-based coverage expansion modifies the thresholds for events A1 and A2 and threshold 2 for event A5 related to coverage-based inter-frequency handover only when measurement-based inter-frequency handover (controlled by the **COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter) or measurement-based inter-frequency redirection (controlled by the **REDIRECTION** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter) is enabled in coverage scenarios.
Low-frequency TDD	NSA/SA selection based on uplink coverage	**LTE_FDD_NSA_SA_UL_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter	*NSA-SA Selection Based on User Experience*	When NSA/SA selection based on uplink coverage is enabled together with user-experience-based coverage expansion, the probability that NSA/SA selection based on uplink coverage takes effect decreases.

4.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.3.4 Networking

- The network is an SA network and uses at least two frequencies.
- There are coverage differences among carriers.

4.3.3.5 Others

None

4.3.4 Operation and Maintenance

4.3.4.1 Data Configuration

4.3.4.1.1 Data Preparation

Before activating this function, configure NR cell frequency relationships and neighbor relationships. For details about related parameters, see **Table 4-14** and the table with the parameters used to configure neighboring cells in *SA Mobility Management in Connected Mode*, respectively.

Table 4-22 describes the parameters used for function activation.

Table 4-22 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Mobility Algorithm Switch	**gNBMobilityCommParam.*MobilityAlgoSwitch***	COV_THLD_ADAPT_SW	You are advised to select this option.

Parameter Name	Parameter ID	Option	Setting Notes
Adaptive Coverage RSRP Threshold	**gNBMobilityCommParam**.***AdaptCovRsrpThld***	N/A	Set this parameter to its recommended value. It is recommended that this parameter be set to a value greater than or equal to the threshold for event A2 related to inter-RAT handovers from NG-RAN to E-UTRAN. Otherwise, the number of outgoing handover executions from NG-RAN to E-UTRAN will increase.
Mobility Function Indication	**NRExternalNCell**.***MobilityFunInd***	COV_THLD_ADAPT_SW	If user-experience-based coverage expansion (controlled by the **COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter) is not enabled on the gNodeB serving an external NR cell, it is recommended that the **COV_THLD_ADAPT_SW** option of the **NRExternalNCell**.***MobilityFunInd*** parameter be deselected. This prevents coverage-based inter-frequency outgoing handovers immediately after UEs are handed over to this cell.

Parameter Name	Parameter ID	Option	Setting Notes
NR Move to E-UTRAN UL SINR Low Threshold	**NRDUCellSrsMeas**.***NrToEutranSinrLowThld***	N/A	Set this parameter to its recommended value.
Service Function Switch	**NRCellAlgoSwitch**.***ServiceFunctionSwitch***	VOLUME_IDENTIFY_SW	You are advised to select this option.
Common UL Large Pkt Ident Thld	**NRDUCellServExp**.***CommonUlLargePktIdentThld***	N/A	Set this parameter to its recommended value.

4.3.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Selecting the COV_THLD_ADAPT_SW option
MOD GNBMOBILITYCOMMPARAM: MobilityAlgoSwitch=COV_THLD_ADAPT_SW-1;
//Configuring the RSRP threshold used in coverage threshold adaptation (It is recommended that this
parameter be set to a value greater than or equal to the threshold for event A2 related to inter-RAT
handovers from NG-RAN to E-UTRAN.)
MOD GNBMOBILITYCOMMPARAM: AdaptCovRsrpThld=-118;
//Deselecting the COV_THLD_ADAPT_SW option for external NR cells (This option can be deselected if user-
experience-based coverage expansion is enabled on the local base station but not enabled on the
neighboring base stations.)
MOD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0,
MobilityFunInd=COV_THLD_ADAPT_SW-0;
//Changing the uplink SINR threshold for NR inter-frequency handovers
MOD NRDUCELLSRSMEAS: NrDuCellId=0, NrToEutranSinrLowThld=-30;
//Selecting the VOLUME_IDENTIFY_SW option if uplink traffic volumes need to be determined for the
threshold adaptation for coverage-based inter-frequency handovers function
MOD NRCELLALGOSWITCH: NrCellId=0, ServiceFunctionSwitch=VOLUME_IDENTIFY_SW-1;
//Changing the threshold for identifying common large-packet UEs in the uplink as required (It is
recommended that the default value be used. The value 80 is used as an example.)
MOD NRDUCELLSERVEXP: NrDuCellId=0, CommonUlLargePktIdentThld=80;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Deselecting the COV_THLD_ADAPT_SW option
MOD GNBMOBILITYCOMMPARAM: MobilityAlgoSwitch=COV_THLD_ADAPT_SW-0;
```

4.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.3.4.2 Activation Verification

If both of the following conditions are met, user-experience-based coverage expansion has taken effect:

- The values of all counters listed in **Table 4-23** increase.

Table 4-23 Counters related to the number of experience-based smart carrier selection executions and the number of uplink-coverage-based inter-frequency handovers

Counter ID	Counter Name
1911833645	**N.MultiFreqSmartSel.Exec**
1911833798	**N.HO.InterFreq.ULCoverage.PrepAttOut**
1911833799	**N.HO.InterFreq.ULCoverage.ExecAttOut**
1911833800	**N.HO.InterFreq.ULCoverage.ExecSuccOut**

- The values of all counters listed in **Table 4-24** decrease.

Table 4-24 Counters related to the number of coverage-based inter-frequency handovers

Counter ID	Counter Name
1911819867	**N.HO.InterFreq.Coverage.PrepAttOut**
1911819873	**N.HO.InterFreq.Coverage.ExecAttOut**
1911819878	**N.HO.InterFreq.Coverage.ExecSuccOut**

4.3.4.3 Network Monitoring

Observe the average downlink UE throughput to check whether the UE data rate increases after this function is enabled.

Average downlink UE throughput = **User Downlink Average Throughput (DU)**

4.4 User-Experience-based NSA SCG Coverage Expansion

4.4.1 Principles

On an NSA network where there are coverage differences among NR carriers, as the coverage capabilities and bandwidths may vary among frequency bands, cells on the frequency bands that have good uplink and downlink coverage but small bandwidths are more likely to be added to secondary cell groups (SCGs), and therefore a larger number of NSA UEs access these cells. This leads to congestion on the frequency bands that have good uplink and downlink coverage but small bandwidths. As a result, the experience of most NSA UEs is affected. The causes of congestion on the NR frequency bands that have good uplink and downlink coverage but small bandwidths are as follows:

- On the NR side, when NSA UEs served by SCG cells on the frequency bands that have large bandwidths but a large difference between uplink and downlink coverage move out of the coverage area of these cells, these cells are removed from the SCGs and the cells on the frequency bands that have good uplink and downlink coverage but small bandwidths are added to the SCGs by means of coverage-based inter-frequency handovers.
- On the NR side, cell-edge NSA UEs served by cells on the frequency bands that have good uplink and downlink coverage but small bandwidths cannot be transferred to those on the frequency bands that have large bandwidths but a large difference between uplink and downlink coverage by means of SCG change. As a result, these UEs remain camping on cells on the frequency bands that have good uplink and downlink coverage but small bandwidths.

NOTE

The SCG herein refers to the PSCell of an NSA DC UE. For details about the PSCell, see *EPC-based NSA Basics*.

To reduce the load of cells on the frequency bands that have good uplink and downlink coverage but small bandwidths and to improve the traffic absorption capability of cells on the frequency bands that have large bandwidths but a large difference between uplink and downlink coverage, the user-experience-based NSA SCG coverage expansion function has been introduced. This function expands the overlapping coverage area between frequency bands. With the assistance of user experience estimation and uplink SINR monitoring, this function optimizes the load distribution across frequency bands on the basis of the NSA carrier combination selection based on user experience function. As a result, the load of cells on the frequency bands that have good uplink and downlink coverage but small bandwidths decreases, and the average downlink throughput of NSA UEs increases on the NR side. For details about the NSA carrier combination selection based on user experience function, see *EPC-based NSA Experience Improvement*. In this document, the sub-1 GHz and non-sub-1 GHz frequency bands are used as examples. The principles for other frequency bands are similar.

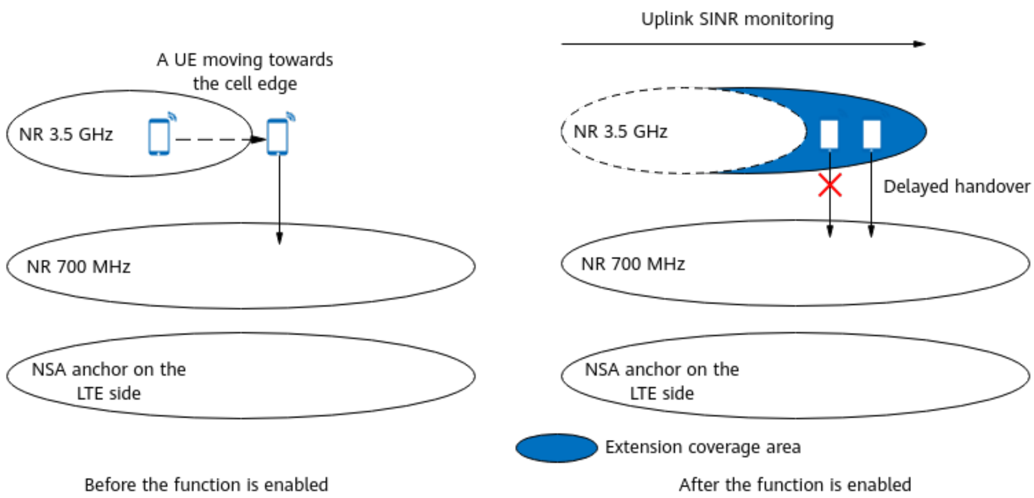
User-experience-based NSA SCG coverage expansion is jointly controlled by the **SCG_ADAPT_COV_THLD_SW** option of the

NsaDcAlgoParam.***NsaDcAlgoExtSwitch*** parameter on the LTE side and the **NSA_COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter on the NR side. Currently, this function is supported only in NSA networking. This function includes the following sub-functions:

- Threshold adaptation for coverage-based inter-frequency handovers

  When this sub-function takes effect, the gNodeB expands the coverage area of cells on non-sub-1 GHz frequency bands to delay handovers of NSA UEs from these cells to those on sub-1 GHz frequency bands, reducing the load of the latter cells. **Figure 4-9** shows an example of an NSA network where 3.5 GHz and 700 MHz NR cells are deployed. In the example, the gNodeB expands the coverage area of the 3.5 GHz NR cell to delay the handovers of UEs from this cell to the 700 MHz NR cell for SCG change. For details about threshold adaptation for coverage-based inter-frequency handovers, see **4.4.1.1 Threshold Adaptation for Coverage-based Inter-Frequency Handovers**.

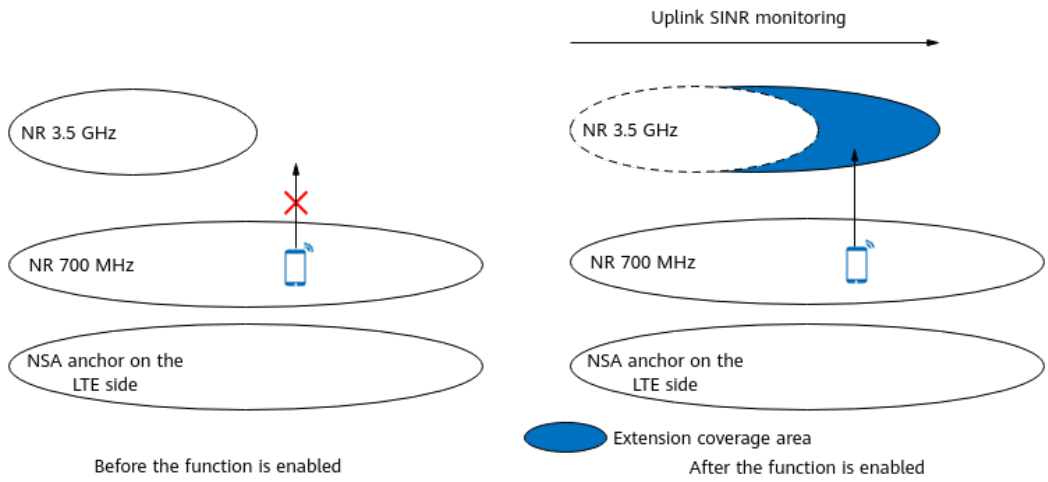
  Figure 4-9 Threshold adaptation for coverage-based inter-frequency handovers



- Handover threshold adaptation for NR SCG change triggered by NSA carrier combination selection based on user experience

  When this sub-function takes effect, the eNodeB instructs the gNodeB to expand the coverage area of cells on non-sub-1 GHz frequency bands. This increases the probability of adding these cells to SCGs and the probability of handovers from cells on sub-1 GHz frequency bands to those on non-sub-1 GHz frequency bands, and therefore reduces the load of cells on sub-1 GHz frequency bands. **Figure 4-10** shows an example of an NSA network where 3.5 GHz and 700 MHz NR cells are deployed. The gNodeB expands the coverage area of the 3.5 GHz NR cell to increase the probability of adding the 3.5 GHz NR cell to the SCG. For details about handover threshold adaptation for NR SCG change triggered by NSA carrier combination selection based on user experience, see **4.4.1.2 Handover Threshold Adaptation for NR SCG Change Triggered by NSA Carrier Combination Selection Based on User Experience**.

Figure 4-10 Handover threshold adaptation for NR SCG change triggered by NSA carrier combination selection based on user experience



4.4.1.1 Threshold Adaptation for Coverage-based Inter-Frequency Handovers

After "threshold adaptation for coverage-based inter-frequency handovers" is enabled, the gNodeB selects target UEs for which coverage can be expanded and then determines whether to initiate inter-frequency handovers for the UEs based on the following information: whether the UEs have reported event A2 related to coverage-based inter-frequency handovers or whether the UEs are identified as UEs under weak uplink coverage.

1. The gNodeB selects target UEs for which coverage can be expanded. For details, see **gNodeB Selecting Target UEs of Coverage Expansion**.
2. The gNodeB determines whether inter-frequency handovers can be initiated using either of the following methods:
   – If a UE has reported event A2 related to coverage-based inter-frequency handovers, the gNodeB further determines whether an inter-frequency handover can be initiated for the UE. For details, see **Coverage-based Inter-Frequency Handover**.
   – If a UE is identified as a UE under weak uplink coverage, the gNodeB further determines whether an inter-frequency handover can be initiated for the UE. For details, see **Uplink-Coverage-based Inter-Frequency Handover**. For details about how the base station determines UEs under weak uplink coverage, see **4.2.1.1 Decision on Whether to Initiate an Uplink-Coverage-based Inter-Frequency Handover**.

gNodeB Selecting Target UEs of Coverage Expansion

The following requirements must be met for a target UE of coverage expansion:

- The UE is an NSA UE for which an SCG has been added.
- The UE is not a VoNR UE. For details about VoNR, see *VoNR*.
- The UE is not running an emergency service.
- The UE has at least one bearer with a QCI for which the **gNBQciBearer**.***NetworkingOptionOptFlag*** parameter is set to **PERMIT** and does not have a bearer with a QCI for which this parameter is set to **FORBID**.

- The **gNBRfspConfig**.***NetworkingOptionOptFlag*** parameter corresponding to the RFSP index of the UE is set to **TRUE**.
- The currently effective threshold for event A2 related to coverage-based inter-frequency handovers is greater than the specified **RSRP threshold used in coverage threshold adaptation** for the UE.

  For non-super uplink UEs: Currently effective threshold for event A2 related to coverage-based inter-frequency handovers = **Threshold for event A2 related to coverage-based inter-frequency handovers** + **Offset to the threshold for event A2 related to inter-frequency handovers**

  For super uplink UEs: Currently effective threshold for event A2 related to coverage-based inter-frequency handovers = **Threshold for event A2 related to coverage-based inter-frequency handovers** + **Offset to the threshold for event A2 related to inter-frequency handovers** + **Offset to the RSRP threshold for SUL coverage-based handovers**
- The UE is being served by a cell that has at least one neighboring frequency for which the **event for triggering coverage-based inter-frequency handovers is event A5**.

Table 4-25 lists the parameters involved in the preceding requirements for target UEs of coverage expansion.

Table 4-25 Parameters related to target UE selection for coverage expansion

Item	Parameter
RSRP threshold used in coverage threshold adaptation	**gNBMobilityCommParam**.***AdaptCovRsrpThld***
Threshold for event A2 related to coverage-based inter-frequency handovers	**NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrpThld***
Offset to the threshold for event A2 related to inter-frequency handovers	**gNBOperatorQciParam**.***InterFreqA2RsrpThldOfs***
Offset to the RSRP threshold for SUL coverage-based handovers	**NRCellMeasConfig**.***SulCovHoRsrpThldOffset***
Event A5 for triggering coverage-based inter-frequency handovers	**NRCellFreqRelation**.***InterFreqHoEventType*** set to **EVENT_A5**

NOTE

- For details about RFSP, see *Flexible User Steering*.
- For details about coverage-based inter-frequency handover for NSA UEs, see *NSA Mobility Management*.

Coverage-based Inter-Frequency Handover

When threshold adaptation for coverage-based inter-frequency handovers is enabled, the differences in coverage-based inter-frequency handovers between target UEs of coverage expansion and common UEs are as follows:

- The gNodeB updates the following thresholds related to coverage-based inter-frequency handovers for target UEs of coverage expansion. For these UEs:
  - Threshold for event A2 related to coverage-based inter-frequency handovers = RSRP threshold used in coverage threshold adaptation (**gNBMobilityCommParam**.***AdaptCovRsrpThld***)
  - Threshold for event A1 related to coverage-based inter-frequency handovers = Value of the **gNBMobilityCommParam**.***AdaptCovRsrpThld*** parameter + 4 dB
  - Threshold 2 for event A5 related to coverage-based inter-frequency handovers = Value of the **gNBMobilityCommParam**.***AdaptCovRsrpThld*** parameter + 4 dB
- If non-gap-assisted coverage-based inter-frequency PSCell change (controlled by the **COV_BASED_HO_NO_GAP_SW** option of the **NRCellFeatureSwitch**.***InterFreqHoExpSwitch*** parameter) is enabled and the gNodeB receives an event A2 measurement report from a target UE of coverage expansion, the gNodeB will use the SCell coverage information carried in the measurement report to select the frequencies to be delivered for event A5 measurements.
  - If the SCell coverage meets the coverage condition, the base station acts as follows:
    - If the SCells are served by the local base station, the base station delivers only the frequencies of the SCells in event A5 measurement configurations.
    - If the SCells are served by another base station, the base station determines the frequencies to be delivered for event A5 measurements based on the following information: (1) the serving cell's setting of the **COV_THLD_ADAPT_SW** option of the **NRExternalNCell**.***MobilityFunInd*** parameter for the SCells and (2) whether the SCell coverage is greater than the currently effective threshold 2 (sum of **NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld2*** and **gNBOperatorQciParam**.***InterFreqA5RsrpThld2Ofs***) for event A5 related to coverage-based inter-frequency handovers.
      - If the option is selected, the base station delivers only the frequencies of the SCells for event A5 measurements.
      - If this option is deselected and the SCell coverage is greater than the currently effective threshold 2 for event A5 related to coverage-based inter-frequency handovers, the base station

delivers only the frequencies of the SCells for event A5 measurements.

  - If this option is deselected and the SCell coverage is less than or equal to the currently effective threshold 2 for event A5 related to coverage-based inter-frequency handovers, the base station does not additionally determine the frequencies for measurements but delivers event A5 measurement configurations for a coverage-based inter-frequency handover as it does for a common UE.

- If the SCell coverage does not meet the coverage condition, the base station does not additionally determine the frequencies for measurements but delivers event A5 measurement configurations for a coverage-based inter-frequency handover as it does for a common UE.

In the preceding information:

- For details about SCells, see *EPC-based NSA Basics*.
- For details about non-gap-assisted coverage-based inter-frequency PSCell change, see *NSA Mobility Management*.
- Coverage condition: SCell RSRP in the event A2 report – **NRCellInterFHoMeaGrp**.***InterFreqA4A5Hyst*** > **gNBMobilityCommParam**.***AdaptCovRsrpThld*** + 4 dB + **NRCellServExp**.***InterHoNoGapRsrpOfs***

- After receiving A5 measurement reports from target UEs of coverage expansion, the gNodeB performs the following operations:
  - If the reported inter-frequency cell with the largest RSRP is served by the same gNodeB, a handover to this cell is allowed.
  - If the reported inter-frequency cell with the largest RSRP is served by another gNodeB, a further check on the setting of the **COV_THLD_ADAPT_SW** option of the **NRExternalNCell**.***MobilityFunInd*** parameter for this inter-frequency cell is required to determine whether a handover will be initiated:
    - If this option is deselected, the gNodeB checks whether the RSRP of this inter-frequency cell is greater than threshold 2 for event A5 related to coverage-based inter-frequency handovers. This threshold equals the sum of the values of the **NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld2*** and **gNBOperatorQciParam**.***InterFreqA5RsrpThld2Ofs*** parameters.
      - If the RSRP of this inter-frequency cell is greater than this threshold, handovers to this cell are allowed.
      - If the RSRP of this inter-frequency cell is less than or equal to the threshold, handovers to this cell are prohibited.
    - If this option is selected, handovers to this inter-frequency cell are allowed.

Uplink-Coverage-based Inter-Frequency Handover

The differences between uplink-coverage-based inter-frequency handover and the function described in **4.2.1.2 Uplink-Coverage-based Inter-Frequency Handover Execution** are as follows:

Threshold 1 for event A5 related to uplink-coverage-based inter-frequency handovers is specified by different parameters, depending on whether the UE involved is a target UE of coverage expansion.

- If the UE is a target UE of coverage expansion, the threshold takes the value of the **NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld1*** parameter.
- If the UE is not a target UE of coverage expansion, the threshold takes the value of the **NRCellInterFHoMeaGrp**.***UlCovInterFreqA5RsrpThld1*** parameter.

4.4.1.2 Handover Threshold Adaptation for NR SCG Change Triggered by NSA Carrier Combination Selection Based on User Experience

After user-experience-based NSA SCG coverage expansion is enabled, the eNodeB checks the relationship between the **RSRP threshold for SCG coverage adaptation** and the **event B1 threshold** used in the "NSA carrier combination selection based on user experience" function, to determine whether to modify the working process of the "NSA carrier combination selection based on user experience" function. For details about "NSA carrier combination selection based on user experience", see *EPC-based NSA Experience Improvement*.

- If the **event B1 threshold** used in the "NSA carrier combination selection based on user experience" function is less than or equal to the **RSRP threshold for SCG coverage adaptation**, the working process of the "NSA carrier combination selection based on user experience" function will not be modified.
- If the **event B1 threshold** used in the "NSA carrier combination selection based on user experience" function is greater than the **RSRP threshold for SCG coverage adaptation**, the gNodeB performs the following additional actions after receiving a request to extend NR SCG coverage from the eNodeB.
  - The gNodeB first determines the UE type and selects any UE that meets the following conditions:
    - The UE is a non-VoNR UE.
    - The **gNBRfspConfig**.***NetworkingOptionOptFlag*** parameter corresponding to the RFSP index of the UE is set to **TRUE**.
  - For any UE meeting the preceding conditions, the gNodeB further checks whether the downlink RSRP and uplink SINR requirements are met:
    - Downlink RSRP requirement:

      Downlink RSRP of the NR SCG ≥ Value of **gNBMobilityCommParam**.***AdaptCovRsrpThld*** configured on the gNodeB serving the SCG cell + 4 dB
    - Uplink SINR requirement:

      Uplink SINR of the NR SCG ≥ **Lower uplink SINR threshold used for evaluation of UE transfer from NR to E-UTRAN** set for the SCG cell – 2 dB

    When both of the preceding requirements are met, the gNodeB allows the coverage of the NR SCG to be expanded and notifies the eNodeB of the expansion. Then, the eNodeB can configure the NR SCG.

Table 4-26 lists the parameters involved in the preceding process.

Table 4-26 Parameters related to handover threshold adaptation for NR SCG change triggered by NSA carrier combination selection based on user experience

Item	Parameter
RSRP threshold for SCG coverage adaptation	**NrScgFreqConfig**.*ScgCovAdaptRsrpThld*
Event B1 Threshold	**NrScgFreqConfig**.*NsaDcB1ThldRsrp*
Lower uplink SINR threshold used for evaluation of UE transfer from NR to E-UTRAN	**NRDUCellSrsMeas**.*NrToEutranSinrLowThld*

4.4.2 Network Analysis

4.4.2.1 Benefits

User-experience-based NSA SCG coverage expansion reduces the load of sub-1 GHz NR frequency bands and increases the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) on the entire NR network.

The benefits are related to the following factors:

- The higher the proportion of transmission duration on sub-1 GHz NR frequency bands, the greater the benefits.

  Proportion of transmission duration on sub-1 GHz NR frequency bands = Transmission duration on sub-1 GHz NR frequency bands/Transmission duration on all NR frequency bands

  The transmission duration is measured using the **N.ThpTime.DL.RmvLastSlot** counter.
- The heavier the load on sub-1 GHz NR frequency bands relative to that on non-sub-1 GHz NR frequency bands, the greater the benefits.
- If both sub-1 GHz and non-sub-1 GHz frequency bands exist on the NR side, and anchors that support addition of cells on both sub-1 GHz and non-sub-1 GHz NR frequency bands to SCGs exist on the LTE side, the benefits are greater.
- If sub-1 GHz and non-sub-1 GHz NR frequency bands overlap, the benefits are greater.

If threshold adaptation for coverage-based inter-frequency handovers and non-gap-assisted coverage-based inter-frequency PSCell change take effect at the same time, the number of unnecessary gap-assisted measurements may be reduced and thereby the average downlink UE throughput may increase.

4.4.2.2 Impacts

The impacts between this function and other functions are described in **4.4.3.2.3 Function Impacts**.

User-experience-based NSA SCG coverage expansion has the following impacts on the network:

- Impacts on the NR side:
  - Due to cell edge extension for cells in SCGs, the coverage-based inter-frequency handover success rate (**N.NsaDc.InterFreq.Coverage.PSCell.Change.Succ** divided by **N.NsaDc.InterFreq.Coverage.PSCell.Change.Att**) decreases and the **SgNB-Triggered SgNB Abnormal Release Rate (CU)** increases on the NR side.
  - This function extends the edges of cells in SCGs based on downlink experience and therefore may cause uplink user experience on the NR side to deteriorate.
  - The NR cell in the SCG serves a larger number of CEUs after the cell's edge is extended. As a result, user experience in the NR cell deteriorates.
- Impacts on counters on the LTE and NR sides:
  - As this function extends the edges of cells in SCGs, there will be changes in the traffic volume and UE distribution on each frequency band on the LTE and NR sides.
    - LTE counters:
      - The number of UEs in the NSA DC state remains unchanged or increases. Specifically, the values of the **L.Traffic.User.NsaDc.PCell.Avg**, **L.Traffic.User.NsaDc.PCell.Max**, **L.Traffic.User.NsaDc.PCell.Avg.PLMN**, and **L.Traffic.User.NsaDc.PCell.Max.PLMN** counters remain unchanged or increase. If there is only a single non-sub-1 GHz frequency band on the NR side, the values of the **L.Traffic.User.NsaDc.PCell.Avg**, **L.Traffic.User.NsaDc.PCell.Max**, **L.Traffic.User.NsaDc.PCell.Avg.PLMN**, and **L.Traffic.User.NsaDc.PCell.Max.PLMN** counters increase significantly. The number of UEs in the NSA DC state changes, causing changes in the values of other counters related to NSA DC UEs.
      - The number of NSA-capable UEs for which no SCG is added remains unchanged or decreases. Specifically, the value of the **L.NsaDc.Capable.5gUser.NoScg.BorderUE.Avg** counter remains unchanged or decreases. If there is only a single non-sub-1 GHz frequency band on the NR side, the value of the **L.NsaDc.Capable.5gUser.NoScg.BorderUE.Avg** counter decreases significantly.
    - NR counters: The number of NSA DC UEs remains unchanged or increases. Specifically, the values of the **N.User.NsaDc.PSCell.Avg** and **N.User.NsaDc.PSCell.Max** counters remain unchanged or increase. In the case of a single frequency of a non-sub-1 GHz frequency band on the NR side, the values of the **N.User.NsaDc.PSCell.Avg** and **N.User.NsaDc.PSCell.Max** counters

increase significantly. As the coverage of the cells in the NR SCG is extended, UE distribution changes and the number of CEUs increases. As a result, the average timing advance (TA) range increases. The counters **N.RA.TA.UE.index0** to **N.RA.TA.UE.index15** can be used to observe the numbers of TA values that fall into different TA ranges in random access. The number of UEs in the NSA DC state changes, causing changes in the values of other counters related to NSA DC UEs.

– As this function extends the coverage of cells in SCGs, the gNodeB delays coverage-based inter-frequency handovers of NSA UEs if more than one frequency is deployed on the NR side. As a result, the value of the NR counter **N.NsaDc.InterFreq.Coverage.PSCell.Change.Att** may decrease.

If threshold adaptation for coverage-based inter-frequency handovers and non-gap-assisted coverage-based inter-frequency PSCell change take effect at the same time, the network camping policy for UEs may change. As a result, the coverage-based inter-frequency handover success rate (**N.NsaDc.InterFreq.Coverage.PSCell.Change.Succ**/**N.NsaDc.InterFreq.Coverage.PSCell.Change.Att**) decreases and the **SgNB-Triggered SgNB Abnormal Release Rate (CU)** increases.

4.4.3 Requirements

4.4.3.1 Licenses

None

4.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NSA carrier combination selection based on user experience	• NR: **NSA_LTE_SEL_OPT_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter • LTE: **NSA_LTE_FDD_SEL_OPT_SW** or **NSA_LTE_TDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter	*EPC-based NSA Experience Improvement*	User-experience-based NSA SCG coverage expansion (controlled by the **SCG_ADAPT_COV_THLD_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoExtSwitch*** parameter on the LTE side and the **NSA_COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter on the NR side) can be enabled only if the NSA carrier combination selection based on user experience function is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User-experience-based NSA SCG coverage expansion	• NR: **NSA_COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter • LTE: **SCG_ADAPT_COV_THLD_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoExtSwitch*** parameter	*Multi-Frequency Smart Selection*	Threshold adaptation for coverage-based inter-frequency handovers takes effect only when all of the following conditions are met: • User-experience-based NSA SCG coverage expansion is enabled. • The **NRCellMobilityConfig**.***MeasTrigQuantity*** parameter (specifying the triggering quantity for event A1/A2 related to coverage-based inter-frequency measurements) is set to **RSRP**. • The **NRCellFreqRelation**.***InterFreqHoEventType*** parameter (specifying the event for triggering coverage-based inter-frequency handovers) is set to **EVENT_A5**. • Coverage-based inter-frequency handover (controlled by the **COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter) is enabled.
Low-frequency TDD	Triggering quantity for events A1 and A2 related to coverage-based inter-frequency measurements	**NRCellMobilityConfig**.***MeasTrigQuantity***	*NSA Mobility Management*	
Low-frequency TDD	Event for triggering inter-frequency handovers	**NRCellFreqRelation**.***InterFreqHoEventType***	*NSA Mobility Management*	
Low-frequency TDD	Coverage-based inter-frequency handover	**COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*NSA Mobility Management*	

4.4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Protection of UEs under weak uplink coverage	**UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	When user-experience-based NSA SCG coverage expansion is enabled, protection of UEs under weak uplink coverage (controlled by the **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) is not supported.
Low-frequency TDD	Triggering quantity for events A1 and A2 related to coverage-based inter-frequency measurements	**NRCellMobilityConfig**.***MeasTrigQuantity***	*NSA Mobility Management*	If the **NRCellMobilityConfig**.***MeasTrigQuantity*** parameter is set to **RSRP_OR_RSRQ** or **RSRP_OR_SINR**, user-experience-based NSA SCG coverage expansion cannot be enabled.

4.4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SgNB-initiated SgNB release	None	*NSA Mobility Management*	If user-experience-based NSA SCG coverage expansion is enabled, the RSRP threshold used in coverage threshold adaptation (specified by the **gNBMobilityComm-Param**.***AdaptCovRsrpThld*** parameter) must be greater than or equal to the threshold for event A2 related to SCG deletion plus 4 dB. This prevents a significant increase in the number of SCG releases (indicated by **L.NsaDc.SgnbTrig.SgNB.NormRel**). ● Threshold for event A2 related to SCG deletion = **NRCellNsaDcConfig**.***PscellA2RsrpThld*** + **NRCellNsaDcConfigGrp**.***PscellA2RsrpThldOffset*** ● You are advised to set the parameters used for calculating the threshold for event A2 related to SCG deletion to their recommended values. – The larger the threshold for event A2 related to SCG

RAT	Function Name	Function Switch	Reference	Description
				deletion, the smaller the coverage extension area of cells in the SCG. To increase the coverage extension area of cells in the SCG, decrease the threshold for event A2 related to SCG deletion. – Decreasing the threshold for event A2 related to SCG deletion may increase the values of **N.NsaDc.InterSgNB.IntraFreq.PSCell.Change.Att** and **N.NsaDc.IntraSgNB.IntraFreq.PSCell.Change.Att**.
Low-frequency TDD	Flexible user steering	**gNBRfspConfig**.***NetworkingOptionOptFlag***	None	If the **gNBRfspConfig**.***NetworkingOptionOpt-Flag*** parameter corresponding to the RFSP index of a UE is set to **FALSE**, user-experience-based NSA SCG coverage expansion does not take effect for the UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coverage-based inter-frequency PSCell change	**COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*NSA Mobility Management*	User-experience-based NSA SCG coverage expansion modifies the thresholds for events A1 and A2 and threshold 2 for event A5 related to coverage-based inter-frequency handovers of NSA UEs only when coverage-based inter-frequency PSCell change is enabled.
Low-frequency TDD	NSA/SA selection based on uplink coverage	**LTE_FDD_NSA_SA_UL_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter	*NSA-SA Selection Based on User Experience*	When NSA/SA selection based on uplink coverage is enabled together with user-experience-based NSA SCG coverage expansion, the probability that NSA/SA selection based on uplink coverage takes effect decreases.

4.4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

On the LTE side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

On the NR side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- On the LTE side: The required main control board is UMPTb or later, and the required baseband processing unit is LBBPd or later.
- On the NR side: All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.4.3.4 Networking

- NSA networking is used, the number of NR frequencies is greater than or equal to 2, and anchors that support the addition of cells on these NR frequencies to SCGs exist on the LTE side.
- The frequencies have different coverage areas and these coverage areas overlap.

4.4.3.5 Others

None

4.4.4 Operation and Maintenance

4.4.4.1 Data Configuration

4.4.4.1.1 Data Preparation

Before activating this function, configure NR cell frequency relationships and neighbor relationships. For details about related parameters, see **Table 4-14** and the table with the parameters used to configure neighboring cells in *SA Mobility Management in Connected Mode*, respectively.

Before activating this function, activate the NSA carrier combination selection based on user experience function. For details about related configurations, see *EPC-based NSA Experience Improvement*.

Table 4-27 describes the parameters used for function activation.

To enable this function, you need to set the corresponding switch parameters in the **NsaDcAlgoParam** MO on the LTE side and the **gNBMobilityCommParam** MO on the NR side.

Table 4-27 Parameters used for activation

NE	Parameter Name	Parameter ID	Option	Setting Notes
gNodeB	Mobility Algorithm Switch	**gNBMobilityCommParam**.***MobilityAlgoSwitch***	NSA_COV_THLD_ADAPT_SW	You are advised to select this option.
gNodeB	Adaptive Coverage RSRP Threshold	**gNBMobilityCommParam**.***AdaptCovRsrpThld***	N/A	Set this parameter based on the network plan. It is recommended that this parameter be set to a value greater than or equal to the threshold for event A2 related to SCG deletion plus 4 dB to prevent too fast SCG releases.
gNodeB	Mobility Function Indication	**NRExternalNCell**.***MobilityFunInd***	COV_THLD_ADAPT_SW	If user-experience-based NSA SCG coverage expansion (controlled by the **NSA_COV_THLD_ADAPT_SW** option of the **gNBMobilityComm-Param**.***MobilityAlgoSwitch*** parameter) is not enabled on the gNodeB serving an external NR cell, it is recommended that the **COV_THLD_ADAPT_SW** option of the **NRExternalNCell**.***MobilityFunInd*** parameter be deselected. This prevents UEs from being handed over from the external NR cell through coverage-based inter-frequency outgoing handovers immediately after these UEs are handed over to this cell.

NE	Parameter Name	Parameter ID	Option	Setting Notes
gNodeB	NR Move to E-UTRAN UL SINR Low Threshold	**NRDUCellSrsMeas**.***NrToEutranSinrLowThld***	N/A	Set this parameter to its recommended value.
gNodeB	Service Function Switch	**NRCellAlgoSwitch**.***ServiceFunctionSwitch***	VOLUME_IDENTIFY_SW	You are advised to select this option.
gNodeB	Common UL Large Pkt Ident Thld	**NRDUCellServExp**.***CommonUlLargePktIdentThld***	N/A	Set this parameter to its recommended value.
eNodeB	NSA DC Algorithm Extension Switch	**NsaDcAlgoParam**.***NsaDcAlgoExtSwitch***	SCG_ADAPT_COV_THLD_SW	You are advised to select this option.
eNodeB	SCG Adaptive Coverage RSRP Threshold	**NrScgFreqConfig**.***ScgCovAdaptRsrpThld***	N/A	It is recommended that this parameter be set to a value greater than or equal to the value of **gNBMobilityComm-Param**.***AdaptCovRsrpThld*** plus 4 dB.

4.4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Configuration on the eNodeB

```
//Selecting the SCG_ADAPT_COV_THLD_SW option
MOD NSADCALGOPARAM: NsaDcAlgoExtSwitch=SCG_ADAPT_COV_THLD_SW-1;
//Setting the RSRP threshold for SCG coverage adaptation (It is recommended that this parameter be set to a value greater than or equal to the gNodeB-side RSRP threshold used in coverage threshold adaptation plus 4 dB.)
MOD NRSCGFREQCONFIG: PccDlEarfcn=1500, ScgDlArfcn=361000, ScgCovAdaptRsrpThld=-114;
```

Configuration on the gNodeB

```
//Selecting the NSA_COV_THLD_ADAPT_SW option
MOD GNBMOBILITYCOMMPARAM: MobilityAlgoSwitch=NSA_COV_THLD_ADAPT_SW-1;
//Setting the AdaptCovRsrpThld parameter (It is recommended that this parameter be set to a value greater than or equal to the threshold for event A2 related to SCG deletion plus 4 dB.)
MOD GNBMOBILITYCOMMPARAM: AdaptCovRsrpThld=-118;
//Deselecting the COV_THLD_ADAPT_SW option for external NR cells (This option can be deselected if user-experience-based coverage expansion is enabled on the local base station but not enabled on the neighboring base stations.)
MOD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, MobilityFunInd=COV_THLD_ADAPT_SW-0;
//Changing the uplink SINR threshold for NR inter-frequency handovers
MOD NRDUCELLSRSMEAS: NrDuCellId=0, NrToEutranSinrLowThld=-30;
//Selecting the VOLUME_IDENTIFY_SW option if uplink traffic volumes need to be determined for the threshold adaptation for coverage-based inter-frequency handovers function
MOD NRCELLALGOSWITCH: NrCellId=0, ServiceFunctionSwitch=VOLUME_IDENTIFY_SW-1;
//Changing the threshold for identifying common large-packet UEs in the uplink as required (It is recommended that the default value be used. The value 80 is used as an example.)
MOD NRDUCELLSERVEXP: NrDuCellId=0, CommonUlLargePktIdentThld=80;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Configuration on the eNodeB

```
//Deselecting the SCG_ADAPT_COV_THLD_SW option
MOD NSADCALGOPARAM: NsaDcAlgoExtSwitch=SCG_ADAPT_COV_THLD_SW-0;
```

Configuration on the gNodeB

```
//Deselecting the NSA_COV_THLD_ADAPT_SW option
MOD GNBMOBILITYCOMMPARAM: MobilityAlgoSwitch=NSA_COV_THLD_ADAPT_SW-0;
```

4.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.4.2 Activation Verification

User-experience-based NSA SCG coverage expansion has taken effect if the following counter value changes are observed on the NR and LTE sides:

- On the NR side:
  - The value of **N.User.NsaDc.PSCell.Avg**, which indicates the number of NSA DC UEs, increases.
  - The values of all counters listed in **Table 4-28** increase.

Table 4-28 Counters related to the number of uplink-coverage-based inter-frequency handovers

Counter ID	Counter Name
1911833798	**N.HO.InterFreq.ULCoverage.PrepAttOut**
1911833799	**N.HO.InterFreq.ULCoverage.ExecAttOut**
1911833800	**N.HO.InterFreq.ULCoverage.ExecSuccOut**

- The number of coverage-based inter-frequency PSCell changes decreases.
  - **N.NsaDc.InterFreq.Coverage.PSCell.Change.Att**, which indicates the number of coverage-based inter-frequency PSCell change attempts
  - **N.NsaDc.InterFreq.Coverage.PSCell.Change.Succ**, which indicates the number of successful coverage-based inter-frequency PSCell changes

- On the LTE side:

  The value of **L.NsaDc.BcSelOpt.BcChange**, which indicates the number of band combination change attempts for NSA DC UEs in band combination selection optimization, increases.

4.4.4.3 Network Monitoring

Observe the average downlink UE throughput to check whether the UE data rate on the NR side increases after this function is enabled.

The average downlink UE throughput is indicated by **User Downlink Average Throughput (DU)**.

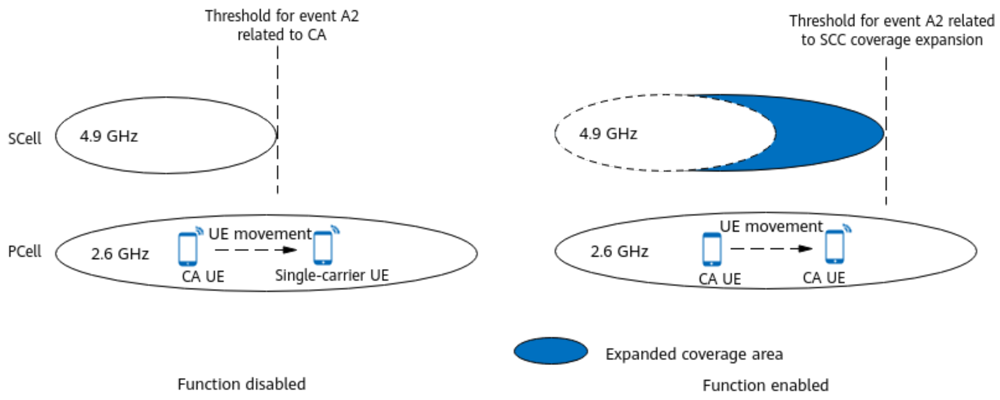
4.5 SCC Coverage Expansion

4.5.1 Principles

On a multi-band network where the frequency difference between frequency bands (for example, the 2.6 GHz and 4.9 GHz frequency bands) is large, the coverage of the higher-band carrier may be discontinuous because its uplink coverage differs considerably from its downlink coverage. In this scenario, the signal quality of the higher-band carrier is poor for a CA UE that is camping on the lower-band carrier and located at the cell edge or a medium distance from the cell center, and therefore the higher-band carrier cannot be configured as an SCC for the UE. As a result, the average downlink throughput of the UE is affected. To increase the average downlink throughput of CA UEs at cell edges, the SCC coverage expansion function is introduced.

The SCC coverage expansion function expands the downlink coverage of SCCs for CA UEs, as shown in **Figure 4-11**. This function increases the number of CA UEs at cell edges and the average downlink throughput of these CA UEs. However, as cell-edge UEs have poor spectral efficiency, the average downlink throughput of CA UEs decreases. To mitigate this, specific conditions must be met for this function to take effect, and policies are devised for experience assurance in SCells. This function is applicable in scenarios where SCCs provide discontinuous coverage and are lightly loaded.

Figure 4-11 Schematic diagram of SCC coverage expansion



The SCC coverage expansion function is applicable on both SA networks and NSA networks.

- On SA networks, this function is controlled by the **SA_SCC_COV_THLD_ADAPT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter on the PCC side.
- On NSA networks, this function is controlled by the **NSA_SCC_COV_THLD_ADAPT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter on the PCC side.

The gNodeB starts the SCC coverage expansion process for a CA UE when SCell configuration is triggered at initial access, an incoming handover, or an incoming RRC connection reestablishment, or when it is triggered based on the results of a periodic UE service evaluation. The process includes two phases:

1. The gNodeB expands the SCC coverage for CA UEs based on the SCC coverage expansion conditions. For details, see **4.5.1.1 SCC Coverage Expansion**.

2. The gNodeB determines whether to stop data split based on the ratio of the scheduling capability of the PCC to that of an SCC. The purpose of this phase is to prevent resources of the SCC from being excessively occupied, thereby protecting the user experience of non-target UEs on the SCC. For details, see **4.5.1.2 Experience Assurance in SCells**.

4.5.1.1 SCC Coverage Expansion

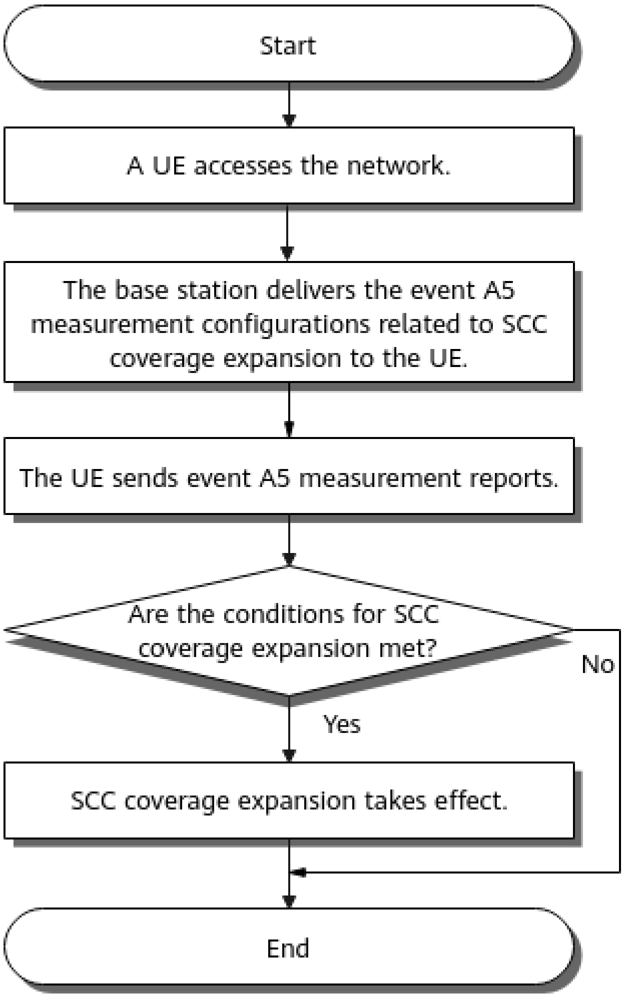
How SCC coverage expansion takes effect varies depending on whether smart carrier selection (controlled by the **SMART_CARR_SEL_SW** option of the **NRDUCellCollabServ**.***MultiCarrMgmtSw*** parameter) is enabled. For details about smart carrier selection, see *Carrier Aggregation* .

- If smart carrier selection is disabled, the process described in **How SCC Coverage Expansion Takes Effect (Mode 1)** is applicable.
- If smart carrier selection is enabled, the process described in **How SCC Coverage Expansion Takes Effect (Mode 2)** is applicable.

How SCC Coverage Expansion Takes Effect (Mode 1)

Figure 4-12 shows the basic process of the SCC coverage expansion function.

Figure 4-12 Basic process of the SCC coverage expansion function



The process is as follows:

1. After a CA UE accesses the network, the gNodeB delivers event A5 measurement configurations related to SCC coverage expansion to the CA UE.

   Threshold 1 for event A5 is fixed at –30 dBm. Threshold 2 for event A5 related to SCC coverage expansion is equal to the threshold for event A2 related to SCC coverage expansion plus 3 dB.

   Threshold for event A2 related to SCC coverage expansion = **NRCellCaMgmtConfig**.***CaA2RsrpThld*** + **gNBCaFrequency**.***CaA2RsrpThldOffset*** + **NRCellCaMgmtConfig**.***SccCovAdaptRsrpThldOffset***

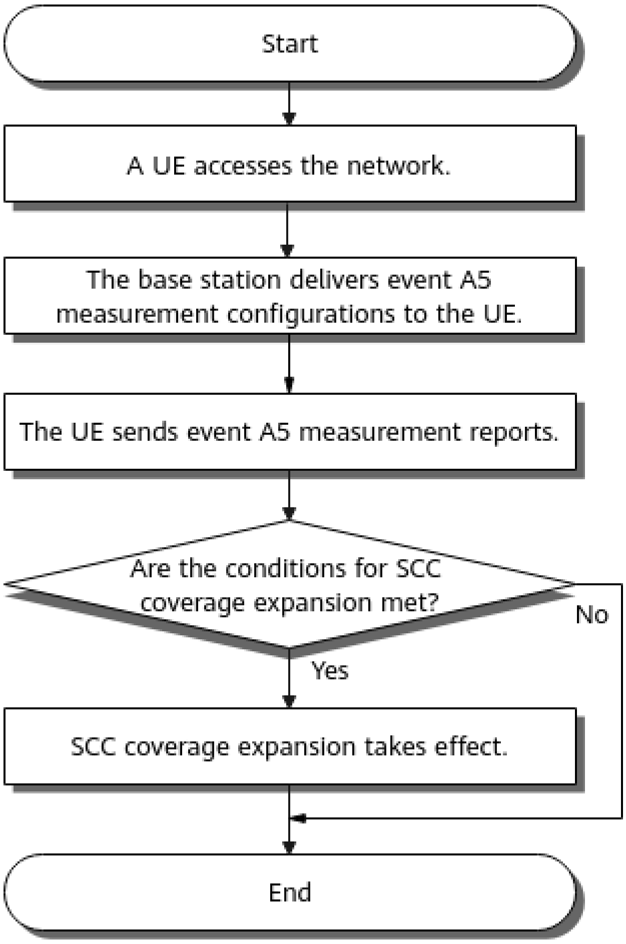
2. The gNodeB checks whether it has received event A5 measurement reports from the CA UE.
   - If the gNodeB does not receive any event A5 measurement report, the process ends.
   - If the gNodeB has received an A5 measurement report, it continues checking within a maximum of 3 seconds whether it has received event A5 measurement reports related to all the candidate frequencies. If it has, the process proceeds to **3** immediately without waiting for the 3-second check period to end; if it has not, the process proceeds to **3** when the 3-second check period ends.
3. Based on the frequency information in the event A5 measurement reports, the gNodeB determines whether any of the inter-frequency carriers meets the SCC coverage expansion conditions.
   - If an inter-frequency carrier meets the conditions, the gNodeB performs SCC coverage expansion for the CA UE. It configures this inter-frequency carrier as an SCC for the CA UE and then delivers event A2 measurement configurations related to SCC coverage expansion to the CA UE.
   - If no inter-frequency carrier meets the conditions, the process ends.

   The conditions for SCC coverage expansion are as follows, and both the conditions must be met:
   - SCC downlink PRB usage < SCC downlink PRB usage threshold (specified by **NRCellCaMgmtConfig**.***SccCovAdaptPrbUsageThld***)
   - SCC downlink CCE allocation failure rate < SCC downlink CCE allocation failure rate threshold (specified by **NRCellCaMgmtConfig**.***SccCovAdaptCceFailThld***)

How SCC Coverage Expansion Takes Effect (Mode 2)

Figure 4-13 shows the basic process of the SCC coverage expansion function.

Figure 4-13 Basic process of the SCC coverage expansion function



1. After a CA UE accesses the network, the gNodeB delivers event A5 measurement configurations to the CA UE.

   Threshold 1 for event A5 is fixed at –30 dBm. Threshold 2 for event A5 is equal to the sum of the **NRCellCaMgmtConfig**.***CaA5RsrpThld2*** and **gNBCaFrequency**.***CaSccA5RsrpThld2Offset*** parameter values.

2. The gNodeB checks whether it has received event A5 measurement reports from the CA UE.
   - If the gNodeB does not receive any event A5 measurement report, the process ends.
   - If the gNodeB has received an A5 measurement report, it continues checking within a maximum of 3 seconds whether it has received event A5 measurement reports related to all the candidate frequencies. If it has, the process proceeds to **3** immediately without waiting for the 3-second check period to end; if it has not, the process proceeds to **3** when the 3-second check period ends.
3. The gNodeB configures an inter-frequency carrier as an SCC for the CA UE based on the frequency information in the event A5 measurement reports. It then delivers event A2 measurement configurations to the CA UE. The threshold for event A2 depends on whether the SCC meets the SCC coverage expansion conditions.

- If the conditions are met, the threshold for event A2 is equal to the threshold for event A2 related to SCC coverage expansion.

  Threshold for event A2 related to SCC coverage expansion = **NRCellCaMgmtConfig**.***CaA2RsrpThld*** + **gNBCaFrequency**.***CaA2RsrpThldOffset*** + **NRCellCaMgmtConfig**.***SccCovAdaptRsrpThldOffset***
- If the conditions are not met, the threshold for event A2 is equal to the sum of the values of **NRCellCaMgmtConfig**.***CaA2RsrpThld*** and **gNBCaFrequency**.***CaA2RsrpThldOffset***.

The conditions for SCC coverage expansion are as follows, and both the conditions must be met:

- SCC downlink PRB usage < SCC downlink PRB usage threshold (specified by **NRCellCaMgmtConfig**.***SccCovAdaptPrbUsageThld***)
- SCC downlink CCE allocation failure rate < SCC downlink CCE allocation failure rate threshold (specified by **NRCellCaMgmtConfig**.***SccCovAdaptCceFailThld***)

4.5.1.2 Experience Assurance in SCells

When a CA UE for which the SCC coverage expansion function has taken effect is in the coverage expansion area and uses the SCC to perform data services, the overall spectral efficiency of the SCC decreases due to poor spectral efficiency of the CA UE. To prevent such CA UEs from excessively preempting the resources of other UEs served by the SCC, CA data split stop based on the scheduling capability ratio is used to guarantee user experience in the SCell.

If the ratio of the scheduling capability of a higher-throughput carrier to that of a lower-throughput carrier is greater than the threshold specified by the **NRDUCellCarrMgmt**.***SplitStopSchCapbRatioThld*** parameter (the value of this parameter must be greater than zero), CA data split stop based on the scheduling capability ratio takes effect. After this function takes effect, data distribution to the carrier with lower scheduling capability will be stopped, thereby improving downlink experience of CA UEs.

4.5.2 Network Analysis

4.5.2.1 Benefits

This function is applicable to multi-carrier networks where cells on high bands provide discontinuous coverage and are lightly loaded. It is recommended that this function be enabled in cells on low bands.

When SCCs provide discontinuous coverage, this function expands the downlink coverage for CA UEs. It increases the average number of UEs that are in the downlink CA state and treat the local cell as their PCell (indicated by **N.User.CA.PCell.DL.Avg**) and slightly increases the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**). The gains depend on the coverage continuity of cells on high bands, the proportions of UEs in the cell center, at the cell edge, and at a medium distance from the cell center in cells on low bands, the downlink load difference between cells on high bands and those on low bands, and the interference level. If cells on high bands provide continuous coverage, the proportion of UEs at the cell edge is low in cells on low bands, the downlink load difference between cells on high bands and those on low bands is small, and the interference level is high, the gains cannot be ensured.

4.5.2.2 Impacts

The impacts between this function and other functions are described in **4.5.3.2.3 Function Impacts**.

- System capacity

  A CA UE with SCells configured has only one RRC connection to the network. Such a UE consumes one sales unit of the license for the number of RRC_CONNECTED UEs. However, the CA UE consumes one hardware resource unit in each of its serving cells. With *n*CC aggregation for all UEs on the network, the maximum allowed number of RRC_CONNECTED UEs decreases to 1/*n* (*n* is an integer). After SCC coverage expansion takes effect, the number of CA UEs on the network increases, consuming more RRC connection resources.

- Resource usage
  - Overall PRB usage of the network

    After SCC coverage expansion takes effect, cell load can be rapidly balanced through carrier management and scheduling to help utilize idle resources and increase the overall downlink PRB usage of the network.

  - PUCCH and PUSCH overheads

    Each CA UE sends the ACK/NACK and CSI related to its SCells in its PCell with the PUCCH. When the PUSCH is not scheduled, the UE sends the information over the PUCCH. When the PUSCH is scheduled, the UE sends the information over the PUSCH. Therefore, the PUCCH overhead or the signaling overhead on the PUSCH increases after SCC coverage expansion takes effect.

  - PDCCH overhead

    As cell-edge UEs usually use high CCE aggregation levels, PDCCH CCE resource consumption increases when these UEs have data transmissions on their SCCs after SCC coverage expansion takes effect. If the SCC CCE usage is high, the downlink CCE allocation failure rate increases.

- Overall downlink throughput of the network

  SCC coverage expansion does not change the network capacity. However, when there are idle resources on the network, enabling SCC coverage expansion increases the overall resource usage on the network and slightly increases downlink throughput. The impact depends on CA UE distribution.

- KPIs
  - After SCC coverage expansion is enabled, the SCC coverage area is expanded so that CA can take effect for cell-edge UEs. As a result, the average number of CA UEs in the relevant cell increases.
  - As the number of CA UEs at cell edges increases after SCC coverage expansion is enabled, the PCC- and SCC-specific average downlink throughput of CA UEs with SCells activated decreases. The average downlink throughput is calculated using the following formula: (**N.CA.SccActiveUser.ThpVol.DL** – **N.CA.ThpVol.DL.LastSlot**)/ **N.CA.ThpTime.DL.RmvLastSlot**.

– Channel quality of UEs that treat a cell as their PCell differs from that of UEs that treat the cell as their SCell. Therefore, SCC coverage expansion will cause CQIs and MCS indexes to change in a cell configured as an SCell for UEs.
– As SCC coverage expansion increases the number of cell-edge UEs for which CA takes effect, the average value of **N.ChMeas.PDSCH.MCS.0** to **N.ChMeas.PDSCH.MCS.28** decreases and the PDCCH CCE usage (indicated by **N.CCE.Used.Avg** and **N.CCE.Avail.Avg**) increases in the relevant SCell.
– After SCC coverage expansion is enabled, cell-edge UEs can measure neighboring cells that meet event A5 requirements and these cells can be successfully added as SCells. This prevents the downlink UE throughput in the PCell from decreasing as a result of periodic gap-assisted event A5 measurements when the UE traffic volume is consistently large.
– As the scheduling capability for CA UEs increases after data is distributed to SCells, the proportion of the data transmission duration in the last slots in which the downlink buffer becomes empty to the total data transmission duration may increase and the average downlink UE throughput (**User Downlink Average Throughput (DU)**) may decrease. The proportion is calculated using the following formula: (**N.ThpTime.DL – N.ThpTime.DL.RmvLastSlot**)/**N.ThpTime.DL**.
– After the "CA data split stop based on the scheduling capability ratio" function takes effect for cell-edge CA UEs for which SCC coverage expansion has taken effect, data distribution to the carriers with weak scheduling capabilities will be stopped. This improves user experience of CA UEs and may reduce the impact on the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**).

4.5.3 Requirements

4.5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-061223	Experience Boosting based on Multi-Band Coordination	NR0S0SCSNR00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA data split stop based on the scheduling capability ratio	**NRDUCellCarrMgmt**.***SplitStopSchCapbRatioThld***	*Carrier Aggregation*	To prevent the downlink experience of non-target UEs on SCCs from significantly deteriorating, the SCC coverage expansion function can be enabled only after the "CA data split stop based on the scheduling capability ratio" function is enabled by setting the **NRDUCellCarrMgmt**.***SplitStopSchCapbRatioThld*** parameter to a value other than **0**.

4.5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag***	*High Speed Mobility*	SCC coverage expansion cannot be enabled for high-speed cells (with **NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**).
Low-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	SCC coverage expansion cannot be enabled for hyper cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	• **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** • **NRDUCellMultiTrp**.***DataTransMode*** set to **BASIC_MODE** or **DPS_MODE**	*Cell Combination*	SCC coverage expansion cannot be enabled for combined cells.
Low-frequency TDD	Joint Transmission Cell Combination	• **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** • **NRDUCellMultiTrp**.***DataTransMode*** set to **JOINT_MODE**	*Cell Combination*	SCC coverage expansion cannot be enabled for combined cells in joint transmission mode.
Low-frequency TDD	Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	When Fusion Cell is enabled, SCC coverage expansion cannot be enabled.

4.5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Selection of the optimal combination of multi-carrier frequencies	**SMART_CARR_SEL_SW** option of the **NRDUCellCollabServ**.***MultiCarrMgmtSw*** parameter	*Carrier Aggregation*	When selection of the optimal combination of multi-carrier frequencies and SCC coverage expansion are both enabled, the base station preferentially delivers inter-frequency event A5 configurations related to selection of the optimal combination of multi-carrier frequencies.
Low-frequency TDD	Inter-FR CA	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	SCC coverage expansion does not take effect for UEs engaged in inter-FR CA.
Low-frequency TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	For CA UEs for which SCC coverage expansion has taken effect, the threshold for event A2 related to SCC coverage expansion is used in event-A2-triggered SCell removal.
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	For CA UEs for which SCC coverage expansion has taken effect, the threshold for event A2 related to SCC coverage expansion is used in event-A2-triggered SCell removal.

4.5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.5.3.4 Networking

- The network uses at least two frequencies.
- **NRCellFreqRelation** MOs are already configured.

4.5.3.5 Others

- Carrier aggregation can take effect normally.
- It is recommended that SCC coverage expansion be disabled during big-event assurance. This prevents network performance deterioration caused by excessive cell-edge UEs for which SCC coverage expansion takes effect.

4.5.4 Operation and Maintenance

4.5.4.1 Data Configuration

4.5.4.1.1 Data Preparation

Table 4-29 describes the parameters used for function activation.

Table 4-29 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	SA_SCC_COV_THLD_ADAPT_SW	Set this option based on operators' requirements.
Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	NSA_SCC_COV_THLD_ADAPT_SW	Set this option based on operators' requirements.
Scc Coverage Adaptation RSRP Threshold Offset	**NRCellCaMgmtConfig**.***SccCovAdaptRsrpThldOffset***	N/A	Set this parameter based on operators' requirements.
Scc Coverage Adaptation PRB Usage Threshold	**NRCellCaMgmtConfig** ***SccCovAdaptPrbUsageThld***	N/A	Set this parameter based on operators' requirements.
Scc Coverage Adaptation CCE Alloc Fail Thld	**NRCellCaMgmtConfig**.***SccCovAdaptCceFailThld***	N/A	Set this parameter based on operators' requirements.
Split Stop Sch Capb Ratio Threshold	**NRDUCellCarrMgmt**.***SplitStopSchCapbRatioThld***	N/A	The value **50** is recommended.

4.5.4.1.2 Using MML Commands

Before using MML commands, refer to **4.5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

On the PCC side:

```
//Enabling CA data split stop based on the scheduling capability ratio
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SplitStopSchCapbRatioThld=50;
```

```
//Enabling the SCC coverage expansion function in SA networking
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=SA_SCC_COV_THLD_ADAPT_SW-1;
//Enabling the SCC coverage expansion function in NSA networking
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=NSA_SCC_COV_THLD_ADAPT_SW-1;
//Setting the offset to RSRP threshold for SCC coverage adaptation
MOD NRCELLCAMGMTCONFIG: NrCellId=0, SccCovAdaptRsrpThldOffset=-6;
//Setting the PRB usage threshold for SCC coverage adaptation
MOD NRCELLCAMGMTCONFIG: NrCellId=0, SccCovAdaptPrbUsageThld=50;
//Setting the CCE allocation failure threshold for SCC coverage adaptation
MOD NRCELLCAMGMTCONFIG: NrCellId=0, SccCovAdaptCceFailThld=8;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling the SCC coverage expansion function in SA networking
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=SA_SCC_COV_THLD_ADAPT_SW-0;
//Disabling the SCC coverage expansion function in NSA networking
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=NSA_SCC_COV_THLD_ADAPT_SW-0;
```

4.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.5.4.2 Activation Verification

If the average number of UEs that are in the downlink CA state and treat the local cell as their PCell (indicated by **N.User.CA.PCell.DL.Avg**) increases, SCC coverage expansion has taken effect.

4.5.4.3 Network Monitoring

Observe the average downlink UE throughput to check whether the UE data rate increases after this function is enabled.

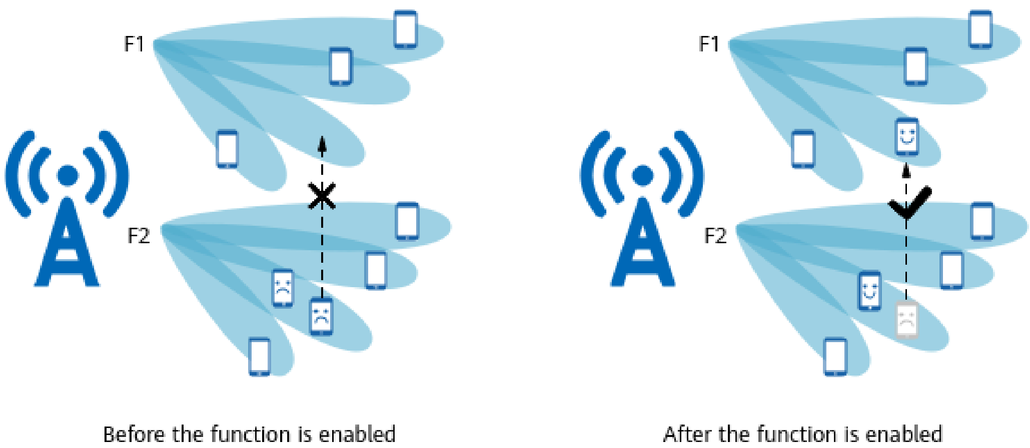
The average downlink UE throughput is indicated by **User Downlink Average Throughput (DU)**.

5 Multi-Frequency Beam Coordination

5.1 Principles

In multi-carrier scenarios, UEs are densely distributed on some sounding reference signal (SRS) beams, which restricts the MU pairing capability and UEs' spectral efficiency and consequently hinders the average downlink UE throughput from increasing. To redistribute UEs across SRS beams and increase the average downlink UE throughput at the network level, the **multi-frequency beam coordination** function shown in **Figure 5-1** is introduced.

Figure 5-1 Multi-frequency beam coordination



The multi-carrier scenarios where multi-frequency beam coordination is applicable vary depending on the **multi-frequency beam coordination enhancement for TDD** function as follows:

- When **multi-frequency beam coordination enhancement for TDD** is disabled:

  Multiple intra-base-station NR TDD massive MIMO cells are deployed using the same RF module and cover the same area. Based on the downlink cell PRB usage, the SRS beam load, and the number of UEs served by SRS beams, the gNodeB optimizes the downlink traffic distribution of UEs among cells and SRS beams, increasing the average downlink UE throughput at the network level.

- When **multi-frequency beam coordination enhancement for TDD** is enabled:

  There are multiple NR TDD massive MIMO cells that have overlapping coverage. Based on the downlink cell PRB usage, the SRS beam load, the number of UEs served by SRS beams, the SRS beam PRB usage, and the SRS beam interference, the gNodeB optimizes the downlink traffic distribution of UEs among cells and SRS beams, increasing the average downlink UE throughput at the network level.

To increase the number of UEs transferred in each period and accelerate inter-cell beam load balancing, **load balancing acceleration for multi-frequency beam coordination** is introduced.

Table 5-1 lists the parameters used to configure the preceding functions.

Table 5-1 Parameters related to function configuration

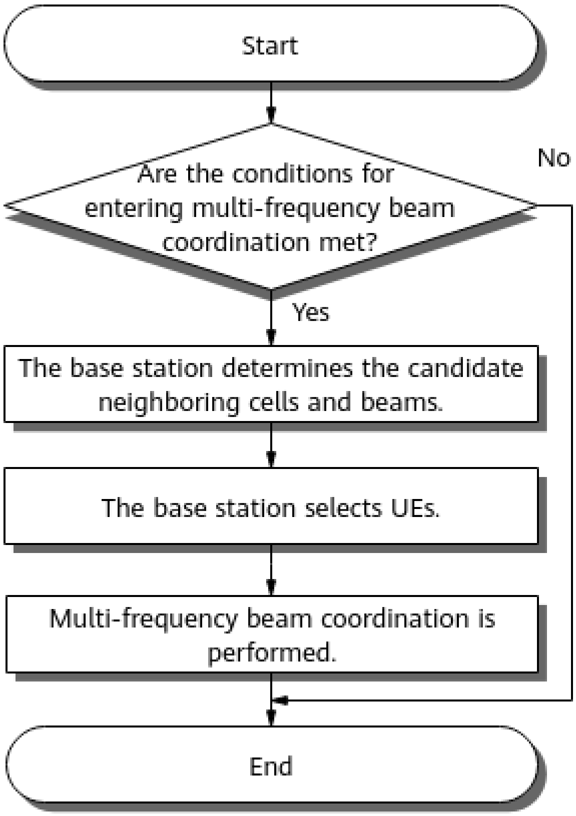
Function	Parameter
Multi-frequency beam coordination	**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter
Multi-frequency beam coordination enhancement for TDD	**MF_BEAM_COORD_TDD_ENH_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter
Load balancing acceleration for multi-frequency beam coordination	**MF_BEAM_COORD_LOAD_BAL_ACC_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter

NOTE

The SRS beam load is calculated based on the number of UEs on the SRS beam, average spectral efficiency of all UEs served by the SRS beam, and cell bandwidth.

Figure 5-2 shows the multi-frequency beam coordination process.

Figure 5-2 Multi-frequency beam coordination process



1. The conditions for entering and exiting multi-frequency beam coordination are described in **5.1.1 Triggering of Multi-Frequency Beam Coordination**.
2. How the gNodeB determines the candidate neighboring cells and beams for multi-frequency beam coordination is described in **5.1.2 Determining Candidate Neighboring Cells and Beams**.
3. How the gNodeB selects UEs for inter-frequency handovers is described in **5.1.3 UE Selection**.
4. How the gNodeB determines whether to instruct the UE to perform an inter-frequency handover is described in **5.1.4 Execution of Multi-Frequency Beam Coordination**.

NOTE

Unless otherwise specified, NR TDD cells mentioned in the following sections of this chapter refer to NR TDD massive MIMO cells.

5.1.1 Triggering of Multi-Frequency Beam Coordination

When multi-frequency beam coordination is enabled in a cell, the gNodeB periodically determines whether the cell enters or exits the multi-frequency beam coordination state. The decision-making period is specified by the **NRCellMlb**.***MlbJudgePeriod*** parameter.

Conditions for Entering the Multi-Frequency Beam Coordination State

The conditions for entering the multi-frequency beam coordination state vary depending on whether multi-frequency beam coordination enhancement for TDD is enabled:

- When multi-frequency beam coordination enhancement for TDD is disabled:

  A cell enters the multi-frequency beam coordination state when both of the following conditions are met throughout a decision-making period:
  - The load of at least one SRS beam is greater than or equal to the sum of the load threshold for beam coordination and the load offset for beam coordination, and the number of UEs served by the beam is at least five.
  - The downlink PRB usage of the cell is greater than or equal to 40%.
- When multi-frequency beam coordination enhancement for TDD is enabled, the serving cell enters the multi-frequency beam coordination state if it has stayed in any of the following states throughout a decision-making period. **Table 5-2** describes the conditions corresponding to each state.

Table 5-2 Serving cell conditions for entering the multi-frequency beam coordination state

Serving Cell State	Corresponding Condition
State 1	● Either of the following conditions is met: – The load of at least one SRS beam is greater than or equal to the sum of the load threshold for beam coordination and the load offset for beam coordination, and the number of UEs served by the beam is at least five. – The interference level of at least one SRS beam is greater than or equal to the interference threshold for beam coordination (–105 dBm), and the number of UEs served by the beam is at least five. ● The downlink PRB usage of the cell is greater than or equal to 40%.
State 2	● The PRB usage of at least one SRS beam is greater than or equal to the sum of the PRB usage threshold for beam coordination and the PRB usage offset for beam coordination (the offset is 5%), and the number of UEs served by the beam is at least five. ● The downlink PRB usage of the cell is greater than or equal to 40%.
State 3	The conditions corresponding to both state 1 and state 2 are met.

The related parameters are as follows:

- The load threshold for beam coordination is specified by the **NRCellMultiFSchCoord**.***BeamCoordLoadThld*** parameter.

- The load offset for beam coordination is specified by the **NRCellMultiFSchCoord**.***BeamCoordLoadOffset*** parameter.
- The PRB usage threshold for beam coordination is specified by the **NRCellMultiFSchCoord**.***BeamCoordPrbUsageThld*** parameter.

Conditions for Exiting the Multi-Frequency Beam Coordination State

The conditions for exiting the multi-frequency beam coordination state vary depending on whether multi-frequency beam coordination enhancement for TDD is enabled:

- When multi-frequency beam coordination enhancement for TDD is disabled:

  A cell exits the multi-frequency beam coordination state when any of the following conditions is met throughout a decision-making period:
  - The load of each SRS beam is less than the load threshold for beam coordination.
  - The number of UEs served by each SRS beam is less than five.
  - The downlink PRB usage of the cell is less than 30%.
- When multi-frequency beam coordination enhancement for TDD is enabled:

  A cell exits the multi-frequency beam coordination state when any of the following conditions is met throughout a decision-making period:
  - All of the following conditions are met:
    - The load of each SRS beam is less than the load threshold for beam coordination.
    - The PRB usage of each SRS beam is less than the PRB usage threshold for beam coordination.
    - The interference level of each SRS beam is less than the interference threshold for SRS beam coordination.
  - The downlink PRB usage of the cell is less than 30%.
  - The number of UEs served by each SRS beam is less than five.

The related parameters are as follows:

- The load threshold for beam coordination is specified by the **NRCellMultiFSchCoord**.***BeamCoordLoadThld*** parameter.
- The PRB usage threshold for beam coordination is specified by the **NRCellMultiFSchCoord**.***BeamCoordPrbUsageThld*** parameter.

NOTE

- In multi-frequency beam coordination, the number of UEs and downlink spectral efficiency are used to evaluate the cell load and SRS beam load.
- If the gNodeB determines in a decision-making period that a cell meets the conditions for entering or exiting the multi-frequency beam coordination state, the cell will enter or exit the state in the next decision-making period.

5.1.2 Determining Candidate Neighboring Cells and Beams

After a cell enters the multi-frequency beam coordination state, the gNodeB first filters the neighboring cells to find out candidate neighboring cells. Based on the filtering result, the gNodeB further determines the candidate neighboring cells and beams.

Filtering Candidate Neighboring Cells

After a cell enters the multi-frequency beam coordination state, the gNodeB filters the neighboring cells to find out candidate neighboring cells. A candidate neighboring cell must meet all the following conditions:

- The candidate neighboring cell and the serving cell are inter-frequency neighboring cells of each other (as specified by the **NRCellFreqRelation**.***SsbFreqPos*** parameter), and the two cells have a neighbor relationship (which is configured using the **NRCellRelation** MO).
- The multi-frequency beam coordination switch (**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter) is turned on for this candidate neighboring cell.
- Handovers to this candidate neighboring cell are allowed:
  - If the serving cell is located on an SA network, the **NRCellRelation**.***NoHoFlag*** parameter is set to **PERMIT_HO** or **SA_PERMIT_NSA_FORBID_HO** for this inter-frequency neighboring cell.
  - If the serving cell is located on an NSA network, the **NRCellRelation**.***NoHoFlag*** parameter is set to **PERMIT_HO** or **SA_FORBID_NSA_PERMIT_HO** for this inter-frequency neighboring cell.
  - If the serving cell is located on an SA-NSA hybrid network, the **NRCellRelation**.***NoHoFlag*** parameter is set to **PERMIT_HO**, **SA_PERMIT_NSA_FORBID_HO**, or **SA_FORBID_NSA_PERMIT_HO** for this inter-frequency neighboring cell.

  NOTE

  The network type (SA or NSA) of the serving cell is specified by the **gNBOperator**.***NrNetworkingOption*** parameter.
- Both the uplink PRB usage and downlink PRB usage of the candidate neighboring cell are less than or equal to 90%.
- The candidate neighboring cell and the serving cell have overlapping coverage. The specific requirements are as follows:
  - When multi-frequency beam coordination enhancement for TDD is disabled, the candidate neighboring cell is an NR TDD cell that is served by the same base station, uses the same RF module, and covers the same area as the serving cell.
  - When multi-frequency beam coordination enhancement for TDD is enabled, the candidate neighboring cell is an NR TDD cell that has overlapping coverage with the serving cell and the downlink CCE allocation failure rate of the candidate neighboring cell is less than or equal to the threshold specified by the **NRCellMultiFSchCoord**.***BeamCoorDlCceThld*** parameter.
- If the uplink-interference-based inter-frequency handover function is enabled, the candidate neighboring cell should not be in the interference state. For details about the uplink-interference-based inter-frequency handover function and how to determine the cell interference status, see *SA Mobility Management in Connected Mode*.

- When the **NRCellMultiFSchCoord**.***UlNiDiffThld*** parameter is set to a value other than **0**, the uplink NI level in the candidate neighboring cell minus that in the serving cell is less than or equal to the threshold specified by this parameter. The counter measuring the uplink NI level is **N.UL.NI.Avg**.

Determining Candidate Neighboring Cells and Beams

The conditions that candidate neighboring cells and beams must meet vary depending on whether multi-frequency beam coordination enhancement for TDD is enabled:

- When multi-frequency beam coordination enhancement for TDD is disabled, a candidate neighboring cell must meet the following condition:

  (SRS beam load of the serving cell – SRS beam load of the candidate neighboring cell)/SRS beam load of the serving cell > Load difference threshold for beam coordination
- When multi-frequency beam coordination enhancement for TDD is enabled, the conditions required for a candidate neighboring cell vary as listed in **5.1.2 Determining Candidate Neighboring Cells and Beams**.

Table 5-3 Conditions required for a candidate neighboring cell

Serving Cell State	Conditions Required for a Candidate Neighboring Cell
State 1	• (SRS beam load of the serving cell – SRS beam load of the candidate neighboring cell)/SRS beam load of the serving cell > Load difference threshold for beam coordination • PRB usage of the candidate neighboring cell < PRB usage threshold for beam coordination
State 2	• (SRS beam PRB usage of the serving cell – SRS beam PRB usage of the candidate neighboring cell)/SRS beam PRB usage of the serving cell > PRB usage difference threshold for beam coordination • Load of the candidate neighboring cell < Load threshold for beam coordination
State 3	• (SRS beam load of the serving cell – SRS beam load of the candidate neighboring cell)/SRS beam load of the serving cell > Load difference threshold for beam coordination • (SRS beam PRB usage of the serving cell – SRS beam PRB usage of the candidate neighboring cell)/SRS beam PRB usage of the serving cell > PRB usage difference threshold for beam coordination

In the preceding information:

- Details about state 1, state 2, and state 3 are described in **Table 5-2**.
- The load difference threshold for beam coordination is specified by the **NRCellMultiFSchCoord**.***BeamCoordLoadDiffThld*** parameter.

- The PRB usage difference threshold for beam coordination is specified by the **NRCellMultiFSchCoord**.***BeamCoordPrbDiffThld*** parameter.

NOTE

If an NR TDD cell served by a MetaAAU is a candidate neighboring cell, the beam load of the cell is affected by vertical beams for UEs and therefore cannot be accurately measured. If this results in an inaccurate range of candidate beams, the user experience of all UEs may be affected.

5.1.3 UE Selection

After determining candidate neighboring cells and beams, the gNodeB selects UEs on beams that meet the conditions for triggering multi-frequency beam coordination. A candidate UE must meet all the following conditions:

- The UE does not support SRS carrier switching or uplink CA. For details, see *Carrier Aggregation*.
- The UE meets the following service type requirements:
  - The UE is not a VoNR UE. For details, see *VoNR*.
  - The UE is not a super uplink UE. For details, see *Super Uplink*.
  - The UE is not a RedCap UE. For details, see *RedCap*.
  - The UE is not with heavy uplink traffic.
  - The UE is not a high-speed UE.
- The UE meets the following status requirements:
  - The UE is in the synchronization state. For details, see *5G Networking and Signaling*.
  - The UE is not in the penalty state.

    NOTE

    A UE is in the penalty state within 120 seconds after any of the following conditions is met:

    - The UE is handed over to the local cell through multi-frequency beam coordination or connected mode MLB. For details about connected mode MLB, see *Mobility Load Balancing in Connected Mode*.
    - The UE is selected by the gNodeB, but no handover is performed for the UE or the handover of the UE fails.
- The UE can be selected as a target UE of multi-frequency beam coordination if the RFSP of the UE is configured on the core network but the corresponding **gNBRfspConfig**.***RfspIndex*** parameter is not set or the **INTER_FREQ_CONNECTED_MLB_SW** option of the **gNBRfspConfig**.***MlbAlgoSwitch*** parameter corresponding to the RFSP of the UE is selected on the gNodeB side. If no RFSP is configured for the UE on the core network, the UE can also be selected as a target UE of multi-frequency beam coordination by default.
- The UE does not have any bearer of a QCI for which the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer**.***MobilityFunInd*** parameter is selected (indicating that unnecessary inter-frequency handovers are prohibited). For details, see *SA Mobility Management in Connected Mode*.

After selecting candidate UEs, the gNodeB further determines the number of UEs to transfer. Then, the gNodeB selects UEs to transfer from the candidate UEs preferentially based on a series of factors such as the downlink MCS, SRS SINR, PDCCH CCE aggregation level, and downlink traffic volume of each UE.

Number of UEs to Transfer

After the candidate neighboring cells and candidate beams are determined, the gNodeB calculates the number of UEs to transfer in the current decision-making period (the number of UEs to transfer cannot exceed 10) as follows:

- When multi-frequency beam coordination enhancement for TDD is disabled, the gNodeB calculates the number of UEs to transfer in the current decision-making period based on the SRS beam load difference between the serving cell and each candidate neighboring cell.
- When multi-frequency beam coordination enhancement for TDD is enabled:
  - If the serving cell is in state 1, the gNodeB calculates the number of UEs to transfer in the current decision-making period based on the SRS beam load difference between the serving cell and each candidate neighboring cell.
  - If the serving cell is in state 2, the gNodeB calculates the number of UEs to transfer in the current decision-making period based on the SRS beam PRB usage difference between the serving cell and each candidate neighboring cell.
  - If the serving cell is in state 3, the gNodeB calculates the number of UEs to transfer in the current decision-making period based on both the SRS beam load difference and the SRS beam PRB usage difference between the serving cell and each candidate neighboring cell.

  For details about state 1, state 2, and state 3, see **Table 5-2**.

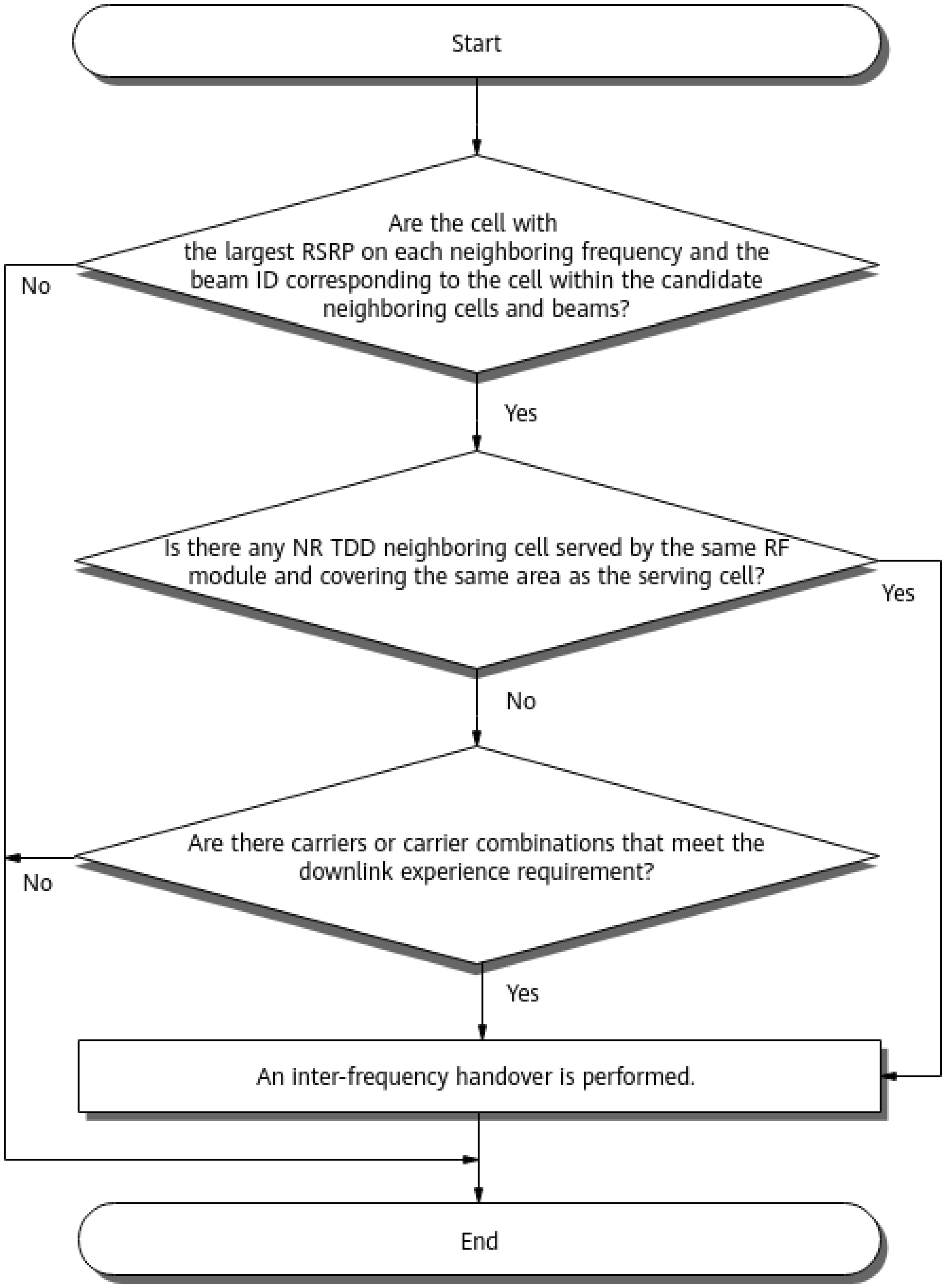
5.1.4 Execution of Multi-Frequency Beam Coordination

When multi-frequency beam coordination enhancement for TDD is disabled, the gNodeB delivers inter-frequency event A5 measurement configurations to the selected UEs. (Threshold 1 for event A5 is –43 dBm, and threshold 2 for event A5 is –140 dBm.) After a UE sends a measurement report to the gNodeB, the gNodeB checks the following information: the configured RSRP threshold 2 for event A5 related to multi-frequency smart selection, the cell with the largest RSRP value on each neighboring frequency in the measurement report, and the beam ID.

- Assume that **RSRP threshold 2 for event A5 related to multi-frequency smart selection** is set to **255**. If the RSRP of a cell is greater than the **RSRP threshold for interference-based inter-frequency handovers**, and the cell and the SRS beam corresponding to the cell are within the determined candidate neighboring cells and beams, the gNodeB instructs the UE to perform an inter-frequency handover.
- Assume that **RSRP threshold 2 for event A5 related to multi-frequency smart selection** is set to a value other than **255**. If the RSRP of a cell is greater than **RSRP threshold 2 for event A5 related to multi-frequency smart selection**, and the cell and the SRS beam corresponding to the cell are within the determined candidate neighboring cells and beams, the gNodeB instructs the UE to perform an inter-frequency handover.
- Otherwise, the procedure ends.

Figure 5-3 shows the process of executing multi-frequency beam coordination when multi-frequency beam coordination enhancement for TDD is enabled.

Figure 5-3 Process of executing multi-frequency beam coordination



1. The gNodeB obtains the RSRP of each neighboring frequency and checks whether the cell with the largest RSRP value on each neighboring frequency and the SRS beam corresponding to the cell are within the determined candidate neighboring cells and beams.
   – The gNodeB obtains the RSRP of neighboring frequencies based on inter-frequency event A5 measurements.

- If the **switch controlling prediction for multi-frequency beam coordination extension** is on, the gNodeB uses RSRP prediction models to predict RSRP of neighboring frequencies, filters out certain neighboring frequencies, delivers event A5 measurement configurations for the neighboring frequencies not filtered out, and collects measurement reports sent by UEs within 3 seconds. For details about RSRP prediction models, see *Virtual Grid-based Multi-Frequency Coordination*. The gNodeB filters out neighboring frequencies as follows:
  - If **RSRP threshold 2 for event A5 related to multi-frequency smart selection** is set to **255**, the gNodeB filters out the neighboring frequencies whose predicted RSRP is less than or equal to the **RSRP threshold for interference-based inter-frequency handovers**.
  - If **RSRP threshold 2 for event A5 related to multi-frequency smart selection** is set to a value other than **255**, the gNodeB filters out the neighboring frequencies whose predicted RSRP is less than or equal to this RSRP threshold 2.
- Otherwise, the gNodeB delivers inter-frequency event A5 measurement configurations to the selected UEs and collects measurement reports sent by the UEs within 3 seconds.

– The gNodeB checks the cell with the largest RSRP value on each neighboring frequency and the corresponding beam ID to determine how the process goes.

- The process proceeds to 2 when either of the following groups of conditions are met: (1) **RSRP threshold 2 for event A5 related to multi-frequency smart selection** is set to **255**, the RSRP of a cell is greater than the **RSRP threshold for interference-based inter-frequency handovers**, and the cell and the SRS beam corresponding to the cell are within the determined candidate neighboring cells and beams. (2) **RSRP threshold 2 for event A5 related to multi-frequency smart selection** is set to a value other than **255**, the RSRP of a cell is greater than **RSRP threshold 2 for event A5 related to multi-frequency smart selection**, and the cell and the SRS beam corresponding to the cell are within the determined candidate neighboring cells and beams.
  - If the serving cell and a candidate neighboring NR TDD cell are not intra-base-station cells that use the same RF module and have the same coverage, the gNodeB uses the intra-frequency SRS beam prediction model to predict the SRS beam ID corresponding to this neighboring cell when the following conditions are met: The intra-frequency SRS beam prediction model for this candidate neighboring cell is available, and the **switch controlling SRS beam prediction for multi-frequency beam coordination extension** is on. For details about intra-frequency SRS beam prediction models, see *Virtual Grid-based Multi-Frequency Coordination*.
  - In other cases, the gNodeB calculates the inter-frequency SRS beam ID based on the SSB beam information in the event A5

measurement reports and the weight mapping between SSB beams.

- Otherwise, the procedure ends.

NOTE

Threshold 1 and threshold 2 for event A5 are –43 dBm and –140 dBm, respectively.

2. The gNodeB checks whether there is any intra-base-station neighboring NR TDD cell served by the same RF module and covering the same area as the serving cell.
   - If there is no such an NR TDD neighboring cell, the process proceeds to **3**.
   - Otherwise, the gNodeB instructs the UE to perform an inter-frequency handover.
3. The gNodeB checks whether there are carriers or carrier combinations that meet the downlink experience requirement.
   - If a carrier or carrier combination of these cells meets the following downlink experience requirement, the gNodeB performs an inter-frequency handover for the UE:

     (Air interface capability of the candidate carrier or carrier combination – Air interface capability of the current carrier or carrier combination)/Air interface capability of the current carrier or carrier combination > **Beam Coord Experience Evaluation Hyst**
   - Otherwise, the procedure ends.

   The downlink experience evaluation method for this function is the same as that for experience-based smart carrier selection described in 4.1.1.1.5 User Experience Evaluation. When UE-specific downlink spectral efficiency needs to be considered, the only difference between the two functions is the switch to turn on.
   - For multi-frequency beam coordination, the switch to turn on is the **switch controlling downlink spectral efficiency prediction for multi-frequency beam coordination**.
   - For experience-based smart carrier selection, the switch to turn on is the **switch controlling spectral efficiency prediction for downlink experience evaluation**.

   The optimal carrier or carrier combination selection in this function is similar to that in experience-based smart carrier selection described in 4.1.1.1.6 Optimal Carrier or Carrier Combination Selection. The only difference between the two functions is the parameter that specifies the downlink air interface capability hysteresis.
   - For multi-frequency beam coordination, the downlink air interface capability hysteresis is specified by the **Beam Coord Experience Evaluation Hyst** parameter.
   - For experience-based smart carrier selection, the downlink air interface capability hysteresis is specified by the **Downlink Experience Evaluation Hysteresis** parameter.

Table 5-4 lists the parameters related to execution of multi-frequency beam coordination.

Table 5-4 Parameters related to execution of multi-frequency beam coordination

Item	Parameter
RSRP threshold 2 for event A5 related to multi-frequency smart selection	**NRCellInterFHoMeaGrp**.***MultiFreqSmtSelA5RsrpThld2***
RSRP threshold for interference-based inter-frequency handovers	**NRCellInterFHoMeaGrp**.***IntrfInterFreqHoRsrpThld***
Switch controlling prediction for multi-frequency beam coordination extension	**MF_BEAM_CO_EXT_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter
Switch controlling SRS beam prediction for multi-frequency beam coordination extension	**MF_BEAM_CO_EXT_SRSBEAM_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter
Beam Coord Experience Evaluation Hyst	**NRCellMultiFSchCoord**.***BeamCoordExpEvaluateHyst***
Switch controlling downlink spectral efficiency prediction for multi-frequency beam coordination	**MF_BEAM_CO_DL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter
Switch controlling spectral efficiency prediction for downlink experience evaluation	**DL_EXP_EVAL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter
Downlink Experience Evaluation Hysteresis	**NRCellCaMgmtConfig**.***DlExpEvalHyst***

5.1.5 Cooperation Between Multi-frequency Beam Coordination and Connected Mode MLB

Multi-frequency beam coordination and connected mode MLB can take effect at the same time only when all of the following conditions are met:

- Multi-frequency beam coordination is enabled.

- Connected mode MLB (controlled by the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter) is enabled.
- The **NRCellMlb**.***MlbTriggerMode*** parameter is set to **CELL_USER_NUM**, **SPEC_EFF_BASED_USER_NUM**, or **RADIO_RESOURCE**, or this parameter is set to **PRB_USAGE_OR_USER_NUM** and the **NRCellMlb**.***MimoLayerNumPolicy*** parameter is set to **BY_CFG**.

The decision-making period for triggering multi-frequency beam coordination and that for triggering connected mode MLB, both of which are specified by the **NRCellMlb**.***MlbJudgePeriod*** parameter, are the same. In a decision-making period:

- If the conditions for triggering multi-frequency beam coordination are met but the load conditions for triggering connected mode MLB are not met, only multi-frequency beam coordination is triggered.
- If the load conditions for triggering connected mode MLB are met but the conditions for triggering multi-frequency beam coordination are not met, only connected mode MLB is triggered.
- If the conditions for triggering multi-frequency beam coordination are met and the load conditions for triggering connected mode MLB are also met, both functions are triggered, and the gNodeB preferentially transfers the UEs served by heavy-load SRS beams of the local cell to qualified candidate neighboring cells for load redistribution. The number of UEs selected does not exceed the total number of UEs selected for MLB within the current period.

NOTE

For details about the conditions for triggering multi-frequency beam coordination, see **5.1.1 Triggering of Multi-Frequency Beam Coordination**. For details about the load conditions for triggering connected mode MLB, see *Mobility Load Balancing in Connected Mode*.

5.1.6 Cooperation Between Multi-Frequency Beam Coordination and Unnecessary Handovers

When the **inter-frequency handover coordination switch** is on, UEs cannot be transferred to high-load beams or cells through unnecessary inter-frequency handovers such as frequency-priority-based handovers. For details about unnecessary inter-frequency handovers, see *SA Mobility Management in Connected Mode*.

The conditions for determining the high-load state of a beam of an NR TDD cell are as follows:

- Assume that multi-frequency beam coordination enhancement for TDD is disabled for the cell. If the SRS beam load of the cell is greater than or equal to the **load threshold for beam coordination**, the beam is considered as a high-load beam.
- Assume that multi-frequency beam coordination enhancement for TDD is enabled for the cell. If the SRS beam load of the cell is greater than or equal to the **load threshold for beam coordination** or the SRS beam PRB usage of the cell is greater than or equal to the **PRB usage threshold for beam coordination**, the beam is considered as a high-load beam.

Assume that the **inter-frequency handover coordination switch** is on and both multi-frequency beam coordination and experience-based smart carrier selection are enabled. When a handover is triggered for experience-based smart carrier selection, the target cell will not be selected from cells in the beam coordination state based on the reported SSB beam measurement information. This prevents ping-pong handovers.

5.1.6 Cooperation Between Multi-Frequency Beam Coordination and Unnecessary Handovers lists the parameters involved in the preceding cooperation.

Table 5-5 Parameters related to cooperation between multi-frequency beam coordination and unnecessary handovers

Item	Parameter
Inter-frequency handover coordination switch	**INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter
Load threshold for beam coordination	**NRCellMultiFSchCoord**.***BeamCoordLoadThld***
PRB usage threshold for beam coordination	**NRCellMultiFSchCoord**.***BeamCoordPrbUsageThld***

5.2 Network Analysis

5.2.1 Benefits

It is recommended that multi-frequency beam coordination be enabled when all the following conditions are met:

- On a multi-carrier network, inter-frequency cells cover the same area.
- UEs are distributed mainly in several beams, and the load difference of the same beam between the co-coverage inter-frequency cells is large.
- The downlink load of the cell is heavy.

After multi-frequency beam coordination takes effect, as the change in the downlink RLC-layer traffic volume affects the average downlink UE throughput, it is recommended that the benefits be observed when the downlink RLC-layer traffic volume remains unchanged before and after the function takes effect. When the downlink RLC-layer traffic volume (indicated by **N.ThpVol.DL**) remains unchanged, the **User Downlink Average Throughput (DU)** on the entire network increases.

It is recommended that multi-frequency beam coordination be enabled for all cells in an area. Otherwise, it is possible that the maximum gains in average downlink UE throughput cannot be obtained.

5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

Multi-frequency beam coordination may have the following impacts on the network:

- UE distribution across frequency bands changes, service distribution across cells and beams also changes, and resource consumption by air interface signaling and the number of handovers increase.
- In downlink heavy-load scenarios, for example, the downlink PRB usage exceeds 70%, the total downlink RLC-layer traffic volume of all cells increases after multi-frequency beam coordination takes effect.
- UE transfer changes the uplink traffic distribution across cells, and the **User Uplink Average Throughput (DU)** slightly decreases.
- In light- or medium-load scenarios, when the uplink RLC-layer traffic volume (indicated by **N.ThpVol.UL**) or downlink RLC-layer traffic volume (indicated by **N.ThpVol.DL**) remains unchanged after this function takes effect, the MAC-layer data transmission duration decreases in relatively heavily loaded cells and increases in relatively lightly loaded cells. Due to load differences between cells, the network-level MAC-layer data transmission duration increases. As a result, the **Cell Uplink Average Throughput (DU)** or **Cell Downlink Average Throughput (DU)** decreases at the network level.

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-070201	Multi-Frequency Spatial Smart Carrier Selection	NR0S00SFBS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDSCH MU-MIMO	**DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	For an NR TDD cell, multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter) can be enabled only when PDSCH MU-MIMO is enabled.
Low-frequency TDD	DL Intra-Freq Spct Eff Mdl Build Allowed	**DL_INTRAF_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	The **MF_BEAM_CO_DL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter can be selected only after the **DL_INTRAF_SE_MODEL_BUILD_ALLOW** or **DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMulti-Carr**.***ModelBuildingSwitch*** parameter is selected. After the **MF_BEAM_CO_DL_SPCT_EFF_PRED_SW** option is selected, the gNodeB can use the prediction results of virtual grid models and considers UE-specific downlink spectral efficiency for downlink air interface capability evaluation.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL Freq Spct Eff Mdl Build Allowed	**DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	The **MF_BEAM_CO_DL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter can be selected only after the **DL_FREQ_SE_MODEL_BUILD_ALLOW** option is selected or the **DL_INTRAF_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMulti-Carr**.***ModelBuildingSwitch*** parameter is selected. After the **MF_BEAM_CO_DL_SPCT_EFF_PRED_SW** option is selected, the gNodeB can use the prediction results of virtual grid models and considers UE-specific downlink spectral efficiency for downlink air interface capability evaluation.
Low-frequency TDD	Load balancing acceleration for multi-frequency beam coordination	**MF_BEAM_COORD_LOAD_BAL_ACC_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	*Multi-Frequency Smart Selection*	Multi-frequency beam coordination enhancement for TDD (controlled by the **MF_BEAM_COORD_TDD_ENH_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter) can be enabled only when load balancing acceleration for multi-frequency beam coordination is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-frequency beam coordination	**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	*Multi-Frequency Smart Selection*	● Multi-frequency beam coordination enhancement for TDD (controlled by the **MF_BEAM_COORD_TDD_ENH_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter) can be enabled only when multi-frequency beam coordination is enabled. ● The **MF_BEAM_CO_DL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter can be selected only after multi-frequency beam coordination is enabled. After the **MF_BEAM_CO_DL_SPCT_EFF_PRED_SW** option is selected, the gNodeB can use the prediction results of virtual grid models and considers UE-specific downlink spectral efficiency for downlink air interface capability evaluation.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Measurement quantity prediction for intra-NR inter-frequency handovers	**NR_IF_HO_MEAS_QTY_PRED_FUN_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	Prediction for multi-frequency beam coordination extension (controlled by the **MF_BEAM_CO_EXT_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) can take effect only when the **NR_IF_HO_MEAS_QTY_PRED_FUN_SW** option is selected. When prediction for multi-frequency beam coordination extension takes effect, the gNodeB can use the RSRP predicted by virtual grid models for target frequency filtering.
Low-frequency TDD	Intra-Freq SRS Beam Model Build Allowed	**INTRAF_SRS_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	SRS beam prediction for multi-frequency beam coordination extension (controlled by the **MF_BEAM_CO_EXT_SRSBEAM_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) can take effect only when the **INTRAF_SRS_MODEL_BUILD_ALLOW** option is selected. When SRS beam prediction for multi-frequency beam coordination extension takes effect, the gNodeB can use virtual grid models to predict target SRS beams during candidate beam selection.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-frequency beam coordination enhancement for TDD	**MF_BEAM_COORD_TDD_ENH_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	*Multi-Frequency Smart Selection*	The **MF_BEAM_CO_EXT_SRSBEAM_PRED_SW** and **MF_BEAM_CO_EXT_PRED_SW** options of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter can be selected only when multi-frequency beam coordination enhancement for TDD is enabled.

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter) cannot be enabled for hyper cells.
Low-frequency TDD	High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag***	*High Speed Mobility*	Multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter) cannot be enabled for high-speed cells (with **NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	When Fusion Cell is enabled, multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter) cannot be enabled.

5.3.2.3 Function Impacts

- Impact relationships with CA-related functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	The gNodeB does not select downlink CA UEs whose SCells perform SRS measurement.
Low-frequency TDD	Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	The gNodeB does not select uplink CA UEs that treat the local cell as their PCell or SCell.
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	The gNodeB does not select downlink CA UEs whose SCells perform SRS measurement.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	The gNodeB does not select uplink CA UEs that treat the local cell as their PCell or SCell.

- Impact relationships with functions related to mobility management

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-number-based connected mode MLB	• **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter • **NRCellMlb**.***MlbTriggerMode*** set to **CELL_USER_NUM**	*Mobility Load Balancing in Connected Mode*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and both multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter) and UE-number-based connected mode MLB are enabled. When a handover is triggered for multi-frequency beam coordination, the target cell will not be selected from cells in the MLB state. This prevents ping-pong handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	• **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter • **NRCellMlb**.***MlbTriggerMode*** set to **SPEC_EFF_BASED_USER_NUM**	*Mobility Load Balancing in Connected Mode*	• Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and both multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter) and spectral-efficiency-based connected mode MLB are enabled. When a handover is triggered for multi-frequency beam coordination, the target cell will not be selected from cells in the MLB state. This prevents ping-pong handovers. • When both spectral-efficiency-based connected mode MLB and multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the

RAT	Function Name	Function Switch	Reference	Description
				NRDUCellFeatureSw.***MultiCarrierSwitch*** parameter) are enabled, the gains of multi-frequency beam coordination decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	5QI-specific UE-number-based connected mode MLB	• **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter • **NRCellMlb**.***MlbTriggerMode*** set to **SPECIFIED_5QI_USER_NUM**	*Mobility Load Balancing in Connected Mode*	• If multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter) and 5QI-specific UE-number-based connected mode MLB are both enabled for a cell, multi-frequency beam coordination does not take effect in the cell. • Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and both multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter) and 5QI-specific UE-number-based connected mode MLB are enabled. When a handover is

RAT	Function Name	Function Switch	Reference	Description
				triggered for multi-frequency beam coordination, the target cell will not be selected from cells in the MLB state. This prevents ping-pong handovers.
Low-frequency TDD	Radio-resource-based connected mode MLB	• **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter • **NRCellMlb**.***MlbTriggerMode*** set to **RADIO_RESOURCE**	*Mobility Load Balancing in Connected Mode*	Assume that the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected and both multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter) and radio-resource-based connected mode MLB are enabled. When a handover is triggered for multi-frequency beam coordination, the target cell will not be selected from cells in the MLB state. This prevents ping-pong handovers.

- Impact relationships with other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Super uplink	**SUPER_UL_TDM_SCH_SW** or **FLEX_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	The gNodeB does not select super uplink UEs.
Low-frequency TDD	UL and DL Decoupling	**NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch***	*UL and DL Decoupling*	The gNodeB does not select UEs for which UL and DL Decoupling has taken effect.
Low-frequency TDD	VoNR	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	The gNodeB does not select VoNR UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NSA carrier combination selection based on user experience	• NR: **NSA_LTE_SEL_OPT_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter • LTE: **NSA_LTE_FDD_SEL_OPT_SW** or **NSA_LTE_TDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter	*EPC-based NSA Experience Improvement*	If both multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter) and NSA carrier combination selection based on user experience are enabled, multi-frequency beam coordination does not take effect for NSA UEs.
Low-frequency TDD	RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	The gNodeB does not select RedCap UEs as target UEs of multi-frequency beam coordination.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink-experience-based inter-frequency handover	**UL_EXP_BASED_INTER_FREQ_HO_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter	*RedCap*	Uplink-experience-based inter-frequency handover does not take effect for RedCap UEs that are handed over to the local cell through multi-frequency beam coordination.
Low-frequency TDD	Intra-frequency MRO under cell-level MRO	• **MRO_SCENARIO_IDENT_SW** option of the **gNBMro**.***ScenarioIdentifySwitch*** parameter • **INTRA_FREQ_MRO_SW** option of the **NRCellMro**.***MroSwitch*** parameter	*MRO*	When both downlink spectral efficiency prediction for multi-frequency beam coordination and intra-frequency MRO are enabled, event A3 reporting in downlink spectral efficiency prediction for multi-frequency beam coordination is affected. As a result, the downlink experience evaluation result of the serving cell may be different from the expected result, which may cause the average downlink UE throughput to decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-level MRO	**UE_MRO_SWITCH** option of the **NRCellMro**.***MroSwitch*** parameter	*MRO*	When both downlink spectral efficiency prediction for multi-frequency beam coordination and UE-level MRO are enabled, event A3 reporting in downlink spectral efficiency prediction for multi-frequency beam coordination is affected for UEs involved in ping-pong handovers and downlink spectral efficiency prediction for multi-frequency beam coordination. As a result, the downlink experience evaluation result of the serving cell may be different from the expected result, which may cause the average downlink UE throughput to decrease.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable 32T32R/64T64R RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Networking

- The network uses at least two frequencies.
- The Xn interface between gNodeBs works normally.

5.3.5 Cells

The **NRDUCellTrp**.***TxRxMode*** parameter must be set to **32T32R** or **64T64R** for NR TDD cells engaged in multi-frequency beam coordination.

5.3.6 Others

The benefits of multi-frequency beam coordination strongly depend on how accurately the load difference and PRB usage difference for beam coordination are evaluated, whereas the evaluation is complex. To obtain the optimal benefits, contact Huawei professional optimization service engineers.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-6 describes the parameters used for function activation. For details about other related parameters, see **Table 5-7**. This section does not describe parameters related to cell establishment.

Table 5-6 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Multi-Carrier Switch	**NRDUCellFeatureSw**.***MultiCarrierSwitch***	MULTI_FREQ_BEAM_COORD_SW	Select this option if multi-frequency beam coordination is required.
Multi-Carrier Switch	**NRDUCellFeatureSw**.***MultiCarrierSwitch***	MF_BEAM_COORD_TDD_ENH_SW	Select this option if multi-frequency beam coordination enhancement for TDD is required.
MLB Judge Period	**NRCellMlb**.***MlbJudgePeriod***	N/A	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
Beam Coordination Load Threshold	**NRCellMultiFSchCoord**.***BeamCoordLoadThld***	N/A	Use the recommended value.
Beam Coordination Load Offset	**NRCellMultiFSchCoord**.***BeamCoordLoadOffset***	N/A	Use the recommended value.
Beam Coordination Load Difference Thld	**NRCellMultiFSchCoord**.***BeamCoordLoadDiffThld***	N/A	Use the recommended value.
Beam Coordination PRB Usage Threshold	**NRCellMultiFSchCoord**.***BeamCoordPrbUsageThld***	N/A	Use the recommended value.
Beam Coordination PRB Usage Diff Thld	**NRCellMultiFSchCoord**.***BeamCoordPrbDiffThld***	N/A	Use the recommended value.
Beam Coordination DL CCE Threshold	**NRCellMultiFSchCoord**.***BeamCoorDlCceThld***	N/A	Use the recommended value.
Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	MF_BEAM_CO_DL_SPCT_EFF_PRED_SW	You are advised to select this option.
Beam Coord Experience Evaluation Hyst	**NRCellMultiFSchCoord**.***BeamCoordExpEvaluateHyst***	N/A	Use the recommended value.
Intrf-based Inter-Freq HO RSRP Thld	**NRCellInterFHoMeaGrp**.***IntrfInterFreqHoRsrpThld***	N/A	Use the recommended value.
Multi-Freq Smart Sel A5 RSRP Thld2	**NRCellInterFHoMeaGrp**.***MultiFreqSmtSelA5RsrpThld2***	N/A	Set this parameter based on operators' requirements.

Parameter Name	Parameter ID	Option	Setting Notes
Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	MF_BEAM_CO_EXT_PRED_SW	Set this option based on operators' requirements. Select this option if the RSRP predicted by virtual grids needs to be used for target frequency filtering during the execution phase of multi-frequency beam coordination.
Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	MF_BEAM_CO_EXT_SRSBEAM_PRED_SW	Set this option based on operators' requirements. Select this option if virtual grids need to be used to predict target SRS beams during candidate beam selection.
Multi-Carrier Switch	**NRDUCellFeatureSw**.***MultiCarrierSwitch***	MF_BEAM_COORD_LOAD_BAL_ACC_SW	You are advised to select this option.
UL NI Difference Threshold	**NRCellMultiFSchCoord**.***UlNiDiffThld***	N/A	Use the recommended value.

Table 5-7 Other related parameters

Parameter Name	Parameter ID	Option	Setting Notes
SSB Frequency Position	**NRCellFreqRelation**.***SsbFreqPos***	N/A	Set this parameter based on operators' requirements.
No Handover Flag	**NRCellRelation**.***NoHoFlag***	N/A	Use the recommended value.
Index to RAT/Frequency Selection Priority	**gNBRfspConfig**.***RfspIndex***	N/A	Set this parameter based on operators' requirements.
MLB Algorithm Switch	**gNBRfspConfig**.***MlbAlgoSwitch***	INTER_FREQ_CONNECTED_MLB_SW	Set this option based on operators' requirements.

Parameter Name	Parameter ID	Option	Setting Notes
Mobility Function Indication	**gNBQciBearer**.***MobilityFunInd***	INTER_FREQ_HO_SUPPRESS_SW	Set this option based on operators' requirements.
Inter-Frequency Handover Switch	**NRCellAlgoSwitch**.***InterFreqHoSwitch***	INTER_FREQ_HO_COLLABORATION_SW	Use the recommended value.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Configurations need to be completed for both the serving cell and neighboring cells.

There are only intra-base-station NR TDD cells using the same RF module and covering the same area.

```
//Selecting the MULTI_FREQ_BEAM_COORD_SW option
MOD NRDUCELLFEATURESW: NrDuCellId=0, MultiCarrierSwitch=MULTI_FREQ_BEAM_COORD_SW-1;
//Setting the decision-making period for multi-frequency beam coordination for the cell
MOD NRCELLMLB: NrCellId=0, MlbJudgePeriod=10;
//Setting the load threshold for beam coordination
MOD NRCELLMULTIFSCHCOORD: NrCellId=0, BeamCoordLoadThld=40;
//Setting the load offset for beam coordination
MOD NRCELLMULTIFSCHCOORD: NrCellId=0, BeamCoordLoadOffset=15;
//Setting the load difference threshold for beam coordination
MOD NRCELLMULTIFSCHCOORD: NrCellId=0, BeamCoordLoadDiffThld=15;
//Setting RSRP threshold 2 for event A5 related to multi-frequency smart selection to, for example, –104
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
MultiFreqSmtSelA5RsrpThld2=-104;
//(Required only when RSRP threshold 2 for event A5 related to multi-frequency smart selection is set to
255) Setting the RSRP threshold for interference-based inter-frequency handovers
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, IntrfInterFreqHoRsrpThld=-104;
//Adding an NR cell frequency relationship
ADD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=8000, FrequencyBand=N77;
//Configuring permission of beam coordination with the neighboring cell
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1, NoHoFlag=PERMIT_HO;
//Disabling beam coordination for UEs with a specific RFSP ID based on the network plan if this RFSP ID is
configured on the core network
ADD GNBRFSPCONFIG: OperatorId=1, RfspIndex=2, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-0;
```



```
//Specifying whether to enable suppression of unnecessary inter-frequency handovers of UEs with bearers
of specific QCIs based on the network plan (In the example, the suppression is enabled.)
MOD GNBQCIBEARER: Qci=1, MobilityFunInd=INTER_FREQ_HO_SUPPRESS_SW-1;
//Selecting the INTER_FREQ_HO_COLLABORATION_SW option
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=INTER_FREQ_HO_COLLABORATION_SW-1;
//Setting the uplink NI difference threshold
MOD NRCELLMULTIFSCHCOORD: NrCellId=0, UlNiDiffThld=0;
```

There are multiple NR TDD cells that have overlapping coverage.

```
//Selecting the MULTI_FREQ_BEAM_COORD_SW option
MOD NRDUCELLFEATURESW: NrDuCellId=0, MultiCarrierSwitch=MULTI_FREQ_BEAM_COORD_SW-1;
//Selecting the MF_BEAM_COORD_LOAD_BAL_ACC_SW option
 MOD NRDUCELLFEATURESW: NrDuCellId=0, MultiCarrierSwitch=
MF_BEAM_COORD_LOAD_BAL_ACC_SW-1;
//Selecting the MF_BEAM_COORD_TDD_ENH_SW option
MOD NRDUCELLFEATURESW: NrDuCellId=0, MultiCarrierSwitch=MF_BEAM_COORD_TDD_ENH_SW-1;
//Setting the decision-making period for multi-frequency beam coordination for the cell
MOD NRCELLMLB: NrCellId=0, MlbJudgePeriod=10;
//Setting the load threshold for beam coordination
MOD NRCELLMULTIFSCHCOORD: NrCellId=0, BeamCoordLoadThld=40;
//Setting the load offset for beam coordination
MOD NRCELLMULTIFSCHCOORD: NrCellId=0, BeamCoordLoadOffset=15;
//Setting the load difference threshold for beam coordination
MOD NRCELLMULTIFSCHCOORD: NrCellId=0, BeamCoordLoadDiffThld=15;
//Setting the PRB usage threshold for beam coordination
MOD NRCELLMULTIFSCHCOORD: NrCellId=0, BeamCoordPrbUsageThld=20;
//Setting the PRB usage difference threshold for beam coordination
MOD NRCELLMULTIFSCHCOORD: NrCellId=0, BeamCoordPrbDiffThld=15;
//Setting the CCE allocation failure rate threshold
MOD NRCELLMULTIFSCHCOORD: NrCellId=0, BeamCoorDlCceThld=8;
//(Recommended when the DL_INTRAF_SE_MODEL_BUILD_ALLOW option is selected) Selecting the
MF_BEAM_CO_DL_SPCT_EFF_PRED_SW option to improve the spectral efficiency prediction accuracy
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=MF_BEAM_CO_DL_SPCT_EFF_PRED_SW-1;
//Setting the hysteresis for user experience evaluation in multi-frequency beam coordination
MOD NRCELLMULTIFSCHCOORD: NrCellId=0, BeamCoordExpEvaluateHyst=-20;
//Selecting the MF_BEAM_CO_EXT_PRED_SW option if the RSRP predicted by virtual grid models needs to
be used for target frequency filtering during the execution phase of multi-frequency beam coordination
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=MF_BEAM_CO_EXT_PRED_SW-1;
//Selecting the MF_BEAM_CO_EXT_SRSBEAM_PRED_SW option for NR TDD cells served by different AAUs if
virtual grid models need to be used to predict target SRS beams during candidate beam selection
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=MF_BEAM_CO_EXT_SRSBEAM_PRED_SW-1;
//Setting RSRP threshold 2 for event A5 related to multi-frequency smart selection to, for example, –104
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
MultiFreqSmtSelA5RsrpThld2=-104;
//(Required only when RSRP threshold 2 for event A5 related to multi-frequency smart selection is set to
255) Setting the RSRP threshold for interference-based inter-frequency handovers
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, IntrfInterFreqHoRsrpThld=-104;
//Adding an NR cell frequency relationship
ADD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=8000, FrequencyBand=N77;
//Configuring permission of beam coordination with the neighboring cell
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1, NoHoFlag=PERMIT_HO;
//Disabling beam coordination for UEs with a specific RFSP ID based on the network plan if this RFSP ID is
configured on the core network
ADD GNBRFSPCONFIG: OperatorId=1, RfspIndex=2, MlbAlgoSwitch=INTER_FREQ_CONNECTED_MLB_SW-0;
//Specifying whether to enable suppression of unnecessary inter-frequency handovers of UEs with bearers
of specific QCIs based on the network plan (In the example, the suppression is enabled.)
MOD GNBQCIBEARER: Qci=1, MobilityFunInd=INTER_FREQ_HO_SUPPRESS_SW-1;
//Selecting the INTER_FREQ_HO_COLLABORATION_SW option
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=INTER_FREQ_HO_COLLABORATION_SW-1;
//Setting the uplink NI difference threshold
MOD NRCELLMULTIFSCHCOORD: NrCellId=0, UlNiDiffThld=0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

There are only intra-base-station NR TDD cells using the same RF module and covering the same area.

```
//Deselecting the MULTI_FREQ_BEAM_COORD_SW option
MOD NRDUCELLFEATURESW: NrDuCellId=0, MultiCarrierSwitch=MULTI_FREQ_BEAM_COORD_SW-0;
```

There are multiple NR TDD cells that have overlapping coverage.

```
//Deselecting the MF_BEAM_COORD_TDD_ENH_SW option
MOD NRDUCELLFEATURESW: NrDuCellId=0, MultiCarrierSwitch=MF_BEAM_COORD_TDD_ENH_SW-0;
//Deselecting the MULTI_FREQ_BEAM_COORD_SW option
MOD NRDUCELLFEATURESW: NrDuCellId=0, MultiCarrierSwitch=MULTI_FREQ_BEAM_COORD_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

If the value of any of the counters listed in **Table 5-8** is not 0, multi-frequency beam coordination has taken effect.

Table 5-8 Counters related to multi-frequency beam coordination

Counter ID	Counter Name
1911837901	**N.HO.InterFreq.BeamCoordination.ExecSuccOut**
1911837904	**N.HO.InterFreq.BeamCoordination.PrepAttOut**
1911837905	**N.HO.InterFreq.BeamCoordination.ExecAttOut**
1911837903	**N.NsaDc.InterFreq.BeamCoordination.PSCell.Change.Succ**
1911837902	**N.NsaDc.InterFreq.BeamCoordination.PSCell.Change.Att**

5.4.3 Network Monitoring

Observe the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) to monitor the rate increase.

6 Macro-Micro Smart Carrier Selection

6.1 Principles

6.1.1 Macro-Micro Intra-Frequency Optimal Carrier Selection

When a UE moves to an overlapping area between outdoor and indoor cells respectively served by macro and micro base stations deployed on the same frequency, the UE may not obtain the optimal downlink experience after selecting the cell with the largest RSRP to camp on. This is because the macro and micro base stations differ greatly in antenna gains. To improve user experience in macro-micro overlapping areas, the macro-micro intra-frequency optimal carrier selection function is introduced.

The macro-micro intra-frequency optimal carrier selection function is controlled by the **INTRA_FREQ_COORD_HO_OPT_SW** option of the **NRCellAlgoSwitch**.***IntraFreqHoAlgoSw*** parameter. For which types of UEs macro-micro intra-frequency optimal carrier selection can take effect is controlled by the **NRCellAlgoSwitch**.***IntraFreqCoordHoMode*** parameter as follows:

- If this parameter is set to **NSA_SA_MODE**, macro-micro intra-frequency optimal carrier selection can take effect for both NSA UEs and SA UEs.
- If this parameter is set to **NSA_ONLY_MODE**, macro-micro intra-frequency optimal carrier selection can take effect only for NSA UEs.
- If this parameter is set to **SA_ONLY_MODE**, macro-micro intra-frequency optimal carrier selection can take effect only for SA UEs.

NOTE

Macro and micro base stations are distinguished based on the following numbers of antennas:

- Macro base stations: 32T32R or 64T64R
- Micro base stations 1T1R, 2T2R, or 4T4R

Other numbers of antennas (such as 1T2R, 2T4R, and 8T8R) cannot be used to distinguish macro and micro base stations, and macro-micro intra-frequency optimal carrier selection is not applicable to base stations with these other numbers of antennas.

Intelligent macro-micro collaboration, a function corresponding to macro-micro intra-frequency optimal carrier selection, is available on the MAE-Optimization.

The intelligent macro-micro coordination function can generate recommended settings for parameters (listed in **Table 6-3**) related to macro-micro intra-frequency optimal carrier selection based on indicators such as the number of handovers and average UE rate. The recommended parameter settings can be automatically or manually delivered to base stations. This function performs the following actions:

- Basic KPI monitoring and assurance
- Automatic feature activation and parameter optimization

NOTE

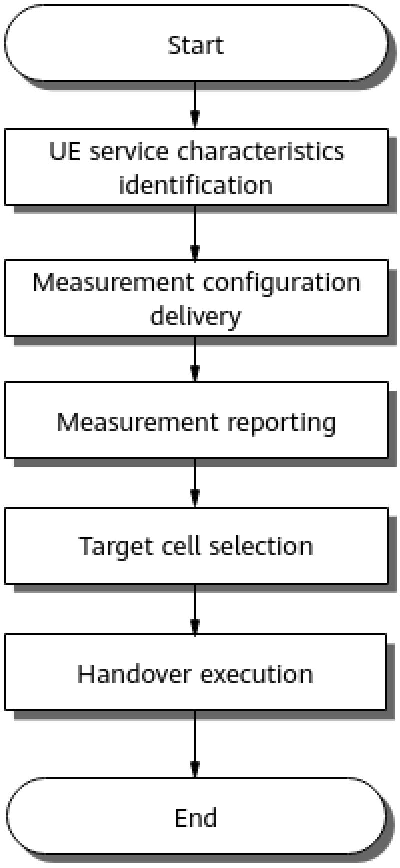
For details about the intelligent macro-micro coordination function on the MAE-Optimization, contact Huawei technical support.

For a fast track through the principles of macro-micro intra-frequency optimal carrier selection, see **(Video)Macro-micro intra-frequency optimal carrier selection**.

6.1.1.1 Basic Process

Figure 6-1 shows the basic process of macro-micro intra-frequency optimal carrier selection.

Figure 6-1 Basic process of macro-micro intra-frequency optimal carrier selection



1. The method for identifying UE service characteristics is described in **UE Service Characteristics Identification**.
2. Measurement configuration delivery by the gNodeB to UEs is described in **Measurement Configuration Delivery**.
3. Measurement reporting from UEs to the gNodeB is described in **Measurement Reporting**.
4. How the gNodeB selects the target cell for a handover is described in **Target Cell Selection**.
5. Handover execution instruction by the gNodeB is described in **Handover Execution**.

UE Service Characteristics Identification

Service characteristics vary with UEs. For a given UE, a carrier or carrier combination may provide the optimal experience in only one direction, uplink or downlink. As such, the service characteristics of UEs need to be identified. Service characteristics identification can be performed for a UE that meets all of the following conditions:

- The UE is not running an emergency service.
- The **gNBRfspConfig**.***NetworkingOptionOptFlag*** parameter corresponding to the RFSP index of the UE is set to **TRUE**.
- The UE has at least one bearer with a QCI for which the **gNBQciBearer**.***NetworkingOptionOptFlag*** parameter is set to **PERMIT**.

- The UE does not have any bearer with a QCI for which the **gNBQciBearer**.***NetworkingOptionOptFlag*** parameter is set to **FORBID**.

For a UE meeting the preceding conditions, the gNodeB identifies its service characteristics based on its uplink and downlink traffic volumes, to find out whether it is a downlink-preferred UE.

- If the uplink traffic volume of the UE is small and the downlink traffic volume of the UE is large, the UE is a downlink-preferred UE.
- If both the uplink and downlink traffic volumes of the UE are large:
  - If the downlink traffic volume is at least 30 times the uplink traffic volume, the downlink services have an absolute priority and the UE is a downlink-preferred UE.
  - If the downlink traffic volume is less than 30 times the uplink traffic volume, the uplink traffic volume is less than 30 times the downlink traffic volume, and the **NRCellCaMgmtConfig**.***CaBandCombSelectionStrat*** parameter is set to **BANDWIDTH_FIRST** or **PEAK_RATE_FIRST**, the UE is a downlink-preferred UE.
- In other cases, the UE is not a downlink-preferred UE.

NOTE

- Immediately after a UE initially accesses a cell or is admitted to a cell through an incoming handover, an incoming RRC connection reestablishment, or incoming RRC connection resumption (transition from the RRC inactive state to the RRC connected state), both the uplink and downlink traffic volumes of the UE are considered small by default. Therefore, macro-micro intra-frequency optimal carrier selection does not take effect for the UE.
- In the entire process of macro-micro intra-frequency optimal carrier selection for a UE, the UE must be always identified as a downlink-preferred UE. If the UE is identified as a non-downlink-preferred UE in a certain phase, the macro-micro intra-frequency optimal carrier selection process will be terminated for the UE.

Measurement Configuration Delivery

- If the **HetNet Coordination Cell Flag** is set to **NOT_CONFIG** for the local cell, the procedure ends.
- The gNodeB delivers event A3 configurations related to intra-frequency handovers and event A3 configurations related to macro-micro optimal carrier selection if all the following conditions are met: (1) The **HetNet Coordination Cell Flag** is set to a value other than **NOT_CONFIG** for the local cell. (2) The UE is identified as a downlink-preferred UE. (3) The **HetNet Coordination Cell Flag** is set to a value other than **NOT_CONFIG** for at least one of the neighboring cells of the local cell. (4) The setting of the **HetNet Coordination Cell Flag** for the neighboring cell is different from that for the local cell. Otherwise, the procedure ends.
  - The A3 offset included in the delivered event A3 configurations related to intra-frequency handovers is the **A3 offset for intra-frequency handovers**.
  - The A3 offset included in the delivered event A3 configurations related to macro-micro optimal carrier selection varies depending on whether **spectral-efficiency-based intra-frequency handover for macro-micro coordination** is enabled.

- If it is enabled: When the downlink intra-frequency spectral efficiency prediction model or downlink-frequency spectral efficiency model of the serving cell is usable, the A3 offset is the **Intra-freq HO Coverage Extension A3 Offset**; otherwise, the A3 offset is the **HetNet Intra-Frequency A3 Offset**. For details about the building of downlink intra-frequency spectral efficiency prediction models, see *Virtual Grid-based Multi-Frequency Coordination*.
- If it is disabled, the A3 offset is the **HetNet Intra-Frequency A3 Offset**.

Table 6-1 lists the parameters involved in the preceding measurement configuration delivery.

Table 6-1 Parameters related to measurement configuration delivery

Item	Parameter
HetNet Coordination Cell Flag	Local cell: **NRCellMultiFResCoord**.***HetNetCoordCellFlag***
	Neighboring cell: **NRCellRelation**.***HetNetCoordCellFlag***
A3 offset for intra-frequency handovers	**NRCellIntraFHoMeaGrp**.***IntraFreqHoA3Offset***
Spectral-efficiency-based intra-frequency handover for macro-micro coordination	**SPC_EFF_INTRA_FREQ_COORD_HO_SW** option of the **NRCellAlgoSwitch**.***IntraFreqHoAlgoSw*** parameter
Intra-freq HO Coverage Extension A3 Offset	**NRCellIntraFHoMeaGrp**.***IntraFreqHoCovExtendA3Ofs***
HetNet Intra-Frequency A3 Offset	**NRCellIntraFHoMeaGrp**.***HetNetIntraFreqA3Offset***

Measurement Reporting

The UE performs measurements based on the event configuration delivered to it and reports measurement results.

Target Cell Selection

After receiving an event A3 measurement report, the gNodeB determines the target cell. For details, see "Target Cell or Frequency Decision" under the section "Basic Functions of Mobility Management in Connected Mode" in *SA Mobility Management in Connected Mode*.

After the target cell is determined, if the **HetNet DL PRB Usage Threshold** is set to a value other than **100** and the PRB usage of the serving cell and that of the target cell are greater than this threshold, the procedure ends. Otherwise, the process goes as follows:

- If the A3 offset for intra-frequency handovers is less than that for macro-micro optimal carrier selection:
  - If the **HetNet Coordination Cell Flag** parameters are set to different values other than **NOT_CONFIG** for the serving cell and the target cell:
    - If the RSRP of the serving cell is less than the **coverage-based event A2 threshold** or the UE is identified as a UE under weak uplink coverage (for details about the identification conditions, see **4.2.1.1 Decision on Whether to Initiate an Uplink-Coverage-based Inter-Frequency Handover**), the base station performs a coverage-based intra-frequency handover.
    - Otherwise, the base station continues to wait for event A3 measurement reports for macro-micro optimal carrier selection. After receiving an event A3 measurement report for macro-micro optimal carrier selection, the base station performs operations as follows:
      - If the A3 offset of the measurement event corresponding to the measurement report is the **Intra-freq HO Coverage Extension A3 Offset**, a handover decision is made as described in **Handover Decision Method 1**.
      - If the A3 offset of the measurement event corresponding to the measurement report is the **HetNet Intra-Frequency A3 Offset**, the base station directly initiates a request for an intra-frequency handover to the target cell.
  - If the **HetNet Coordination Cell Flag** parameters are set to the same value for the serving cell and the target cell or to a value other than **NOT_CONFIG** for either the serving cell or the target cell, the base station performs a coverage-based intra-frequency handover.
- If the A3 offset for intra-frequency handovers is greater than that for macro-micro optimal carrier selection:
  - If the RSRP of the target cell is less than **coverage-based event A5 threshold 2**, the base station continues to wait for event A3 measurement reports for intra-frequency handovers.
  - Otherwise:
    - If the offset of the measurement event corresponding to the measurement report is the **Intra-freq HO Coverage Extension A3 Offset**, a handover decision is made as described in **Handover Decision Method 2**.
    - If the offset of the measurement event corresponding to the measurement report is the **HetNet Intra-Frequency A3 Offset**, the base station directly initiates a request for an intra-frequency handover to the target cell.
- If the A3 offset for intra-frequency handovers is equal to that for macro-micro optimal carrier selection:
  - If the **HetNet Intra-Frequency A3 Offset** is used as the A3 offset for macro-micro optimal carrier selection:
    - If the A3 offset for intra-frequency handovers is less than the **Intra-freq HO Coverage Extension A3 Offset**, the gNodeB performs operations as follows after receiving an event A3 measurement report for intra-frequency handovers:

If the **HetNet Coordination Cell Flag** parameters are set to different values other than **NOT_CONFIG** for the serving cell and the target cell, a handover decision is made as described in **Handover Decision Method 1**.

If the **HetNet Coordination Cell Flag** parameters are set to the same value for the serving cell and the target cell or to a value other than **NOT_CONFIG** for either the serving cell or the target cell, the base station performs a coverage-based intra-frequency handover.

- If the A3 offset for intra-frequency handovers is greater than the **Intra-freq HO Coverage Extension A3 Offset**, the gNodeB performs differently as follows after receiving an event A3 measurement report for macro-micro optimal carrier selection: If the RSRP of the target cell is less than coverage-based event A5 threshold 2, the gNodeB continues to wait for event A3 measurement reports for intra-frequency handovers; otherwise, a handover decision is made as described in **Handover Decision Method 2**.
- If the A3 offset for intra-frequency handovers is equal to the **Intra-freq HO Coverage Extension A3 Offset**, the base station performs a coverage-based intra-frequency handover.

– If the **Intra-freq HO Coverage Extension A3 Offset** is used as the A3 offset for macro-micro optimal carrier selection, the base station performs a coverage-based intra-frequency handover.

Table 6-2 lists the parameters involved in the target cell selection procedure.

Table 6-2 Parameters related to target cell selection

Item	Parameter
HetNet DL PRB Usage Threshold	**NRCellMobilityConfig**.***HetNetDlPrbUsageThld***
Coverage-based event A2 threshold	For super uplink UEs: **NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrpThld*** + **gNBOperatorQciParam**.***InterFreqA2RsrpThldOfs*** + **NRCellMeasConfig**.***SulCovHoRsrpThldOffset*** – **NRCellInterFHoMeaGrp**.***InterFreqA1A2Hyst***
	For non-super-uplink UEs: **NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrpThld*** + **gNBOperatorQciParam**.***InterFreqA2RsrpThldOfs*** – **NRCellInterFHoMeaGrp**.***InterFreqA1A2Hyst***
Coverage-based event A5 threshold 2	**NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld2*** + **gNBOperatorQciParam**.***InterFreqA5RsrpThld2Ofs*** – **NRCellInterFHoMeaGrp**.***InterFreqA4A5Hyst*** – **NRCellRelation**.***CellIndividualOffset***
For details about other parameters involved in target cell selection, see **Table 6-1**.	

NOTE

For details about super uplink UEs, see *Super Uplink*.

Handover Execution

The gNodeB instructs the UE to perform a handover.

6.1.1.2 Handover Decision Methods

Handover Decision Method 1

The base station checks whether the virtual grid models (downlink intra-frequency spectral efficiency prediction models or downlink-frequency spectral efficiency models) of the serving cell and target cell are usable.

- If the virtual grid model of one cell is unusable, the base station initiates a request for an intra-frequency handover to the target cell.
- If the virtual grid models of both cells are usable, the base station further checks whether the switch for spectral-efficiency-based handover of UEs in extension coverage areas (specified by the **EXTEND_COV_SPCT_EFF_HO_SW** option of the **NRCellAlgoSwitch**.***IntraFreqHoAlgoSw*** parameter) is turned on.
  - If it is off, the base station initiates a request for an intra-frequency handover to the target cell.
  - If it is on, the base station further checks the downlink air interface capability.
    - If the air interface capability of the serving cell is greater than that of the target cell, the base station continues to wait for event A3 measurement reports for macro-micro optimal carrier selection.
    - Otherwise, the base station initiates a request for an intra-frequency handover to the target cell.

Handover Decision Method 2

The base station checks whether the virtual grid models (downlink intra-frequency spectral efficiency prediction models or downlink-frequency spectral efficiency models) of the serving cell and target cell are usable.

- If the virtual grid model of one cell is unusable, the base station initiates a request for an intra-frequency handover to the target cell.
- If the virtual grid models of both cells are usable, the base station further checks the downlink air interface capability.
  - If the downlink air interface capability of the target cell is greater than that of the serving cell, the base station initiates a request for an intra-frequency handover to the target cell.
  - Otherwise, an intra-frequency handover will not be initiated.

NOTE

- For details about virtual grid models, see *Virtual Grid-based Multi-Frequency Coordination*.
- For details about how the base station evaluates the downlink air interface capability, see descriptions of downlink air interface capability in **4.1.1.1.5 User Experience Evaluation**.
- The downlink air interface capability evaluation is related to the hysteresis for HetNet downlink experience evaluation. If the downlink air interface capability of the serving cell exceeds that of a neighboring cell by an amount greater than the hysteresis, the downlink air interface capability of the serving cell is considered to be higher than that of the neighboring cell. Otherwise, the air interface capabilities of the two cells are considered to be the same. The hysteresis for HetNet downlink experience evaluation is specified by the **NRCellCaMgmtConfig**.***HetNetDlExpEvalHyst*** parameter.
- To prevent macro-micro intra-frequency optimal carrier selection from causing ping-pong handovers, measures against ping-pong handovers are added. For details, see **6.1.1.3 Measures Against Ping-Pong Handovers**.

6.1.1.3 Measures Against Ping-Pong Handovers

For SA UEs, a 15-second timer can be used to prevent ping-pong handovers. This timer-based measure is used before an intra-frequency handover is performed.

- Assume that the target cell is the cell that the UE last camped on. If the 15-second timer for this target cell has not expired, an intra-frequency handover to this target cell will not be initiated. If the 15-second timer for this target cell has expired, an intra-frequency handover to this target cell will be directly initiated.
- Assume that the target cell is not the cell that the UE last camped on. An intra-frequency handover to this target cell will be directly initiated.

For NSA UEs, the 15-second timer used to prevent ping-pong handovers is not supported.

For all UEs, the basic anti-ping-pong function depends on the A3 offset setting and hysteresis parameter setting for the macro-micro intra-frequency optimal carrier selection function.

- If the macro-micro intra-frequency optimal carrier selection function uses the A3 offset for coverage extension for intra-frequency handover (specified by the **NRCellIntraFHoMeaGrp**.***IntraFreqHoCovExtendA3Ofs*** parameter), the following measures against ping-pong handovers can be used:
  - When the A3 offset for coverage extension for intra-frequency handover is set for the serving cell and target cell, ensure that the sum of the offset values for the serving cell and target cell is greater than or equal to 0. This prevents ping-pong handovers at the edge of the macro-micro overlapping area.
  - During user experience evaluation, the **NRCellCaMgmtConfig**.***HetNetDlExpEvalHyst*** parameter can prevent ping-pong handovers caused by rate fluctuation that occurs after previous handovers.
- If the macro-micro intra-frequency optimal carrier selection function uses the HetNet intra-frequency A3 offset (specified by the **NRCellIntraFHoMeaGrp**.***HetNetIntraFreqA3Offset*** parameter), the following measure against ping-pong handovers can be used:

When the HetNet intra-frequency A3 offset is set for the serving cell and target cell, ensure that the sum of the offset values for the serving cell and target cell is greater than or equal to 0. This prevents ping-pong handovers at the edge of the macro-micro overlapping area.

6.2 Network Analysis

6.2.1 Benefits

The macro-micro intra-frequency optimal carrier selection function is applicable to macro-micro intra-frequency HetNets. It is recommended that this function be enabled in light-load scenarios where the rates of UEs in the overlapping area between macro and micro cells differ greatly. After this function is enabled, the downlink throughput of UEs in macro-micro intra-frequency overlapping areas increases.

Assume that the macro-micro intra-frequency optimal carrier selection function is in effect based on intra-frequency spectral efficiency grid models, which are dynamically updated with the traffic model change on the live network. If the accuracy of the models becomes lower than expected due to the traffic model change on the live network, the gNodeB considers that the models have aged. Before new intra-frequency spectral efficiency grid models are launched, the maximum gains in the average downlink UE throughput may not be obtained in macro-micro overlapping areas.

6.2.2 Impacts

The impacts between this function and other functions are described in **6.3.2.3 Function Impacts**.

The macro-micro intra-frequency optimal carrier selection function may have the following impacts on the network:

- The distribution of UEs in macro and micro cells changes. When the rates of UEs served by the macro cell are higher than those of UEs served by the micro cell in the macro-micro overlapping area, the **Downlink Traffic Volume (DU)** of the macro base station increases and that of the micro base station decreases. When the rates of UEs served by the macro cell are lower than those of UEs served by the micro cell in the macro-micro overlapping area, the **Downlink Traffic Volume (DU)** of the macro base station decreases and that of the micro base station increases.
- When the coverage in the macro-micro overlapping area is poor, the **Intra-Frequency Handover Out Success Rate (CU)** and **Service Call Drop Rate (CU, Inactive)** fluctuate.
- The number of A3 measurement reports increases, which increases the number of air interface signaling messages and signaling radio bearer (SRB) resource consumption.
- When intra-frequency spectral efficiency grid models are used by the macro-micro intra-frequency optimal carrier selection function and the accuracy of the models is low, the downlink traffic volume of UEs in macro-micro overlapping areas (indicated by **N.Volume.DL.HetNetOverlap**) and the downlink user-perceived rates of these UEs are lower than when the models

are not used by the macro-micro intra-frequency optimal carrier selection function.

6.3 Requirements

6.3.1 Licenses

The macro-micro intra-frequency optimal carrier selection function requires the following license.

Feature ID	Feature Name	Model	Sales Unit
FOFD-071901	Macro-Micro Smart Carrier Selection	NR0S0MMSCS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coverage-based intra-frequency handover	**COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***IntraFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	The macro-micro intra-frequency optimal carrier selection function works on the basis of coverage-based intra-frequency handover to find cells capable of providing better user experience for downlink-preferred UEs. Therefore, this function can take effect only after coverage-based intra-frequency handover is enabled. If macro-micro intra-frequency optimal carrier selection is enabled and coverage-based intra-frequency handover is disabled when a UE accesses the network and coverage-based intra-frequency handover is enabled afterwards, macro-micro intra-frequency optimal carrier selection does not take effect for the UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL Intra-Freq Spct Eff Mdl Build Allowed	**DL_INTRAF_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	The **MF_BEAM_CO_DL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter can be selected only when the **DL_INTRAF_SE_MODEL_BUILD_ALLOW** or **DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMulti-Carr**.***ModelBuildingSwitch*** parameter is selected.
Low-frequency TDD	DL Freq Spct Eff Mdl Build Allowed	**DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	The **MF_BEAM_CO_DL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter can be selected only when the **DL_INTRAF_SE_MODEL_BUILD_ALLOW** or **DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMulti-Carr**.***ModelBuildingSwitch*** parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL Intra-Freq Spectral Eff Predict Sw	**DL_INTRA_FREQ_SPCT_EFF_PRED_SW** option of the **NRCellSmartMultiCarr**.***NrMultiCarrierAlgoSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	A gNodeB can use the prediction results of virtual grid models and considers UE-specific downlink spectral efficiency during downlink air interface capability evaluation for macro-micro intra-frequency optimal carrier selection only when all the following conditions are met: ● The **DL_INTRA_FREQ_SPCT_EFF_PRED_SW** option is selected. ● The **DL_INTRAF_SE_MODEL_BUILD_ALLOW** or **DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter is selected. ● The **MF_BEAM_CO_DL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-Freq Coordinated Handover Opt Sw	**INTRA_FREQ_COORD_HO_OPT_SW** option of the **NRCellAlgoSwitch**.***IntraFreqHoAlgoSw*** parameter	*Multi-Frequency Smart Selection*	The **SPC_EFF_INTRA_FREQ_COORD_HO_SW** option of the **NRCellAlgoSwitch**.***IntraFreqHoAlgoSw*** parameter can be selected only when the **INTRA_FREQ_COORD_HO_OPT_SW** option is selected.
Low-frequency TDD	Spct-Eff-based Intra-Freq Coord HO Sw	**SPC_EFF_INTRA_FREQ_COORD_HO_SW** option of the **NRCellAlgoSwitch**.***IntraFreqHoAlgoSw*** parameter	*Multi-Frequency Smart Selection*	The **EXTEND_COV_SPCT_EFF_HO_SW** option of the **NRCellAlgoSwitch**.***IntraFreqHoAlgoSw*** parameter can take effect only when the **SPC_EFF_INTRA_FREQ_COORD_HO_SW** option is selected.

6.3.2.2 Mutually Exclusive Functions

None

6.3.2.3 Function Impacts

- Macro-micro intra-frequency optimal carrier selection

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Interference-isolation-oriented inter-frequency handover	**INTRF_ISO_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	When the interference-isolation-oriented inter-frequency handover function takes effect, UEs in macro-micro overlapping areas are transferred to inter-frequency cells, reducing the number of UEs for which macro-micro intra-frequency optimal carrier selection can be triggered.

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.3.4 Networking

Macro-micro intra-frequency optimal carrier selection:

- This function is supported only on networks with intra-frequency TDD cells deployed. In addition, macro and micro cells must be lightly loaded and have overlapping coverage areas.
- The Xn interface between gNodeBs works properly.

6.3.5 Others

The activation of macro-micro intra-frequency optimal carrier selection requires the intelligent macro-micro coordination function of the MAE-Optimization to be enabled. The intelligent macro-micro coordination function is available in MAE V100R023C10 or later.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

It is recommended that the intelligent macro-micro coordination function of the MAE-Optimization be used to activate macro-micro intra-frequency optimal carrier selection. For detailed operations, contact Huawei technical support. **Table 6-3** describes the parameters used for activating the macro-micro intra-frequency optimal carrier selection function. This section does not describe parameters related to cell establishment.

Table 6-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Intra-Freq Coord Handover Mode	**NRCellAlgoSwitch**.***IntraFreqCoordHoMode***	N/A	Use the recommended value.
Intra-Frequency Handover Algorithm Switch	**NRCellAlgoSwitch**.***IntraFreqHoAlgoSw***	INTRA_FREQ_COORD_HO_OPT_SW	Set this option based on the network plan. Select this option if the macro-micro intra-frequency optimal carrier selection function needs to be enabled.

Parameter Name	Parameter ID	Option	Setting Notes
Intra-Frequency Handover Algorithm Switch	**NRCellAlgoSwitch**.***IntraFreqHoAlgoSw***	SPC_EFF_INTRA_FREQ_COORD_HO_SW	Set this option based on the network plan. Select this option if the spectral-efficiency–based intra-frequency handover for macro-micro coordination function needs to be enabled. This option can be selected only when the **INTRA_FREQ_COORD_HO_OPT_SW** option of the **NRCellAlgoSwitch**.***IntraFreqHoAlgoSw*** parameter is selected.
Intra-Frequency Handover Algorithm Switch	**NRCellAlgoSwitch**.***IntraFreqHoAlgoSw***	EXTEND_COV_SPCT_EFF_HO_SW	You are advised to select this option.
HetNet Coordination Cell Flag	**NRCellMultiFResCoord**.***HetNetCoordCellFlag***	N/A	Use the recommended value.
HetNet Coordination Cell Flag	**NRCellRelation**.***HetNetCoordCellFlag***	N/A	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
HetNet Intra-Frequency A3 Offset	**NRCellIntraFHoMeaGrp**.***HetNetIntraFreqA3Offset***	N/A	Set this parameter based on operators' requirements. To prevent ping-pong handovers, it is recommended that the sum of the values of **NRCellIntraFHoMeaGrp**.***HetNetIntraFreqA3Offset*** for the macro and micro cells that have neighbor relationships be greater than or equal to 0.
Intra-freq HO Coverage Extension A3 Offset	**NRCellIntraFHoMeaGrp**.***IntraFreqHoCovExtendA3Ofs***	N/A	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
Measurement Policy Switch	**NRCellAlgoSwitch**.***MeasPolicySwitch***	INDR_OUTDR_OVERLAP_MEAS_SW	This option specifies whether to enable measurement for indoor-outdoor overlapping area evaluation. It is recommended that this option be selected when macro-micro intra-frequency optimal carrier selection or interference-isolation-oriented inter-frequency handover is enabled. If this option is selected, A3 measurement configurations are delivered for evaluation of whether UEs have moved to indoor-outdoor overlapping areas. If this option is deselected, this function does not take effect.
HetNet DL Experience Evaluation Hysteresis	**NRCellCaMgmtConfig**.***HetNetDlExpEvalHyst***	N/A	Use the recommended value.

6.4.1.2 Using MML Commands

Before using MML commands, refer to **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Configuring the macro-micro intra-frequency optimal carrier selection function

It is recommended that the intelligent macro-micro coordination function of the MAE-Optimization be used to activate macro-micro intra-frequency optimal carrier selection, reducing manual operations and improving cell parameter optimization efficiency. When this function is manually activated, related parameters must be configured for both the macro and micro cells, with the only configuration difference between the macro and micro cells lying in the settings of the parameter specifying the HetNet coordination cell flag. The configuration for the micro cell is used for example.

```
//Setting the HetNet coordination cell flag for the serving and neighboring cells
MOD NRCELLMULTIFRESCOORD: NrCellId=0, HetNetCoordCellFlag=MICRO_CO_CELL;
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0,
HetNetCoordCellFlag=MACRO_CO_CELL;
//Setting the intra-frequency coordinated handover mode. The following uses NSA_SA_MODE as an
example.
MOD NRCELLALGOSWITCH: NrCellId=0, IntraFreqCoordHoMode=NSA_SA_MODE;
//Selecting the INTRA_FREQ_COORD_HO_OPT_SW, SPC_EFF_INTRA_FREQ_COORD_HO_SW, and
EXTEND_COV_SPCT_EFF_HO_SW options
MOD NRCELLALGOSWITCH: NrCellId=0,
IntraFreqHoAlgoSw=INTRA_FREQ_COORD_HO_OPT_SW-1&SPC_EFF_INTRA_FREQ_COORD_HO_SW-1&EXTE
ND_COV_SPCT_EFF_HO_SW-1;
//Setting the HetNet intra-frequency A3 offset and the A3 offset for coverage expansion for intra-frequency
handover
MOD NRCELLINTRAFHOMEAGRP: NrCellId=0, IntraFreqHoMeasGroupId=0, HetNetIntraFreqA3Offset=2,
IntraFreqHoCovExtendA3Ofs=2;
//Setting the hysteresis for HetNet downlink experience evaluation
MOD NRCELLCAMGMTCONFIG: NrCellId=0, HetNetDlExpEvalHyst=20;
//Selecting the INDR_OUTDR_OVERLAP_MEAS_SW option
MOD NRCELLALGOSWITCH: NrCellId=0, MeasPolicySwitch=INDR_OUTDR_OVERLAP_MEAS_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Disabling the macro-micro intra-frequency optimal carrier selection function

```
//Deselecting the INTRA_FREQ_COORD_HO_OPT_SW, SPC_EFF_INTRA_FREQ_COORD_HO_SW, and
EXTEND_COV_SPCT_EFF_HO_SW options
MOD NRCELLALGOSWITCH: NrCellId=0,
IntraFreqHoAlgoSw=INTRA_FREQ_COORD_HO_OPT_SW-0&SPC_EFF_INTRA_FREQ_COORD_HO_SW-0&EXTE
ND_COV_SPCT_EFF_HO_SW-0;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

If any of the counters listed in **Table 6-4** returns a non-zero value after the macro-micro intra-frequency optimal carrier selection function is enabled, this function has taken effect.

Table 6-4 Counters related to macro-micro intra-frequency optimal carrier selection

Counter ID	Counter Name
1911840433	**N.HO.IntraFreq.HetNet.Coordination.ExecSuccOut**
1911840434	**N.HO.IntraFreq.HetNet.Coordination.ExecAttOut**
1911840435	**N.HO.IntraFreq.HetNet.Coordination.PrepAttOut**

6.4.3 Network Monitoring

Macro-micro intra-frequency optimal carrier selection:

- For UEs performing full buffer services, observe the benefits of this function by monitoring individual UEs.

  The procedure is detailed as follows: On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management** > **NR** > **User Performance Monitoring** > **User Common Monitoring**, and check the changes of the downlink RLC throughput of individual UEs.
- For eMBB UEs, observe the benefits of this function by monitoring the average downlink data rate of UEs in the macro-micro overlapping areas.

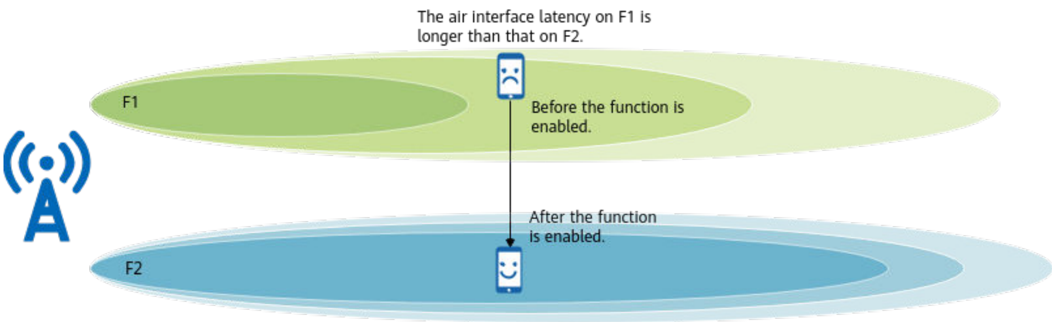
  The average downlink data rate of UEs in the macro-micro overlapping areas (Mbit/s) is the result of 1000 x (**N.Volume.DL.HetNetOverlap – N.Volume.DL.LastSlot.HetNetOverlap**)/ **N.Time.DL.RmvLastSlot.HetNetOverlap**.

7 Latency-based Optimal Carrier Selection

7.1 Principles

In multi-carrier scenarios where the coverage varies with carriers, the air interface latency also varies with carriers for a Cloud X UE at a given UE location. Against this backdrop, the latency-based optimal carrier selection function is introduced to select the latency-optimal carrier and thereby reduce the downlink air interface latency of Cloud X services, as illustrated in **Figure 7-1**. This function supports different latency-based carrier selection policies, and the optimal-carrier selection policy varies depending on the configured latency-based carrier selection policy.

Figure 7-1 Schematic diagram of latency-based optimal carrier selection



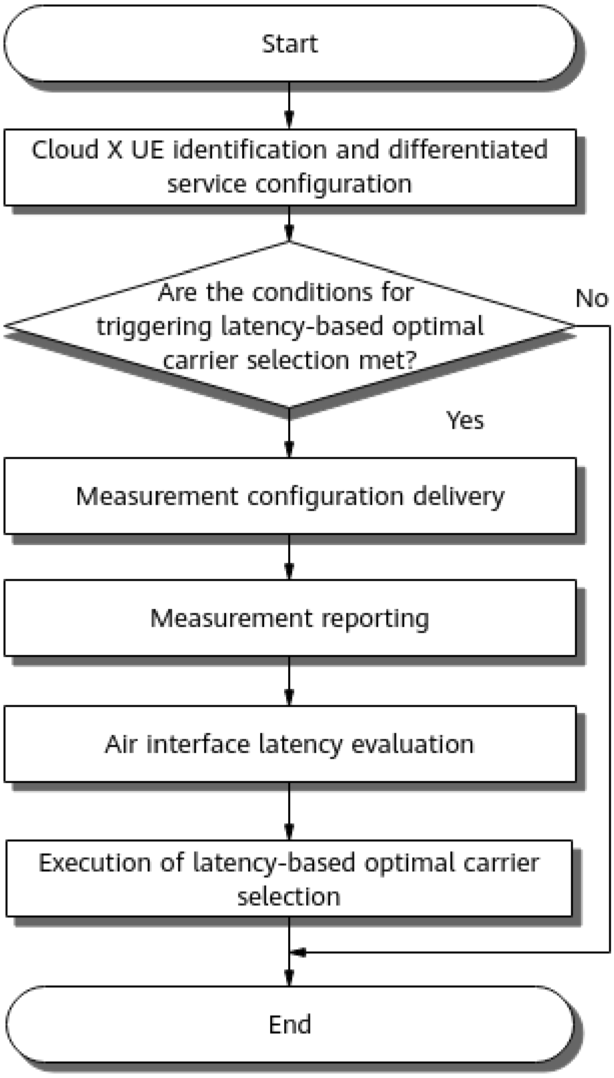
7.1.1 Basic Function of Latency-based Optimal Carrier Selection

After latency-based optimal carrier selection is enabled, the gNodeB selects a carrier allowing for short air interface latency for a UE based on factors including coverage, air interface latency, bandwidth, and load.

This function is controlled by the **DELAY_BASED_CARR_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter.

The basic process of latency-based optimal carrier selection is shown in **Figure 7-2**.

Figure 7-2 Basic process of latency-based optimal carrier selection



1. How the base station identifies Cloud X UEs based on QCIs and differentiated service configuration are described in **7.1.1.1 Cloud X UE Identification and Differentiated Service Configuration**.
2. The scenarios where latency-based optimal carrier selection can be triggered are described in **7.1.1.2 Triggering of Latency-based Optimal Carrier Selection**.
3. How the gNodeB selects target frequencies for measurement is described in **7.1.1.3 Measurement Configuration Delivery**.
4. Measurement reporting from UEs to the gNodeB is described in **7.1.1.4 Measurement Reporting**.
5. How the gNodeB evaluates air interface latency is described in **7.1.1.5 UE-specific Air Interface Latency Evaluation**.
6. How the gNodeB determines whether to perform an inter-frequency handover is described in **7.1.1.6 Execution of Latency-based Optimal Carrier Selection**.

7.1.1.1 Cloud X UE Identification and Differentiated Service Configuration

A gNodeB identifies Cloud X UEs based on QCIs. When the **gNBQciBearer**.***ServiceType*** parameter is set to **EMBB_CLOUD_VR** for a QCI, services with the QCI are identified as Cloud VR services. When the **gNBQciBearer**.***ServiceType*** parameter is set to **EMBB_CLOUD_GAMING** for a QCI, services with the QCI are identified as cloud gaming services. For details about basic Cloud X functions, see *Ultimate Cloud X Experience*.

Operators can require differentiated service experience for different UEs. Related parameters can be set as follows to achieve differentiated service experience for Cloud X UEs:

1. The **NRCellOpPolicy**.***OperatorId*** parameter can be set to different values to differentiate operators, and the **NRCellOpPolicy**.***gNBDiffUserExpGroupId*** parameter is used to identify the differentiated user experience group for a specific operator.
2. The differentiated user experience group ID can be used to associate a specific operator with a gNodeB differentiated experience group configured using the **gNBDiffUserExpGroup** MO. In this MO, the **gNBDiffUserExpGroup**.***gNBCloudXParamGroupId*** parameter can be set to specify a gNodeB Cloud X service parameter group for the operator.
3. The gNodeB Cloud X service parameter group ID can be used to associate the operator with a gNodeB Cloud X service parameter group configured using the **gNBCloudXParamGroup** MO. In this MO, the threshold and policy parameters described in **Table 7-1** are used to configure different thresholds and policies for latency-based optimal carrier selection.

Table 7-1 Parameters related to thresholds and policies used in latency-based optimal carrier selection

Parameter Name	Parameter ID	Description
Delay-based Carrier Selection Policy	DelayBasedCarrSelPolicy	This parameter specifies the latency-based carrier selection policy.
DL Delay-based Carrier Selection Hysteresis	DlDelayCarrSelHys	This parameter specifies the hysteresis for downlink latency-based carrier selection.
Delay-based Carr Sel DL CCE Failure Rate Thld	DlyCarrSelDlCceFailRateThd	This parameter specifies the threshold expressed as the downlink CCE allocation failure rate used in downlink latency-based carrier selection.

Parameter Name	Parameter ID	Description
Delay-based Carr Sel DL PRB Usage Thld	DlyCarrSelDlPrbUsageThld	This parameter specifies the threshold expressed as the downlink PRB usage used in downlink latency-based carrier selection.
Delay-based Carr Sel DL Frame Loss Rate Thld	DlyCarrSelDlFrmLosRateThd	This parameter specifies the threshold expressed as the downlink frame loss rate used in latency-based carrier selection.
Delay-based Carr Sel DL Frame Loss Rate Hyst	DlyCarrSelDlFrmLossRateHys	This parameter specifies the downlink frame loss rate hysteresis used in latency-based carrier selection.

NOTE

In the air interface latency evaluation for an identified Cloud X UE, virtual grid models for both the serving and candidate cells need to be used to predict the downlink spectral efficiency. Therefore, if the base station finds that the virtual grid model for the serving cell has not come online after identifying a Cloud X UE, the process ends and latency-based optimal carrier selection is stopped for the UE. For details about how the base station uses virtual grid models to predict downlink spectral efficiency, see **7.1.1.5 UE-specific Air Interface Latency Evaluation**.

7.1.1.2 Triggering of Latency-based Optimal Carrier Selection

Latency-based optimal carrier selection can be triggered for a UE that meets all of the following conditions:

- The UE is an SA UE.
- The UE is not a VoNR UE. For details about VoNR, see *VoNR*.
- The UE is not a RedCap UE. For details about RedCap, see *RedCap*.

For a UE that meets the preceding triggering conditions, the gNodeB determines whether the UE meets the conditions for starting or stopping latency-based optimal carrier selection based on the frame loss rate within 10 consecutive seconds.

- If the frame loss rate is greater than or equal to the sum of the downlink frame loss rate threshold used in latency-based carrier selection and the downlink frame loss rate hysteresis used in latency-based carrier selection, latency-based optimal carrier selection can be started.

- If the frame loss rate is less than or equal to the downlink frame loss rate threshold used in latency-based carrier selection, latency-based optimal carrier selection can be stopped.

In the preceding information:

- The downlink frame loss rate threshold used in latency-based carrier selection is specified by the **gNBCloudXParamGroup**.***DlyCarrSelDlFrmLosRateThd*** parameter.
- The downlink frame loss rate hysteresis used in latency-based carrier selection is specified by the **gNBCloudXParamGroup**.***DlyCarrSelDlFrmLossRateHys*** parameter.

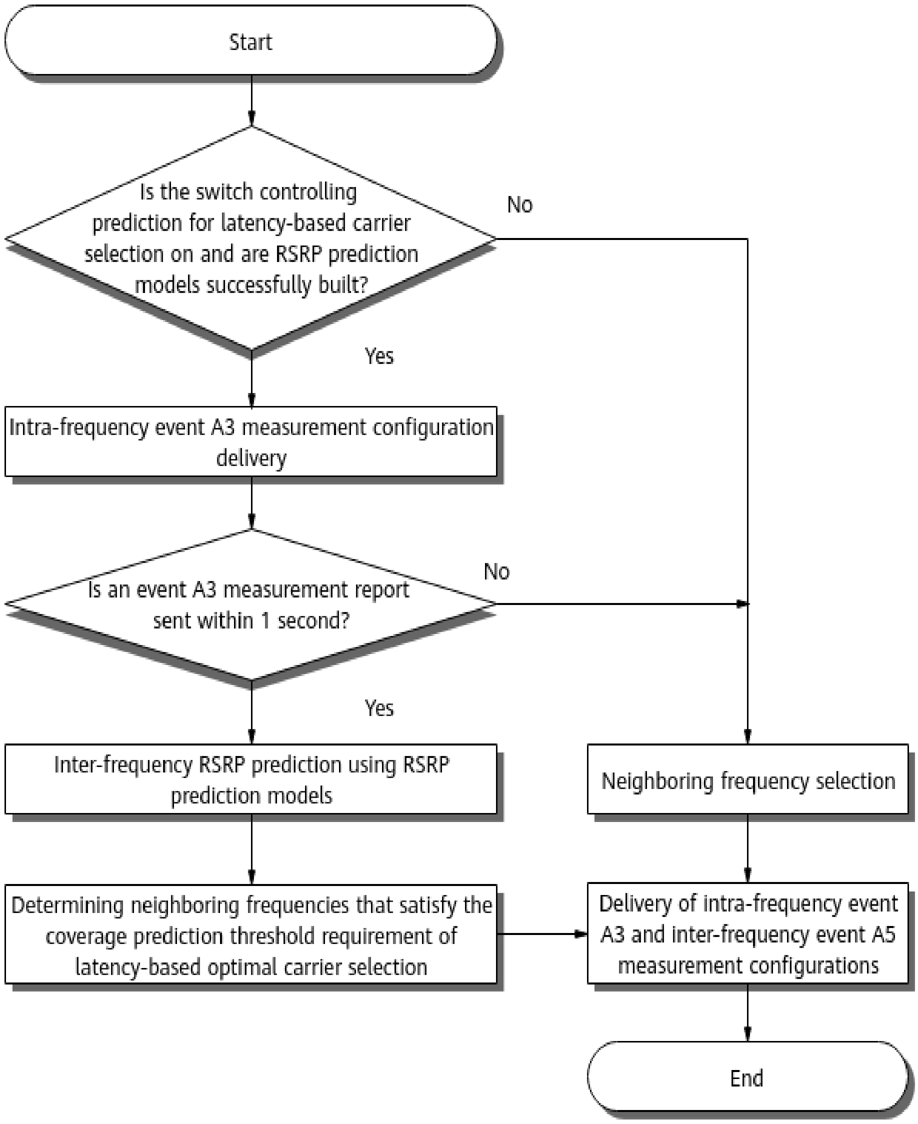
NOTE

The frame loss rate indicates the proportion of the number of frames whose delay exceeds the threshold specified by the **gNBQciBearer**.***CloudXFrmLossDelayThld*** parameter to the total number of transmitted frames within a period.

7.1.1.3 Measurement Configuration Delivery

When latency-based optimal carrier selection is triggered for a UE, the gNodeB selects neighboring frequencies for measurement based on the setting of the switch controlling prediction for latency-based carrier selection and RSRP prediction models. It then delivers inter-frequency event A5 measurement configurations related to the frequencies to be measured all together to the UE. (Threshold 1 for event A5 is –43 dBm, and threshold 2 for event A5 is –140 dBm.) **Figure 7-3** shows the process. The switch controlling prediction for latency-based carrier selection is specified by the **DELAY_BASED_CARR_SEL_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter. For details about RSRP prediction models, see *Virtual Grid-based Multi-Frequency Coordination*.

Figure 7-3 Process of measurement configuration delivery



The process is as follows:

1. Checking the setting of the switch controlling prediction for latency-based carrier selection and the building status of RSRP prediction models
   - If the switch is on and RSRP prediction models are successfully built, the gNodeB delivers intra-frequency event A3 measurement configurations and starts a 1s timer.
   - If the switch is off or no RSRP prediction model is successfully built, the process proceeds to **3**.
2. Checking whether the UE sends an event A3 measurement report within 1s
   - If the UE sends an event A3 measurement report within 1s, the gNodeB retrieves the RSRP values of the serving cell and intra-frequency neighboring cells from the measurement report and inputs them into the corresponding RSRP prediction models to predict the RSRP values of inter-frequency neighboring cells. The gNodeB then filters out the neighboring frequencies of cells whose RSRP values are less than the coverage

prediction threshold for latency-based optimal carrier selection. The gNodeB can select a maximum of seven neighboring frequencies for measurement. After the selection, the process proceeds to **4**.

   – If the UE does not send an event A3 measurement report within 1s, the process proceeds to **3**.

3. Selecting neighboring frequencies based on certain principles
   – The base station selects the NR frequencies specified by the **NRCellFreqRelation**.***SsbDescMethod*** and **NRCellFreqRelation**.***SsbFreqPos*** parameters for measurement configuration delivery.
   – The base station selects a maximum of seven frequencies in ascending order of frequency numbers.
4. Delivering intra-frequency event A3 configurations and inter-frequency event A5 configurations

   The gNodeB delivers intra-frequency event A3 measurement configurations and inter-frequency event A5 measurement configurations related to the frequencies to be measured all together to the UE.

NOTE

The coverage prediction threshold for latency-based optimal carrier selection is specified by the **NRCellInterFHoMeaGrp**.***DlDlyInterFreqA5RsrpThld2*** parameter.

7.1.1.4 Measurement Reporting

The UE performs measurements based on the received event configurations and reports measurement results.

The base station determines whether to start air interface latency evaluation for the UE based on whether the UE successfully sends event A5 measurement reports.

- If the UE fails to send event A5 measurement reports for all neighboring frequencies contained in the event A5 measurement configurations within the time in which the gNodeB waits for the reports, the gNodeB deletes the inter-frequency event A5 measurement configurations for all the neighboring frequencies so that the UE stops the current inter-frequency event A5 measurements and the process ends.
- If the UE sends event A5 measurement reports for all neighboring frequencies contained in the event A5 measurement configurations within the time in which the gNodeB waits for the reports, air interface latency evaluation will start. For details about air interface latency evaluation, see **7.1.1.5 UE-specific Air Interface Latency Evaluation**.
- If the UE fails to send an event A5 measurement report for a neighboring frequency contained in the event A5 measurement configurations within the time in which the gNodeB waits for the reports, the gNodeB deletes the inter-frequency event A5 measurement configuration for this neighboring frequency so that the UE stops the current inter-frequency event A5 measurements. Then, air interface latency evaluation is started for the UE. For details about air interface latency evaluation, see **7.1.1.5 UE-specific Air Interface Latency Evaluation**.

When air interface latency evaluation is allowed for a UE, the base station determines the information to input into the corresponding virtual grid models based on whether the UE successfully sends an event A3 measurement report.

- If the UE sends an event A3 measurement report within the time in which the base station waits for the report, the base station inputs the SSB beam information contained in the measurement report into the corresponding virtual grid model (an intra-frequency spectral efficiency prediction model or a downlink-frequency spectral efficiency model) for UE-specific downlink spectral efficiency prediction of the serving cell, as described in **7.1.1.5 UE-specific Air Interface Latency Evaluation**.
- If the UE fails to send an event A3 measurement report within the time in which the base station waits for the report, the base station retrieves the SSB beam information of the serving cell from the event A5 measurement report with the largest serving cell RSRP among the received event A5 measurement reports, and then inputs the retrieved information into the corresponding virtual grid model (an intra-frequency spectral efficiency prediction model or a downlink-frequency spectral efficiency model) for UE-specific downlink spectral efficiency prediction of the serving cell, as described in **7.1.1.5 UE-specific Air Interface Latency Evaluation**.

NOTE

- The length of time in which the gNodeB waits for intra-frequency event A3 measurement reports is the same as that of time in which the gNodeB waits for inter-frequency event A5 measurement reports.
- The time in which the gNodeB waits for inter-frequency A5 measurement reports is related to the number of measured neighboring frequencies.

7.1.1.5 UE-specific Air Interface Latency Evaluation

The gNodeB evaluates the air interface latency in the UE's serving cell and in the cells selected based on the received measurement reports and then selects an optimal carrier based on the evaluation results. The selected cells must meet specific conditions and come from the cells each with the highest RSRP value on the corresponding neighboring frequencies in inter-frequency event A5 measurement reports.

The cells involved in the air interface latency evaluation vary depending on UE states as follows:

- For a single-carrier UE, if the RSRP of the cell with the largest RSRP on a neighboring frequency is greater than RSRP threshold 2 for triggering event A5 related to downlink-delay-based inter-frequency handovers (specified by the **NRCellInterFHoMeaGrp**.***DlDlyInterFreqA5RsrpThld2*** parameter), the cell can be selected as a candidate cell. Air interface latency in this candidate cell and that in the serving cell will be evaluated.
- For a multi-carrier UE, if the RSRP of the cell with the largest RSRP on a neighboring frequency is greater than RSRP threshold 2 for triggering event A5 related to downlink-delay-based inter-frequency handovers (specified by the **NRCellInterFHoMeaGrp**.***DlDlyInterFreqA5RsrpThld2*** parameter), the cell can be selected as a candidate cell. Air interface latency in this candidate cell and that in the serving PCell will be evaluated.

Candidate Cell Selection

A candidate cell must meet all of the following conditions:

- The **CLOUD_VR_SERVICE_SWITCH** option of the **NRCellAlgoSwitch**.***CloudVrSwitch*** parameter is selected.
- The downlink PRB usage is less than or equal to the downlink PRB usage threshold for latency-based carrier selection (specified by the **gNBCloudXParamGroup**.***DlyCarrSelDlPrbUsageThld*** parameter).
- The downlink CCE allocation failure rate is less than or equal to the downlink CCE allocation failure rate threshold for latency-based carrier selection (specified by the **gNBCloudXParamGroup**.***DlyCarrSelDlCceFailRateThd*** parameter).

Aspects Considered in Air Interface Latency Evaluation

The air interface latency in the serving cell and that in a candidate cell are evaluated from the following aspects:

- Downlink frame size: The base station measures the sizes of downlink frames of Cloud X UEs per second.
- UE-specific downlink spectral efficiency: Assume that the switch controlling the use of virtual-grid-predicted downlink spectral efficiency for downlink experience evaluation (specified by the **DL_EXP_EVAL_SPCT_EFF_PRED_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter) is on. The gNodeB retrieves the SSB beam information of the cell with the largest RSRP value on a frequency from the corresponding measurement report and inputs the information into the virtual grid model for this frequency to obtain the UE-specific downlink spectral efficiency in the cell. The virtual grid model should be a downlink intra-frequency spectral efficiency prediction model or a downlink-frequency spectral efficiency model. For details about virtual grid model building, see *Virtual Grid-based Multi-Frequency Coordination*.
- Downlink target IBLER:
  - If the **DL_IBLER_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter is selected, the downlink target IBLER is specified by the **gNBDUSchParamGroup**.***DlTargetIbler*** parameter.
  - If the **DL_IBLER_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter is deselected, the downlink target IBLER is specified by the **NRDUCellPdsch**.***DlTargetIbler*** parameter.
- Equivalent downlink cell bandwidth: This bandwidth is calculated based on factors such as the downlink cell bandwidth, slot configuration, and UE-supported downlink bandwidth.
- Downlink load factor: This factor is calculated based on the number of UEs to be scheduled in the downlink per second.
- Downlink CCE allocation failure rate: This rate is calculated based on the total number of downlink CCE allocation times and the number of successful downlink CCE allocation times. Downlink CCE allocation failures affect the downlink scheduling delay of UEs.

7.1.1.6 Execution of Latency-based Optimal Carrier Selection

The carrier with the optimal latency is selected based on the air interface latency evaluation result.

An inter-frequency handover will be performed for the UE if the UE still meets the conditions for starting latency-based optimal carrier selection and the following condition is met: (Estimated latency in the serving cell – Estimated latency in the candidate cell)/Estimated latency in the serving cell > Downlink latency-based carrier selection hysteresis (specified by the **gNBCloudXParamGroup**.***DlDelayCarrSelHys*** parameter). Otherwise, the handover will not be performed.

7.1.2 Policies of Latency-based Optimal Carrier Selection

The latency-based carrier selection policy is specified by the **gNBCloudXParamGroup**.***DelayBasedCarrSelPolicy*** parameter.

- If this parameter is set to **DELAY_FIRST**, the delay-first policy will be used. With this policy, the base station selects optimal carriers for all UEs based on latency. For details, see **7.1.2.1 Delay-First Policy**.
- If this parameter is set to **NEAR_POINT_FIXED_ANCHOR**, the cell-center fixed anchor policy is used. With this policy, the base station performs latency-based optimal carrier selection only for cell-edge UEs, but does not trigger inter-frequency handovers of cell-center UEs for latency-based optimal carrier selection. For details, see **7.1.2.2 Cell-Center Fixed Anchor Policy**.

7.1.2.1 Delay-First Policy

The delay-first policy takes effect if latency-based optimal carrier selection is enabled and the **gNBCloudXParamGroup**.***DelayBasedCarrSelPolicy*** parameter is set to **DELAY_FIRST**. With this policy, the base station selects an optimal carrier for any UE based on latency, no matter whether the UE is a cell-center or cell-edge UE. For details about optimal carrier selection in this policy, see the basic process described in **7.1.1 Basic Function of Latency-based Optimal Carrier Selection**.

7.1.2.2 Cell-Center Fixed Anchor Policy

The cell-center fixed anchor policy takes effect if latency-based optimal carrier selection is enabled and the **gNBCloudXParamGroup**.***DelayBasedCarrSelPolicy*** parameter is set to **NEAR_POINT_FIXED_ANCHOR**.

The cell-center fixed anchor policy differentiates between cell-center UEs and cell-edge UEs. Therefore, cell-center UEs and cell-edge UEs must be identified before carriers are selected.

Cell-Center UE and Cell-Edge UE Identification

Event A1 and event A2 configurations are delivered to RRC_CONNECTED UEs in order to identify UEs as cell-center or cell-edge UEs. The process is as follows:

1. After a UE accesses a cell, the gNodeB identifies it as a cell-edge UE by default and delivers an event A1 configuration to it.

2. After receiving an event A1 report, the gNodeB identifies the UE as a cell-center UE, deletes the event A1 configuration, and delivers an event A2 configuration.
3. After receiving an event A2 report, the gNodeB identifies the UE as a cell-edge UE, deletes the event A2 configuration, and delivers the event A1 configuration.

Carrier Selection for Cell-Center UEs and Cell-Edge UEs

When latency-based optimal carrier selection is triggered for a UE, the selection is performed based on its cell-center or cell-edge status:

- If the UE is a cell-center UE, an inter-frequency handover will not be performed for the UE for latency-based optimal carrier selection.
- If the UE is a cell-edge UE, an optimal carrier is selected based on the delay-first policy as described in **7.1.1 Basic Function of Latency-based Optimal Carrier Selection**.

7.2 Network Analysis

7.2.1 Benefits

After latency-based optimal carrier selection is enabled, the gNodeB considers a variety of factors including coverage, air interface latency, bandwidth, and load when it selects carriers allowing for optimal latency experience for UEs. This shortens the user-perceived latency of Cloud X UEs and increases the number of UEs that meet the downlink frame loss rate threshold of Cloud X services. This function brings greater benefits to networks with more frequencies, larger coverage difference between frequency bands, larger load difference between cells, or larger UE traffic volumes.

7.2.2 Impacts

The impacts between this function and other functions are described in **7.3.2.3 Function Impacts**.

Latency-based optimal carrier selection has the following impacts on the network:

- Inter-band UE distribution changes. The number of signaling messages over the air interface and the number of handovers increase. The service drop rate, access success rate, and handover success rate fluctuate.
- When the number of Cloud X UEs increases, more downlink resources will be occupied, and the resources available for other UEs will decrease. As a result, the throughput of eMBB services decreases.
- In light-load scenarios where multiple frequencies are used and the cells working on these frequencies differ greatly in their bandwidths, if the positive gain of latency-based optimal carrier selection is less than the negative impact brought by the increase in measurement overheads, the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) and average uplink UE throughput (indicated by **User Uplink Average Throughput (DU)**) decrease.

- Periodic latency evaluation for UEs increases the CPU usage of the main control board.
- Latency-based optimal carrier selection depends on successfully built virtual grid models. If the characteristics input into a virtual grid model when the model is used are greatly different from the characteristics of samples used for model training (for example, the samples used for model training include data of neighboring cells, but neighboring cells cannot be measured when the model is used), the UE-specific downlink spectral efficiency predicted by the model is inaccurate, affecting the latency evaluation accuracy. If the downlink spectral efficiency of a UE varies greatly between frequencies due to factors such as frequency deployment and UE mobility, the latency evaluation result may be greatly inaccurate.
- Because cloud gaming services occupy a small number of RBs, air interface capability improvement of carriers does not significantly reduce the latency and the latency difference between carriers is small for cloud gaming services. Therefore, latency-based optimal carrier selection is not recommended for the cloud gaming service scenario. Otherwise, the average uplink and downlink UE throughputs may decrease.

7.3 Requirements

7.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-091209	Latency-based Optimal Carrier Selection (NR)	NR0S00DSCS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL Intra-Freq Spct Eff Mdl Build Allowed	**DL_INTRAF_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	Latency-based optimal carrier selection requires that the **DL_INTRAF_SE_MODEL_BUILD_ALLOW** or **DL_FREQ_SE_MODEL_BUILD_ALLOW** option be selected.
Low-frequency TDD	DL Freq Spct Eff Mdl Build Allowed	**DL_FREQ_SE_MODEL_BUILD_ALLOW** option of the **NRCellSmartMultiCarr**.***ModelBuildingSwitch*** parameter	*Virtual Grid-based Multi-Frequency Coordination*	
Low-frequency TDD	Latency-based optimal carrier selection	**DELAY_BASED_CARR_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	The switch controlling prediction for latency-based carrier selection can be turned on only after latency-based optimal carrier selection is enabled.
Low-frequency TDD	Cloud X service	**CLOUD_VR_SERVICE_SWITCH** option of the **NRCellAlgoSwitch**.***CloudVrSwitch*** parameter	*Ultimate Cloud X Experience*	Latency-based optimal carrier selection can take effect only after the Cloud X service switch is turned on.

7.3.2.2 Mutually Exclusive Functions

None

7.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Frequency-priority-based inter-frequency handover	**FREQ_PRIORITY_BASED_HO** option of the **NRCellAlgoSwitch.*InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode*	• After a UE is handed over to a cell for latency-based optimal carrier selection, frequency-priority-based inter-frequency handover does not take effect for the UE in this cell. • For a UE for which latency-based optimal carrier selection has not taken effect, frequency-priority-based inter-frequency handover takes effect when the relevant handover conditions are met.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA-UE-specific frequency-priority-based inter-frequency handover	**CA_FREQ_PRI_BASED_INTERF_HO_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*CA Experience Improvement*	● After a UE is handed over to a cell for latency-based optimal carrier selection, CA-UE-specific frequency-priority-based inter-frequency handover does not take effect for the UE in this cell. ● For a UE for which latency-based optimal carrier selection has not taken effect, CA-UE-specific frequency-priority-based inter-frequency handover takes effect when the relevant handover conditions are met.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NSA carrier combination selection based on user experience	• NR: **NSA_LTE_SEL_OPT_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter • LTE: **NSA_LTE_FDD_SEL_OPT_SW** or **NSA_LTE_TDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter	*EPC-based NSA Experience Improvement*	When both latency-based optimal carrier selection and NSA carrier combination selection based on user experience are enabled, latency-based optimal carrier selection does not take effect for NSA UEs.
Low-frequency TDD	VoNR	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	Latency-based optimal carrier selection does not take effect for VoNR UEs.
Low-frequency TDD	RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	Latency-based optimal carrier selection does not take effect for RedCap UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	EPLMN-based intra-RAT inter-PLMN handover	**INTRA_RAT_HO_WITH_GNB_EPLMN_SW** option of the **gNodeBParam**.***EqvPlmnAlgoSwitch*** parameter	*SA Mobility Management in Connected Mode*	When both EPLMN-based intra-RAT inter-PLMN handover and latency-based optimal carrier selection are enabled, the list of PLMNs to which a UE can be handed over contains only the serving PLMN.

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

Only the UMPTg, UMPTe, UMPTga, UMPTha, and UMPThb series boards support this function. UMPTe series boards support this function only in NR-only scenarios. The other series boards support this function in both NR-only and multimode co-MPT scenarios.

All NR-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.3.4 Networking

- The network uses at least two frequencies.
- The Xn interface between gNodeBs is normal so that user experience evaluation information can be transmitted.

7.3.5 Others

- The switch for latency-based optimal carrier selection does not need to be turned on in candidate inter-frequency cells if latency-based optimal carrier selection already takes effect in a single cell, as UEs in this cell can be proactively handed over to other cells based on this function.
- If a cell with latency-based optimal carrier selection disabled is activated first, for example, when its serving base station is reset, the cell cannot serve as a candidate carrier for latency evaluation within 20 minutes after the activation.

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

Table 7-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment and basic Cloud X functions.

Table 7-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
gNodeB Diff User Experience Group ID	**NRCellOpPolicy**.***gNBDiffUserExpGroupId***	N/A	Set this parameter based on the network plan. If this parameter is set to a value other than **255**, ensure that an object has been created for the gNodeB differentiated user experience group identified by this parameter.

Parameter Name	Parameter ID	Option	Setting Notes
gNodeB Diff User Experience Group ID	**gNBDiffUserExpGroup**.***gNBDiffUserExpGroupId***	N/A	Set this parameter to the same value as the **NRCellOpPolicy**.***gNBDiffUserExpGroupId*** parameter.
User Grade by QCI	**gNBDiffUserExpGroup**.***UserGradeByQci***	N/A	Set this parameter based on the network plan. If the **gNBDiffUserExpGroup**.***gNBDiffUserExpGroupId*** parameter is set to the same value for different gNodeB differentiated user experience groups (configured using **gNBDiffUserExpGroup** MOs), the **gNBDiffUserExpGroup**.***UserGradeByQci*** parameter in these MOs must be set to different values.
gNodeB Cloud X Parameter Group ID	**gNBDiffUserExpGroup**.***gNBCloudXParamGroupId***	N/A	Set this parameter based on the network plan. If this parameter is set to a value other than **255**, ensure that an object has been created for the gNodeB Cloud X parameter group identified by this parameter.
Service Type	**gNBQciBearer**.***ServiceType***	N/A	It is recommended that this parameter be set to **EMBB_CLOUD_VR** to support Cloud VR services.

Parameter Name	Parameter ID	Option	Setting Notes
CloudX Resource Control DL PRB Threshold	**NRDUCellDlSch**.***CloudXResCtrlDlPrbThld***	N/A	Use the recommended value.
CloudX Frame Loss Delay Thld	**gNBQciBearer**.***CloudXFrmLossDelayThld***	N/A	Use the recommended value.
gNodeB Cloud X Parameter Group ID	**gNBCloudXParamGroup**.***gNBCloudXParamGroupId***	N/A	Set this parameter to the same value as the **gNBDiffUserExpGroup**.***gNBCloudXParamGroupId*** parameter.
Cloud X QCI	**gNBCloudXParamGroup**.***CloudXQci***	N/A	Set this parameter based on the network plan.
Delay-based Carrier Selection Policy	**gNBCloudXParamGroup**.***DelayBasedCarrSelPolicy***	N/A	Use the recommended value.
DL Delay-based Carrier Selection Hysteresis	**gNBCloudXParamGroup**.***DlDelayCarrSelHys***	N/A	Use the recommended value.
Delay-based Carr Sel DL CCE Failure Rate Thld	**gNBCloudXParamGroup**.***DlyCarrSelDlCceFailRateThd***	N/A	Use the recommended value.
Delay-based Carr Sel DL Frame Loss Rate Thld	**gNBCloudXParamGroup**.***DlyCarrSelDlFrmLosRateThd***	N/A	Use the recommended value.
Delay-based Carr Sel DL Frame Loss Rate Hyst	**gNBCloudXParamGroup**.***DlyCarrSelDlFrmLossRateHys***	N/A	Use the recommended value.
Delay-based Carr Sel DL PRB Usage Thld	**gNBCloudXParamGroup**.***DlyCarrSelDlPrbUsageThld***	N/A	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	DELAY_BASED_CARR_SEL_SW	Select this option.
Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.***MultiFreqAlgoSwitch***	DELAY_BASED_CARR_SEL_PRED_SW	If latency-based optimal carrier selection is enabled, select this option for frequency selection.
DL Delay-based Inter-Freq A5 RSRP Thld2	**NRCellInterFHoMeaGrp**.***DlDlyInterFreqA5RsrpThld2***	N/A	Use the recommended value.

7.4.1.2 Using MML Commands

Before using MML commands, refer to **7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring bearers for Cloud VR services
MOD GNBQCIBEARER: Qci=0, ServiceType=EMBB_CLOUD_VR;
//(TDD) Configuring a gNodeB Cloud X parameter group (the delay-first policy of latency-based optimal
carrier selection is used as an example)
ADD GNBCLOUDXPARAMGROUP: gNBCloudXParamGroupId=0, CloudXQci=0,
DelayBasedCarrSelPolicy=DELAY_FIRST, DlDelayCarrSelHys=30, DlyCarrSelDlCceFailRateThd=20,
DlyCarrSelDlFrmLosRateThd=5, DlyCarrSelDlFrmLossRateHys=2, DlyCarrSelDlPrbUsageThld=45;
//Configuring a gNodeB differentiated user experience group
ADD GNBDIFFUSEREXPGROUP: PolicyIndex=0, gNBDiffUserExpGroupId=0, UserGradeByQci=0,
gNBCloudXParamGroupId=0;
//Configuring an operator-specific policy for the NR cell
ADD NRCELLOPPOLICY: NrCellId=0, OperatorId=0, gNBDiffUserExpGroupId=0;
//Configuring the downlink PRB usage threshold for Cloud X resource control
MOD NRDUCELLDLSCH: NrDuCellId=0, CloudXResCtrlDlPrbThld=100;
//Configuring the frame loss delay threshold for Cloud VR services
MOD GNBQCIBEARER: Qci=0, CloudXFrmLossDelayThld=10;
//Turning on the DELAY_BASED_CARR_SEL_SW
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=DELAY_BASED_CARR_SEL_SW-1;
//Turning on the DELAY_BASED_CARR_SEL_PRED_SW to reduce invalid inter-frequency measurements when
the switch determining whether to allow virtual grid model building is on
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=DELAY_BASED_CARR_SEL_PRED_SW-1;
//Configuring RSRP threshold 2 for event A5 related to downlink-delay-based inter-frequency handovers
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, DlDlyInterFreqA5RsrpThld2=-104;
```


Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the DELAY_BASED_CARR_SEL_PRED_SW
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=DELAY_BASED_CARR_SEL_PRED_SW-0;
//Turning off the DELAY_BASED_CARR_SEL_SW
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=DELAY_BASED_CARR_SEL_SW-0;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

If any of the counters listed in **Table 7-3** returns a non-zero value, latency-based optimal carrier selection has taken effect.

Table 7-3 Counters related to latency-based optimal carrier selection

Counter ID	Counter Name
1911849169	**N.HO.InterFreq.DelayCarrierSel.PrepAttOut**
1911849168	**N.HO.InterFreq.DelayCarrierSel.ExecAttOut**
1911849167	**N.HO.InterFreq.DelayCarrierSel.ExecSuccOut**

7.4.3 Network Monitoring

After this function is enabled, the latency of Cloud X services can be reduced to achieve short latency and a decreased number of service freezing occasions.

- Use drive tests to observe the gains for a single UE. Specifically, observe the freezing and latency of Cloud X services on the UE side.
- Use the video service KPI evaluation capability on the core network side or use a third-party video service KPI evaluation device to obtain video KPIs and MTP delay.

8 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

NOTE

You can find the EXCEL files of parameter reference for the software version used on the live network from the product documentation delivered with that version.

FAQ: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-020100.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

9 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

10 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

11 Reference Documents

- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *SA Mobility Management in Connected Mode*
- *NSA Mobility Management*
- *EPC-based NSA Basics*
- *EPC-based NSA Experience Improvement*
- *Carrier Aggregation*
- *CA Experience Improvement*
- *Flexible User Steering*
- *Multi-Operator Sharing*
- *VoNR*
- *High Speed Mobility*
- *Hyper Cell*
- *Cell Combination*
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *NSA-SA Selection Based on User Experience*
- *Mobility Load Balancing in Connected Mode*
- *Virtual Grid-based Multi-Frequency Coordination*
- *Super Uplink*
- *ANR*
- *PCI Conflict Detection and Self-Optimization*
- *MIMO (TDD)*
- *5G Networking and Signaling*
- *RedCap*
- *MRO*
- *Energy Conservation and Emission Reduction*
- *Ultimate Video Experience*
- *Multi-Frequency Smart Aggregation*

- *Fusion Cell (Low-Frequency TDD)*
- *License Control Item Lists*
- *Ultimate Cloud X Experience*
- *Scalable Bandwidth*
- *UL and DL Decoupling*